Bringing Services Closer to People? Distance to Administrators and the Reach of the State*

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Abstract

As populations grow, extending the reach of the state to a larger and more dispersed population becomes increasingly challenging. Many developing countries have responded by creating new subnational administrative units, bringing local administrators physically closer to citizens. Leveraging an episode of district proliferation in Uganda, this paper studies whether geographical proximity to local government administrative centers improves public service delivery and changes citizens' relationship with the state. To establish causality, I exploit a policy implementation rule that generated exogenous variation in households' proximity to new district headquarters. I find that households living closer to these centers have better access to electricity, roads, health facilities, and water sources. Beyond service delivery, proximity also shifts engagement from contentious to participatory channels and reduces direct experiences of corruption. These effects are stronger in politically aligned districts, where elected and appointed district leaders share political incentives. However, improvements in service delivery and citizen engagement do not necessarily translate into more favorable perceptions of the government. Even where these effects are strongest, proximity does not increase trust in the government or improve democratic perceptions. Finally, extending administrative reach comes at a fiscal cost. Each district split raises transfers to the pre-split territory by roughly 14 percent, with most of the increase concentrated in administrative expenditures rather than service delivery budgets.

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1 Introduction

As populations grow, governments face a fundamental challenge: how to maintain the reach of the state over a larger number of citizens while extending it to communities where government presence remains limited. This is particularly relevant in rural areas of developing countries, where distance can make it harder for governments to learn about local needs, deliver public services, and interact with citizens. These pressures are especially salient in sub-Saharan Africa. The region's population is projected to increase by two-thirds by 2050 (UN, 2024), and its rural population is still expanding (UN, 2018). Governments have responded in part by multiplying subnational administrative units. Since 1990, almost half of sub-Saharan African countries have increased their number of administrative units by at least 20 percent (Grossman & Lewis, 2014).[1] Rather than necessarily devolving new powers, these reforms reorganize the state spatially, creating new administrative centers closer to previously remote populations (Grossman & Lewis, 2014), often with the explicit objective of improving public service delivery (GoU, 1995).

The key margin changed by these reforms is the physical distance between citizens and local administrators. Empirical work shows that distance from political and administrative centers can weaken institutional penetration, political accountability, and public service provision (Michalopoulos & Papaioannou, 2014; Campante & Do, 2014; Asher et al., 2018). Administrative unit creation changes this relationship directly by establishing new centers of government, turning citizens' distance to local administration into a policy margin. Theories of decentralization suggest that bringing government closer can improve information about heterogeneous local preferences and strengthen citizen engagement (Tiebout, 1956; Oates et al., 1972; Alesina & Spolaore, 1997; Faguet, 2023). Shorter distances may also facilitate interactions with government, increase policymakers' responsiveness, and improve access to public services. Yet proximity need not translate into better government. Creating smaller units may eliminate economies of scale and raise administrative costs (Oates et al., 1972), intensify local elite capture (Bardhan & Mookherjee, 2000), or generate administrative capacity constraints (Cohen, 2024). Moreover, in hybrid regimes, bringing government closer may extend central control rather than empower citizens (Martinez-Bravo et al., 2022), particularly where elected local governments coexist with centrally appointed officials (Martinez-Bravo, 2014). Whether deliberately reducing the distance between citizens and administrators strengthens the reach of the state, and how citizens respond to it, therefore remains an open empirical question.

This paper studies whether geographical distance to local government administrative centers

[1] The pattern is not specific to Africa. Indonesia's districts rose from 302 in 1999 to 514 in 2014 (Bazzi & Gudgeon, 2021), Brazil's municipalities from 4,124 in 1989 to 5,507 in 1997 (Dahis & Szerman, 2026), India's districts from 466 in 1991 to 640 in 2011 (Rajan & Malghan, 2022), and Vietnam's provinces from 40 to 64 (Law, 2026). The United States went through the same process while its population was rural and growing: the number of counties rose from 292 in 1790 to 2,713 in 1900 (U.S. Bureau of the Census, 1994). Appendix C reports the series by country (Table A1).

affects the provision of public goods and services in the context of administrative unit proliferation and subdivision in Uganda. This country provides a relevant setting for several reasons. First, Uganda has gone through a large decentralization process, resulting in a rapid expansion of sub-national administrative units (e.g., the number of districts has grown from 38 in 1991 to 136 in 2020). Second, the national government established rules for the geographical location of new administrative headquarters, providing exogenous variation in distance to administrative centers. Third, the availability of geospatial data, combined with comprehensive data on households' access to and satisfaction with public goods provision, offers a unique opportunity to gain a broad overview of the effects of distance to administrators on public service delivery in a redistricting context.

To establish causality, I use an instrumental variable approach leveraging an implementation rule in the process of district creation, which introduced exogenous variation in distance to the new district headquarters. Specifically, governmental guidelines established that district headquarters should be located in the geographical middle of newly created districts (Green, 2010). Building on a strategy used by Campante & Do (2014) for the US, I use the distance from each household's enumeration area to its district's centroid, interacted with an indicator for districts created between 1991 and 2020, as an instrument for the household's distance to the district capital. To implement this strategy, I construct a granular spatial dataset that combines *(i)* rich household microdata on access to public services from the 2021 National Service Delivery Survey (NSDS) and the 2024 Afrobarometer, *(ii)* geospatial data and administrative records identifying newly created districts between 1991 and 2020, *(iii)* geocoded distances from each household's enumeration area to its district's headquarters and centroid, and *(iv)* administrative records of local-government budgets from the Ministry of Finance, Planning and Economic Development. I complement the main analysis with the 2015 NSDS and the 2017 Afrobarometer to examine how the effects of proximity evolved over time, as new district institutions became more established. Together, these data allow me to study how distance to the district's administrative center affects households' access to public goods and services, as well as their relationship with the state.

The findings show that geographical proximity to new district headquarters significantly improves access to public goods and services. Households closer to the district capital have better access to electricity and paved roads, shorter travel times to health facilities, and higher access to drinking water. The effects are strongest for centrally provided services, reflecting both direct infrastructure investments near new headquarters and the coordinating role of district administrators in channeling national programs to nearby communities. These effects also vary with the political organization of district government. In politically aligned districts, where the elected district chairperson and the centrally appointed Resident District Commissioner (RDC) belong to the same party, proximity produces larger gains in centrally provided infrastructure, particularly roads. By contrast, the effects on district-provided services differ little with political alignment.

Various tests provide empirical support for the identification strategy. First, I examine whether the identifying variation is independent of pre-existing local conditions. The geographic centroids of new districts, which determine the instrument, are not systematically located in more developed sub-counties (as measured by census living standards, night-time lights, and population density), consistent with district boundaries not being strategically drawn to include more developed areas. Moreover, parishes later selected as district headquarters show no differential trends in cropland or built-up area before their designation. I also address non-random household exposure to the geometry generated by redistricting (Borusyak & Hull, 2023). Controlling for the expected instrument across feasible counterfactual redistricting configurations leaves the estimates stable or larger, while randomization inference over 1,000 counterfactual configurations supports the first stage and the main service results. Second, I examine whether the instrument captures broader geographical centrality rather than proximity to the district administration itself. The first stage is unchanged when controlling for proximity to major urban centers, and instrumented proximity to district headquarters has no systematic effect on households' knowledge of or distance to lower-tier local governments (LCI, LCII, and LCIII), supporting the interpretation that the identifying variation is specific to the district level. The findings are also robust to placebo outcomes, alternative measures of proximity and the instrument, excluding households near headquarters towns, restricting the sample to newly created districts, weak-instrument-robust inference, and corrections for multiple hypothesis testing. Two complementary exercises provide further evidence. The main service results also hold in the 2024 Population and Housing Census, using data from more than 900,000 households. In addition, a geographic regression discontinuity design comparing enumeration areas across newly created district boundaries provides consistent evidence using a different source of identifying variation.

I leverage a unique dataset on service delivery to study two potential mechanisms behind these results: citizen engagement and accountability. On engagement, proximity increases state presence by raising citizens' knowledge of district level institutions and reducing reliance on traditional alternatives. As the state becomes more accessible, contentious engagement falls significantly: households closer to headquarters are less likely to join protests or contact the media. The evidence on participatory engagement is more suggestive, but the coefficients consistently point toward greater involvement in local decision making. These changes are concentrated in politically aligned districts, where proximity produces a stronger shift away from contentious engagement and toward formal interactions with district institutions. On accountability, proximity reduces direct corruption experiences even as awareness of corruption remains high, with the strongest reductions again appearing in aligned districts. Satisfaction with the RDC also increases with proximity, particularly where the RDC and elected district chairperson are politically aligned.

Despite these improvements in services, engagement, and accountability, proximity does not translate into higher trust in government or more favorable democratic perceptions. Satisfaction

with democracy, perceived freedom to vote and organize, rule of law evaluations, and trust in local and national institutions are all largely unaffected. Importantly, these null effects persist in aligned districts, even though this is where the gains in services and citizen-state interaction are most pronounced. In non-aligned districts, proximity is instead associated with lower perceived freedom to vote and declining trust in local government and the ruling party. These results suggest that improvements in government presence and performance do not necessarily translate into greater political legitimacy, and that the political environment conditions how citizens respond to a more proximate state.

I also examine how the effects of proximity evolve as new district administrations become established. Using two rounds of the National Service Delivery Survey, I document that the service delivery effects strengthened between 2015 and 2021. While only agricultural inputs responded to proximity in 2015, by 2021 significant gains had emerged for health, water, and electricity access. Complementary remote-sensing evidence shows that economic activity and built-up area around new headquarters also increase gradually after their establishment. Together, these patterns are consistent with the effects of administrative proximity strengthening as new district institutions consolidate.

Finally, I quantify the fiscal cost of this proliferation strategy using the approved local-government budget estimates published by the Ministry of Finance, Planning and Economic Development for fiscal years 2012/13 to 2023/24 (Ministry of Finance, Planning and Economic Development, 2024). I aggregate intergovernmental transfers to fixed territories defined by the 1991 district borders and estimate a staggered difference in differences model showing that real transfers rise by roughly 14 percent once a territory is first split. This increase is concentrated in duplicated administrative payroll rather than in conditional service delivery grants. The composition of these additional resources suggests that the service gains are not mechanically driven by larger service delivery budgets. At the same time, it shows that bringing government closer through district creation requires a fiscally costly expansion of the administrative apparatus.

This paper contributes to three strands of literature. First, it contributes to the literature on state capacity and the spatial reach of the state. While a large literature studies the determinants and consequences of state capacity (Besley & Persson, 2009; Finan et al., 2017; Acemoglu et al., 2015; Dell et al., 2018; Nathan, 2023), a related literature emphasizes how distance from political and administrative centers can weaken the state's ability to reach citizens (Michalopoulos & Papaioannou, 2014; Campante & Do, 2014). Closest to this paper are studies examining how distance to administrative centers shapes state presence and local development. Mastrorocco & Teso (2026) show that shorter travel times to Washington increased federal presence and bureaucratic professionalization in the nineteenth-century United States, while Müller-Crepon (2023) documents that shorter travel times to capitals are associated with better development outcomes across Africa. At the local level, Asher et al. (2018) use discontinuities across Indian district borders to show that administrative remoteness reduces access to public goods

and worsens economic outcomes, while related work examines how distance to local state institutions shapes traditional authority across Africa (Henn, 2023). Another strand studies the consequences of gaining administrative capital status, showing that it can shift economic activity toward newly designated centers (Kulka & Smith, 2024; Bluhm et al., 2026). While these studies establish the importance of administrative geography, they primarily examine the consequences of distance to existing centers or the concentration of economic activity following capital designation. This paper contributes by bringing causal evidence from a natural experiment on the effect of geographical distance to administrative centers, within a context of rapid district proliferation, on access to public goods and services. I further go beyond existing work by identifying two channels through which proximity operates: citizen engagement with local government and accountability, in a setting where redistricting is widespread and policy-relevant for many African countries.

Second, this paper contributes to the literature on decentralization and administrative unit proliferation. District creation aim to bring local government structures closer to citizens without devolving additional powers to local governments, distinguishing it from decentralization reforms that change the distribution of authority across levels of government (Cheema & Rondinelli, 2007). Existing work documents the political incentives behind the creation of new administrative units, including patronage, recentralization, and the convergence of national and local interests (Green, 2010; Lewis, 2018; Grossman & Lewis, 2014). Evidence on the consequences of proliferation is mixed. Cross-country evidence shows that service provision increases with the number of administrative units, but at decreasing rates (Grossman et al., 2017). Country studies find that district splits improved service delivery in Brazil (Dahis & Szerman, 2026), but not in Burkina Faso (Billing, 2019), and did not reduce conflict in Indonesia (Bazzi & Gudgeon, 2021). Related work shows that smaller local governments provide more public goods in India (Narasimhan & Weaver, 2024). In Uganda, Cohen (2024) finds that the 2009–10 district splits increased frontline staffing but slowed improvements in infrastructure and economic development in newly created districts relative to districts that did not split. Unlike these studies, which estimate the overall effects of district creation, this paper contributes by isolating the spatial margin directly changed by administrative unit creation. I further examine how these benefits evolve as new administrations become established and document the additional fiscal resources required to expand the administrative apparatus.

Lastly, this paper contributes to the literature on the political economy of decentralization and local accountability, particularly in settings where elected local governments coexist with centrally appointed officials. Central governments can design decentralization reforms to retain political control (Martinez-Bravo et al., 2022), while elected and appointed local officials respond to different political incentives (Martinez-Bravo, 2014). Political alignment across levels of government also shapes resource allocation and policy implementation (Brollo & Nannicini, 2012; Asher & Novosad, 2017). In Uganda, Raffler (2022) finds that lowering the cost of political oversight improves service quality only in local governments not aligned with the

ruling party. I show that the effects of administrative proximity on centrally provided infrastructure, citizen engagement, and some measures of corruption are stronger where the elected district chairperson is politically aligned with the appointed Resident District Commissioner. Whereas political oversight appears more effective where local politicians have incentives to check the central government, the benefits of administrative proximity are stronger where district leaders share political incentives, consistent with a greater role for coordination. I further show that improvements in service delivery and changes in citizen engagement do not translate into greater trust in government or more favorable democratic perceptions, even in politically aligned districts. In non-aligned districts, some of these attitudes instead deteriorate, consistent with evidence that improvements in government performance do not necessarily generate greater political support in non-democratic settings (Besley et al., 2026). These findings suggest that extending the administrative reach of the state can improve public service delivery without necessarily strengthening political legitimacy.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 summarizes the conceptual framework guiding the empirical analysis, with a formal model presented in the Appendix A. Section 4 details the data construction and displays descriptive statistics for the main variables. Section 5 outlines the empirical framework. Section 6 presents the main results, while sections 7 – 10 detail the mechanisms, effects on democratic perceptions, dynamic effects, and fiscal costs, respectively. Section 11 concludes.

2 Institutional Background

The decentralization process in Uganda began in 1987 with the establishment of a five-tier administrative structure consisting of villages (LCI), parishes (LCII), sub-counties (LCIII), counties (LCIV), and districts (LCV), with districts being the largest administrative unit. While each level has distinct operational responsibilities, local government councils, which serve as the highest political authority within their jurisdictions, operate specifically at the district and sub-county levels. These councils typically consist of a chairperson, vice-chairperson, and five councilors, though additional members may be included, such as representatives of women, youth, and people with disabilities. They are responsible for planning and implementing public service-related projects within their jurisdictions. At the district level, two key figures share administrative responsibilities: the Local Council V (LCV) chairperson, who is directly elected by local voters and leads the district council, and the Resident District Commissioner (RDC), who is appointed by the President and represents the central government within the district. While the LCV is accountable to local constituents, the RDC is accountable upward to the national government, and the degree of political alignment between these two figures can shape how effectively district resources are coordinated and services delivered.

The constitutional framework for decentralization and devolution of powers in Uganda was established in the 1995 Ugandan Constitution and the 1997 Local Governments Act. Notably,

the 1995 constitution granted the central government the authority to create sub-national units. Article 179 states that: "Parliament may: (a) alter the boundaries of districts; and (b) create new districts [...] Any measure for the alteration of the boundaries of the creation of districts or administrative units shall be based on the necessity for effective administration and the need to bring services closer to the people" (GoU, 1995). This marked a turning point in the decentralization policy, as, since 1997, district creation increased substantially from 39 districts in 1994 to 112 in 2014, reaching a total of 136 districts in 2020.[2]

The 1997 Local Governments Act define the division of service delivery responsibilities between district governments and the central state (GoU, 1997). Districts are mandated to provide primary education, primary health care through health centers and district hospitals, rural water supply for domestic use, agricultural extension services, district and feeder roads, and community development services (Ahmad et al., 2009). The central government is responsible for the provision of services with national reach such as the national and trunk-road network, electricity transmission and rural electrification, regional and national referral hospitals, and large-scale water infrastructure (Mushemeza, 2019). This institutional structure motivates the categorization of outcomes by the level of government primarily responsible for delivery, distinguishing between services provided by district governments and those under central mandate.

The legal procedure for creating new local government units involves several stages. First, local residents, through their council representative, present a request to the district council. If approved, a resolution to create a new district is submitted to the Ministry of Local Government. After approval, the proposal is referred to the Cabinet for its evaluation. If the Cabinet supports the request, it is then submitted to Parliament for final approval (Awortwi & Helmsing, 2014). Upon successful completion of the process, the district issuing the request is split, allowing the splitter district to establish a new administrative system. Once a district is approved, the new administrative unit is authorized to recruit over 200 technical and administrative staff, a process that has been linked to patronage networks (Green, 2010) and electoral incentives, including vote-buying (Awortwi & Helmsing, 2014). Importantly, the governmental guidelines establish that new districts should locate their headquarters in the geographical middle of the newly created district (Green, 2010).[3]

According to the Ugandan Constitution, the main objectives of creating new administrative units are political devolution and improving public service delivery. Political devolution aims to empower leaders at lower administrative levels, enabling them to manage local affairs more effectively (Mushemeza, 2019). This process is legitimatized by local elections held every four years, which have witnessed significant participation in recent periods. Simultaneously, the

[2]The literature does not have a consensus on the underlying motives for splitting a district. While Green (2010) argues that there is no concrete evidence that districts have been created in deprived areas, Grossman & Lewis (2014) suggests that districts with politically marginalized minorities were more likely to be split.

[3]Fieldwork interviews with district officials across twelve districts confirm that the centrality norm is a consistent practice. Officials in every district visited stated that the council deliberately chose a central location to guarantee equal access for local leaders and citizens attending meetings at the headquarters.

creation of new administrative units seeks to enhance public service delivery through a bottom-up approach, where local governments provide opportunities for local participation, leading to better utilization of available resources (Mushemeza, 2019).

3 Conceptual Framework

As described in the previous section, the creation of new districts in Uganda introduces two administrative figures with different accountability relationships operating simultaneously at the district level: the elected LCV chairperson, who is accountable to local voters, and the centrally appointed RDC, who represents the national government. The geographical placement of the new district headquarters determines the physical distance between households and these administrators and the administrative services they are in charge of. This section summarizes the conceptual framework that guides the empirical analysis, describing how proximity to the administrative center and political alignment between the LCV and RDC jointly shape service delivery, citizen engagement, and accountability. A formal model, building on Martinez-Bravo et al. (2022), developing these predictions is presented in Appendix A, with proofs in Appendix B.

Provision of public goods and services. Administrative effort translates into household-level access to public goods and services. Administrators choose an overall level of investment for the district, but the household-level return to these investments depends on distance to the headquarters. Infrastructure radiates outward from the administrative center: electricity grids, paved roads, health facilities, and water sources are more accessible near the headquarters. While the marginal welfare gains from public goods may be largest in remote, underserved areas, the spatial decay of delivery means these gains are the hardest to realize. This argument motivates creating new headquarters closer to previously remote areas.

The degree of political alignment between the LCV and RDC affects the total effort directed toward locally valued services. When both administrators share political affiliation, their incentives are aligned: the elected LCV channels local preferences into service delivery decisions while the appointed RDC coordinates the allocation of centrally financed programs, and both pull in the same direction. When they are not aligned, the RDC's career concerns and political loyalty rewards may be larger than local welfare, reducing the effort the RDC devotes to services valued locally. Importantly, distance and alignment interact. Proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. This complementarity predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

Citizen engagement. The quality of public service delivery depends not only on administrative effort but also on the information that administrators receive about local needs. A central insight of the decentralization literature is that citizens have an informational advantage regarding local conditions and the performance of local officials (Björkman & Svensson, 2009). How

effectively citizens can exercise this advantage depends on how they choose to engage with the state, which is shaped by the costs they face.

Citizens can engage through participatory channels, such as attending district council meetings, meeting the RDC or the LCV, or using formal feedback mechanisms, which transmit detailed information about local priorities directly to decision-makers but require traveling to the headquarters. Alternatively, they can engage through contentious channels, like joining protests, contacting the media, or organizing collective pressure campaigns, which are noisier but can be pursued locally at costs that do not depend on distance. This cost structure implies that proximity reduces the relative cost of participatory engagement, shifting citizens from contentious to formal channels. The pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it toward channels that are more informative for administrators. Political alignment further shapes this choice. When the RDC internalizes local welfare, the state is more responsive to citizen input, increasing the returns to participatory engagement. In non-aligned districts, the lower responsiveness of the state reduces these returns, and citizens may not shift away from contentious channels even when the headquarters is nearby. Together, these mechanisms predict that proximity should reduce contentious engagement and increase participatory engagement, and that this shift should be more pronounced in aligned districts.

Accountability. A further channel through which proximity affects service delivery is the monitoring of local officials. Citizens who live closer to the headquarters interact more frequently with district institutions and are better positioned to observe and report misconduct, raising the expected cost of corruption for administrators. Two forces reinforce each other in deterring corrupt behavior: external monitoring by citizens, which increases with proximity, and internal alignment with local welfare, which increases with political congruence between the LCV and RDC. In aligned districts, the RDC internalizes the welfare cost of resource diversion, making corruption less attractive even when citizen monitoring is imperfect. These mechanisms predict that proximity should reduce corruption experiences and that the reduction should be stronger in politically aligned districts.

Taken together, the framework identifies three reinforcing channels (provision, engagement, and accountability) through which proximity to the administrative center improves public service delivery. All three channels are amplified by political alignment between the elected and appointed district leaders. The main results in Section 6 test the distance predictions, while Section 7 examines the engagement and accountability channels. The alignment predictions are tested alongside each set of results, using the specification described in Section 5, which splits the sample by political congruence between the LCV chairperson and the RDC.

4 Data

To study the effects of the geographical distance between citizens and public administrators in the Ugandan redistricting context, I combine various sources of geolocalized secondary data,

as well as earth observation data. This section describes the data sources and construction as well as the main variables of interest.

4.1 Sources and Dataset Construction

4.1.1 Household Survey Data

The main data source is the 2021 National Service Delivery Survey (NSDS), conducted by the Uganda Bureau of Statistics on behalf of the Ministry of Public Services to assess the efficiency and effectiveness of service delivery from the perspectives of both providers and recipients (UBOS, 2022). The survey covers 10,246 households across 1,088 enumeration areas in all 136 districts as of 2020 and is, by design, representative of each district. Of these households, 9,313 form the sample for analysis after excluding those with insufficient geolocalization data and households belonging to Kampala and its metropolitan area Wakiso.[4] The survey provides rich household and community-level information on education, health, water, electricity, roads, and agricultural inputs, covering services provided by both district and central governments. I complement the analysis using the 2015 wave of the NSDS to examine how the effects of proximity evolve over time. Detailed variable definitions and index construction are described in Appendix D.

Measures of citizen engagement, corruption experiences, and democratic perceptions come from the tenth round of the Afrobarometer survey, conducted in Uganda in 2024. The Afrobarometer is a nationally representative survey covering 2,400 respondents in 105 districts and more than 270 enumeration areas, providing detailed information on citizens' interactions with government institutions, experiences of bribery and petty corruption, satisfaction with democracy, perceptions of the rule of law, and trust in local and national institutions. While the NSDS offers broader household coverage and richer service delivery measures, the Afrobarometer provides attitudinal outcomes that are not available in the NSDS, making the two surveys complementary.

Finally, I use the 10 percent samples of the 2014 and 2024 National Population and Housing Censuses, collected by UBOS. The 2024 sample covers 1.07 million households (4.69 million individuals) and contains information on housing conditions, access to basic services, asset ownership, education, health, and civil registration. Since the census does not provide household coordinates, I assign households to the centroid of their parish when parish identification

[4]These households are excluded as the capital city service delivery is governed by a separate institutional framework under the Kampala Capital City Authority. Additionally, as noted by Cohen (2024), Kampala is an extreme outlier across a variety of service access indicators. In addition, for eleven enumeration areas, the survey's administrative district label does not match the district in which the enumeration area's coordinates are located. For nine of these cases, the coordinates fall within a neighboring district that was carved from the same 1991 mother district, so I use the district implied by the spatial coordinates. The remaining two enumeration areas are dropped because their coordinates lie implausibly far (more than 100 kilometers) from the district reported in the survey.

is available, and to the sub-county centroid otherwise (74% identified at the parish level, 26% at the sub-county), clipping the unit polygons to the shoreline before taking the centroid. I use the 2024 census to replicate the main service results in a larger, nationally representative sample (Appendix E.5), and the 2014 census to measure living standards before the creation of districts established from 2014 onwards.

4.1.2 Distance Variables

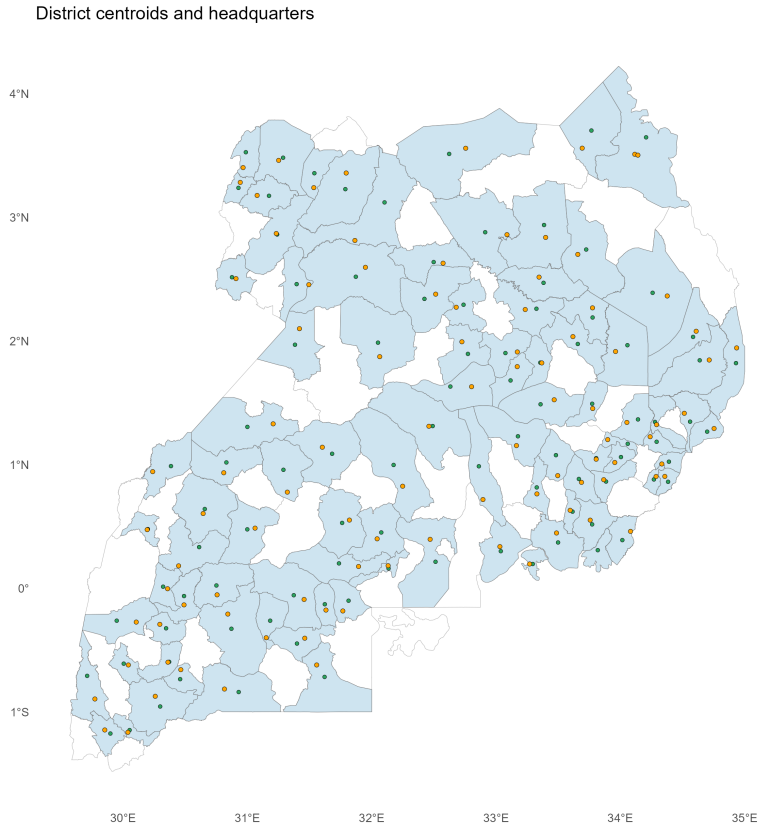
To assess the impact of a household's proximity to the district's administrative center on public service delivery, I leverage a government policy mandating district headquarters to be located near the district's geographical center. Within these guidelines, however, the specific placement of headquarters may also be influenced by factors such as road access, proximity to other constituencies, and the availability of infrastructure. These factors could introduce bias when analyzing the effects of proximity on service provision. To address this, I compute the straight-line distance from each household's enumeration area to the district's centroid, which serves as an instrument for household proximity to the district headquarters. Figure 1 illustrates the spatial relationship between headquarters and district centroids for newly created districts.

Two key proximity variables are constructed from the household questionnaire. *SR Proximity Household – District Headquarters* is defined as the negative log of the NSDS self-reported distance (in kilometers) from the household to the district capital. *Proximity Household – District Centroid* is the negative log of the straight-line distance between the household's enumeration area geocoordinates and the district's centroid. Relying on proximity calculations using log distance throughout ensures that higher values of all proximity variables correspond to households located closer to the reference point, so that a positive coefficient in the regression analysis is consistent with the expected beneficial effects of administrative proximity.

4.1.3 Evolution of Districts

I combine geospatial data with national district registries to document geographical changes associated with decentralization and district splits between 1991 and 2020. The geospatial data, obtained from UBOS, includes the geographical borders of districts for different years, enabling the identification of district splits and distinguishing between mother and splitter districts. The geographical data is complemented with administrative records on the creation of districts for years in which district's borders are not available. This dataset enables the construction of the variable $NewDistrict_{1991-2020}$, an indicator variable that takes the value of one for newly created districts between 1991 and 2020.

FIGURE 1: SPATIAL DISTRIBUTION OF NEW DISTRICTS' HEADQUARTERS



Notes: The map displays new districts (shaded) and old districts (both mother and never split districts) for the period 1991-2020. Green dots mark district centroids and orange dots district headquarters.

4.1.4 Remote Sensing Data

I combine household and community survey data with three remote sensing products to capture dynamic development patterns across land use, economic activity, and population over time. First, I use land classification data from the MODIS Land Cover Type product, which provides annual observations at a 500-meter resolution from 2001 to 2023.[5] From these data I construct parish-level variables capturing the share of land under cultivated cropland and cropland mixed with natural vegetation.[6] Second, to proxy for local economic activity and urbanization, I use the harmonized DMSP-OLS/VIIRS nighttime lights for 2020. I aggregate mean radiance to the parish level. Third, I measure local population using the Gridded Population of the World (GPW v4.11) from CIESIN/SEDAC, which provides population count and density estimates at

[5]The MODIS product classifies land cover using the IGBP scheme into 17 categories. I focus on cultivated cropland (>60% cultivated) and cropland mixed with natural vegetation (40–60% cultivation), as these classes together account for the majority of rural land cover in Uganda and provide the most informative measure of agricultural intensification and local economic activity around newly established administrative centers. Changes in built-up area are studied separately using the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S), available at 100-meter resolution in five-year epochs from 1990 to 2025, whose long span provides genuine pre-periods for the headquarters cohorts.

[6]Parishes are the lowest administrative unit with stable boundary data.

a 30-arc-second (1 km) resolution for 2020. These parish-level datasets are used as evidence of non-strategic placement of district headquarters.

4.1.5 Qualitative Fieldwork

The empirical analysis is further supported by fieldwork that was conducted in Uganda from July to August 2024. The fieldwork covered twelve districts spanning both old (created in or before 2010) and newly created (after 2010) administrative units. In each district, I conducted semi structured interviews with district-level officials, including Chief Administrative Officers, district planners, finance officers, and Resident District Commissioners (RDCs), as well as with Local Council 1 (LC1) leaders and households in towns located both near to and far from district headquarters. This design comparing officials and citizens across old and new districts and close and far towns was intended to capture qualitative variation along the dimensions exploited in the empirical analysis. The evidence gathered informs the institutional background (Section 2), motivates several robustness checks (Appendix E), and informs the analysis of the engagement, accountability, and political alignment channels.

4.2 Descriptive Statistics

Table 1 contains summary statistics, detailing information on distance variables, the main service delivery-related outcomes, and mechanisms.

In the NSDS, households report living, on average, 16 kilometers from their district headquarters. Using the enumeration area's geocoordinates, the average straight-line distance to the district headquarters is approximately 12 kilometers, while the corresponding distance to the district centroid is 14.6 kilometers. Under the 1991 district boundaries, these same households would have been, on average, 25.5 kilometers from their district headquarters, approximately 60 percent farther. Nearly half of the households in the sample belong to districts created between 1991 and 2020, reflecting the substantial redistricting that took place over this period.

Turning to the main outcomes, access to services is limited. While 71 percent of households with primary school age children report school attendance, and 86 percent of households that vaccinated their children in the past year did so at a government facility, infrastructure access is considerably lower. Only 16 percent of households have access to improved roads and 47 percent have electricity. Access to improved water sources during the dry season reaches 83 percent, with households traveling 0.8 kilometers on average to their water source. The outcome indices used in the main analysis aggregate these and other related measures. They are centered at zero by construction, with standard deviations between 0.41 and 0.78, reflecting meaningful variation in service access across households.[7] Among the mechanism variables, 25 percent of households report involvement in LCV meetings and 10 percent report having met

[7] Appendix Table A2 lists the components included in each index.

TABLE 1: Summary Statistics

	N	Mean	SD	Min	Max
Panel A: Proximity Variables					
Distance HH – District Headquarters (km)	8640	16.03	14.59	0.10	90.00
Distance HH – District Headquarters 1991 (km)	8640	25.51	22.45	0.04	105.02
Distance EA – District Headquarters (km)	9315	12.31	10.45	0.05	84.90
Distance EA – District centroid (km)	9315	14.58	8.30	0.28	57.07
New district (1991–2020)	9503	0.50	0.50	0.00	1.00
Panel B: Main Outcomes (NSDS 2021)					
School attendance (prim. age)	4726	0.71	0.41	0.00	1.00
Gov. health facility (immunization)	2361	0.86	0.34	0.00	1.00
Improved water source (dry s.)	8738	0.83	0.37	0.00	1.00
Water distance (dry s., km)	7474	0.79	0.92	0.00	5.00
Improved roads	8731	0.16	0.37	0.00	1.00
HH has electricity	8752	0.47	0.50	0.00	1.00
Education index (district)	7811	0.00	0.73	-2.53	0.92
Health index (district)	8400	-0.01	0.49	-3.37	2.21
Water index (district)	9472	0.00	0.47	-2.29	1.19
Roads index (district)	9471	0.00	0.41	-1.42	2.26
Agricultural inputs index (district)	8752	0.03	0.63	-0.47	2.40
Education index (central government)	6009	0.00	0.73	-2.26	0.83
Health index (central government)	9315	0.03	0.78	-0.89	2.43
Water index (central government)	9436	-0.01	0.53	-7.89	0.62
Roads index (central government)	9471	-0.06	0.73	-2.52	1.08
Electricity index (central government)	9482	-0.03	0.61	-2.26	1.92
Panel C: Mechanisms (NSDS 2021)					
Involved in LCV meetings	8752	0.25	0.43	0.00	1.00
Household meets with RDC	8752	0.10	0.30	0.00	1.00
Victim of nepotism	8752	0.03	0.18	0.00	1.00
Victim of absenteeism	8752	0.07	0.26	0.00	1.00
Satisfied with RDC performance	8752	0.33	0.47	0.00	1.00
Family uses community services	8728	0.06	0.23	0.00	1.00
Knows magistrate court	8729	0.38	0.48	0.00	1.00
Panel D: Mechanisms (Afrobarometer Round 10)					
Civic and collective action	2182	0.36	0.48	0.00	1.00
Engagement in protest	2184	0.04	0.19	0.00	1.00
Bribe for medical care	1604	0.41	0.49	0.00	1.00
Trust in local government	2132	0.51	0.50	0.00	1.00
Trust in ruling party	2184	0.59	0.49	0.00	1.00
Panel E: Local leader's political alignment (LCV and RDC)					
Aligned in 2016 - 2021	9524	0.74	0.44	0.00	1.00

Notes: Summary statistics for the main variables used in the analysis. Panels A–C use NSDS 2021 data. Panel D uses Afrobarometer Round 10 data.

the RDC, while direct experiences of corruption are relatively low: 3 and 7 percent of households report being victims of nepotism and absenteeism respectively. Satisfaction with the RDC is reported by 33 percent of households, and 38 percent of households know their magistrate court, suggesting moderate but heterogeneous awareness of district-level institutions.

In the Afrobarometer sample, 36 percent of respondents report engaging in collective civic action, while protest participation is considerably lower at 4 percent. Bribery for medical care is reported by 41 percent of respondents who sought health services, pointing to substantial corruption in healthcare access. Trust in local government stands at around 50 percent, while trust in the ruling party is somewhat higher at 59 percent. Finally, 74 percent of households in the sample are located in politically aligned districts, where the LCV chairperson and RDC share the same political affiliation.

5 Empirical Framework

Measuring the effects of geographical proximity to government administrative centers is challenging because the location of these centers is typically endogenous. Administrative centers may be located in areas with higher population density, greater economic activity, better road access, or higher existing service provision. At the same time, household remoteness may be correlated with unobserved household and local characteristics that also affect the outcomes of interest. As a result, distance to administrative centers may capture factors other than administrative proximity itself.

To circumvent these challenges, this paper leverages quasi-random variation in households' proximity to district headquarters generated by Uganda's redistricting process and the rules governing the placement of new administrative centers. According to the governmental guidelines, the headquarters of a newly created district should be located near its geographical center (Green, 2010). As a result, the geographical shape of the new district helps determine where its headquarters is located and, consequently, how close households are to it. Following the strategy used by Campante & Do (2014) for the United States, I use households' proximity to the district centroid as a source of exogenous variation in proximity to the headquarters. Specifically, I interact proximity to the district centroid with an indicator for districts created between 1991 and 2020, which isolates the variation generated by the placement rule in newly created districts while accounting for differences in pre-split remoteness. Formally, the instrumental variable first-stage and second-stage are given by:

$$\begin{aligned} ProximityHH_HQ_{i,d} = \gamma_0 &+ \gamma_{CN}(ProximityHH_Centroid \times NewDistrict_{1991-2020})_{i,d} \\ &+ \gamma_C ProximityHH_Centroid_i + \gamma_N NewDistrict_{1991-2020,d} \\ &+ \mathbf{X}'_{i,d}\gamma_X + \upsilon_{i,d} \end{aligned} \tag{1}$$

$$Y_{i,d} = \alpha + \beta \widehat{ProximityHH_HQ}_{i,d} + \beta_C ProximityHH_Centroid_i + \beta_N NewDistrict_{1991-2020,d} + \mathbf{X}'_{i,d}\beta_X + \varepsilon_{i,d}, \quad (2)$$

where $ProximityHH_Centroid_i$ is the log proximity of household i's enumeration area to the district centroid, $NewDistrict_{1991-2020,d}$ indicates whether district d was created between 1991 and 2020, their interaction is the excluded instrument, $ProximityHH_HQ_{i,d}$ is the household's self-reported proximity to the district headquarters, and $\mathbf{X}_{i,d}$ is a vector of controls.[8] Finally, $\upsilon_{i,d}$ is the error term clustered at the parish level.

The second-stage estimates in Equation 2 capture the causal effects of households' proximity to the district headquarters on public service delivery outcomes, $Y_{i,d}$, including education, health, water, electricity, roads, and agricultural inputs. Throughout the analysis, the key variable of interest is *proximity* to the district headquarters, so that the coefficient of interest, β, has a straightforward interpretation: a positive estimate indicates that households located closer to the district headquarters experience better public service delivery outcomes, consistent with the government's stated objective of bringing services closer to citizens through district creation.

The conceptual framework in Section 3 distinguishes between two administrators with different accountability structures operating at the district level. The elected LCV chairperson is responsible for coordinating district-mandated services, while the appointed RDC is responsible for channeling centrally financed programs. To map this distinction to the empirical analysis, I construct outcome indices separately for services under district mandate (primary education, district health facilities, rural water supply, district and feeder roads, and agricultural extension and inputs) and services under central government mandate (secondary education, national and referral hospitals, large-scale water infrastructure, the national roads network, and electricity) (GoU, 1997, Second Schedule, Part 2). This distinction allows me to examine whether the effects of administrative proximity differ across levels of government and whether these effects vary with political alignment between the elected and appointed district leaders.

Heterogeneity by political alignment. I classify a district as aligned when the LCV chairperson and the RDC belonged to the same party following the 2016 elections.[9] For each set of outcomes, a companion table reports the 2SLS estimates of the effect of proximity in non-aligned districts, the interaction between proximity and alignment, and the sum of these two coefficients (which gives the effect in aligned districts). For outcomes for which the framework predicts the sign of the interaction term, the tables also report the predicted sign and the corresponding one-sided p-value (significance stars correspond to two-sided tests). In addition, the

[8] Controls include district ever split between 1991-2020, Lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), presence of lakes other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. Lake controls account for districts where water bodies occupy a large share of the territory, which affects the location and interpretation of the geographical centroid.

[9] In the sample, 101 of the 134 districts are classified as aligned.

tables report the Kleibergen–Paap F-statistic for each group's first stage separately. It ranges between 9 and 17 in the NSDS, but falls to between 3 and 9 in the aligned Afrobarometer sample, with the lowest values for the bribery items, whose samples are restricted to respondents who sought the corresponding service. The Afrobarometer panels therefore also report Anderson–Rubin 95 percent confidence intervals, which remain valid under weak instruments. Appendix Tables A14 and A15 report these confidence intervals for every coefficient in the alignment tables.

Identifying assumptions. For the instrumental variables estimates to be interpreted causally, three main assumptions must hold: relevance, independence, and the exclusion restriction.

The relevance assumption holds as long as policymakers comply with the governmental guidelines. If this is the case, households' proximity to a district's centroid should increase the likelihood of being closer to the district's headquarters. This assumption is directly testable and is examined in Section 5.1.

The independence assumption rests on the arbitrariness of the new district's centroid. As the location of a district's centroid depends exclusively on the geographical shape of the new administrative unit, it is unlikely to be related to unobservable characteristics that determine public service delivery outcomes, conditional on the set of controls. In other words, new districts' centroids are located as good as randomly, since they only depend on the respective district's shape.

For the exclusion restriction to hold, the interaction between a household's distance to the district's centroid and the indicator for districts created between 1991 and 2020, conditional on the set of controls, should affect public service delivery outcomes only through the household's distance to the district headquarters. This appears plausible, as a household's distance to the new district's centroid is unlikely to affect any relevant outcome other than through access to administration and related services (Campante & Do, 2014) once its distance to the pre-split district headquarters is controlled for. Restricting the identifying variation to newly created districts addresses concerns associated with pre-existing remoteness (i.e., households that were remote both before and after belonging to a new district).

A set of controls helps strengthen the plausibility of the exclusion restriction. All specifications control for whether a district was split between 1991 and 2020. As the size of a district could affect both the decision to split it and the distance to its centroid, specifications control for district area at the beginning of the period (as of 1991). Controlling for households' distance to district capitals as of 1991 captures potential confounders arising from differential proximity to pre-existing urban centers. Finally, controls for the presence of water bodies, elevation, and macro-regions capture potential confounders coming from geographical factors.

Threats to validity. The main threat is that district boundaries are drawn so that the centroid, and hence the headquarters, falls near a favored group, in which case the centroid would no longer be an arbitrary geometric point. Three institutional features make this unlikely. District creation follows a multi-stage legal procedure in which the initial petition depends on the cooperation of local council representatives, limiting both Parliament's ability to select constituencies and any single actor's ability to shape the outcome (Section 2). New district borders are built from county borders drawn during colonial times, which constrains the feasible shapes. And qualitative and anecdotal evidence indicates that headquarters placement is jointly constrained by the centrality norm and the availability of public land, with new offices often built from scratch (Green, 2010). Section 5.2 complements these institutional arguments with empirical evidence on strategic placement and addresses a separate concern arising from non-random household exposure to the geometry generated by redistricting. Section 5.3 tests the exclusion restriction directly.

5.1 Relevance: First Stage

The spatial correlation between district centroids and headquarters suggests that policymakers comply with the placement rule (Figure 1). Table 2 confirms this relationship. In districts created between 1991 and 2020, a 10 percent increase in proximity to the district centroid is associated with a 4.29 percent increase in proximity to the district headquarters. The first-stage F-statistics are around 28 in the NSDS and 19 in the Afrobarometer.

TABLE 2: FIRST-STAGE: PROXIMITY HOUSEHOLD - DISTRICT'S HEADQUARTERS

	Log SR Proximity Household - District Headquarters	
	NSDS 2021	Afrobarometer R10
Log Proximity Household - District's centroid × New District 91-20	0.429***	0.474***
	(0.080)	(0.107)
Log Proximity Household - District's centroid	-0.051	0.231***
	(0.063)	(0.064)
New district between 1991-2020	1.696***	2.099***
	(0.220)	(0.322)
Observations	8,417	2,184
Number of Parishes	967	271
IV F-stat Kleibergen-Paap rk Wald	28.75	19.45
Controls	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Regression of self-reported household's distance to the district's capital on household's distance to the district's centroid, an indicator for new districts between 1991-2020, and the interaction between the former two. Both distance variables are in logarithms. Controls include: district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. The instrument for Household's distance to the district's capital is the interaction of distance from the household to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Figure 2 displays the identifying variation behind these estimates. After residualizing on the full set of controls, it plots binned means of self-reported proximity to the headquarters against proximity to the centroid, separately for new and old districts. The gradient is strong and positive where the placement rule binds and absent in old districts. Appendix Figure A2

presents the corresponding reduced-form relationships for the main outcome indices.

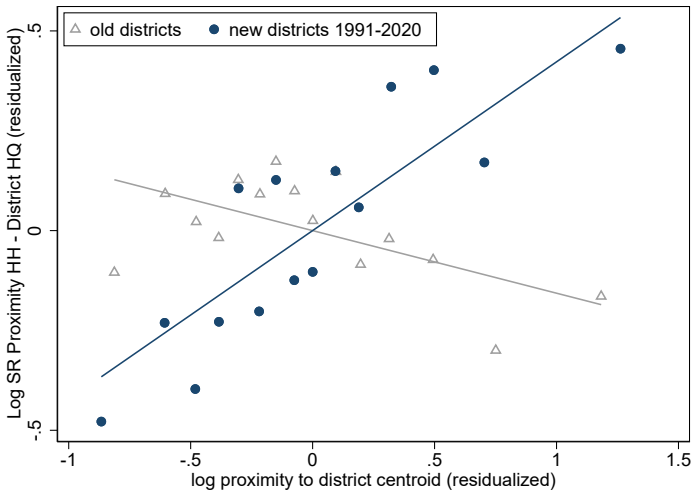


FIGURE 2: First stage binned scatter of residualized self-reported proximity to the district headquarters against residualized proximity to the district centroid for both households in new districts (1991–2020) and old districts. Both variables are residualized on the main specification's control set (survey-weighted). Each dot is a ventile of the residualized centroid proximity.

5.2 Independence: Non-Strategic District Creation and Household Exposure

I first analyze whether district boundaries and headquarters were strategically placed. In particular, I test whether the geographic centroids of new districts fall in more developed areas, and whether parishes selected as headquarters were already on different development trajectories. I then address a separate concern: even if district boundaries are not strategically drawn, households may receive systematically different instrument values because of their location within the original district. To account for this non-random exposure, I follow a design-based recentering approach using feasible counterfactual redistricting configurations.

5.2.1 Centroid Placement

The instrument relies on the geographic centroid of each new district. If district boundaries were drawn to favor centrally located groups, the centroid would fall in areas that were already more populated or developed, and centroid proximity would proxy for pre-existing development. I examine this possibility using 24 districts created between 2014 and 2020, for which pre-creation characteristics are available at the sub-county level. I use three measures: an index of household living standards from the 2014 census (housing conditions, services, assets, and adult education), night-time lights in 2013,[10] and population density in 2010.[11] Table A3

[10]The Annual global VIIRS nighttime lights come from the Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines.

[11]Data obtained from the Gridded Population of the World Database constructed by NASA.

reports logit estimates of whether a sub-county contains the district centroid, with and without district fixed effects. The results show that centroids are not systematically located in more developed sub-counties. If anything, these areas have lower living standards and population density (coefficients of -0.40 and -0.51 per standard deviation, both significant at the 1 percent level), with similar results when including district fixed effects. The coefficient for night-time lights is small and insignificant (0.15, standard error 0.19). Figure 3 illustrates this pattern for Omoro, created from Gulu in 2016, whose centroid falls in Lalogi, one of its least developed sub-counties.

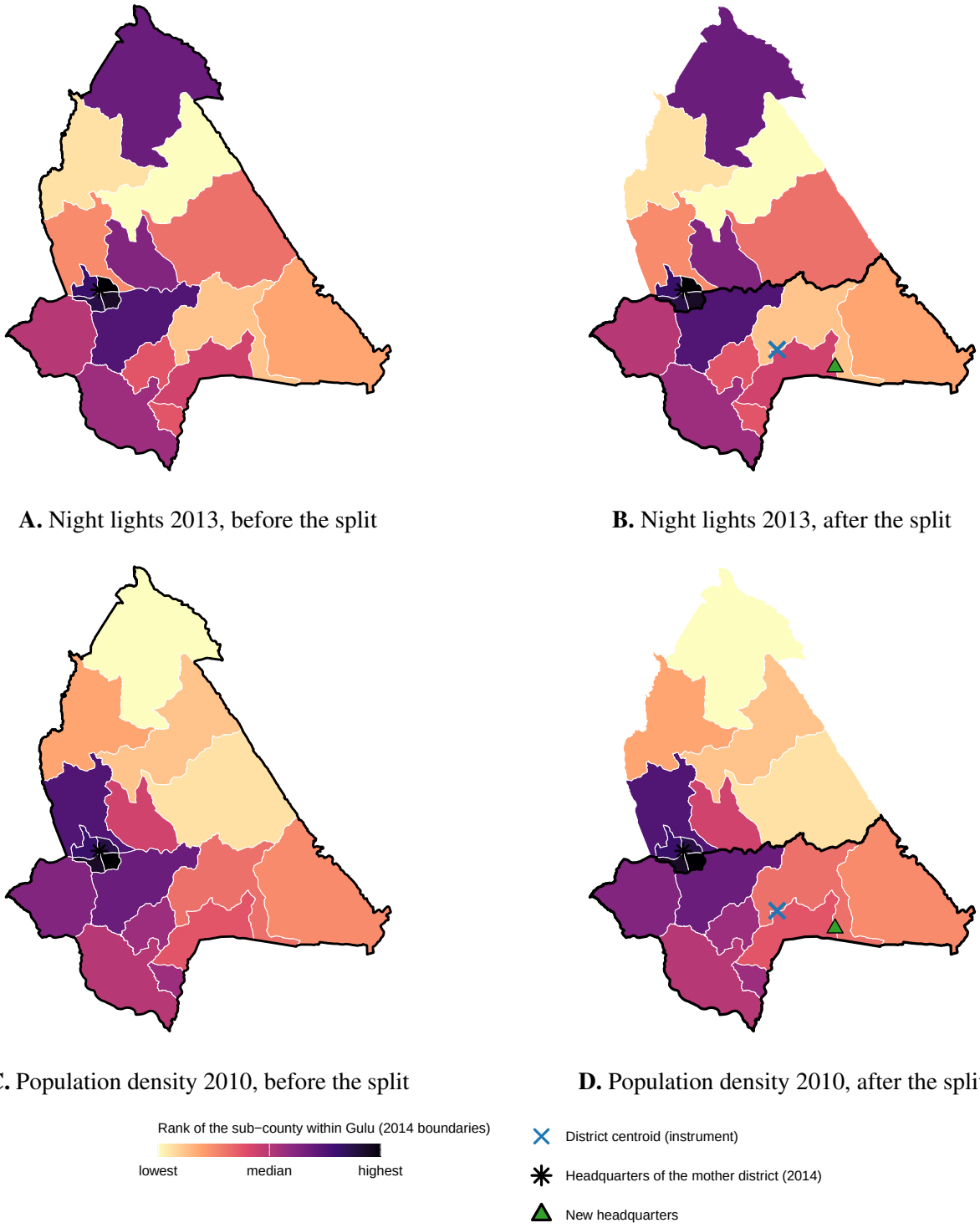
5.2.2 Pre-trends in Land Use

I complement the centroids' placement results by analyzing whether parishes that become district headquarters were already developing differently before their designation. Using MODIS land classification data (2001–2023, 500m resolution) and the staggered difference-in-differences estimator of de Chaisemartin & D'Haultfœuille (2020) at the parish level, Figure 4 shows no differential pre-trends in cultivated cropland between future headquarters parishes and not-yet-treated parishes. I find a similar pattern for built-up area using the GHS-BUILT-S product.[12] In both cases, the outcomes begin to rise only after headquarters creation, consistent with local development following headquarters establishment rather than preceding it. Section 9.2 returns to these post-designation dynamics.

5.2.3 Non-random Exposure and Design-based Inference

Even if district creation is plausibly non-strategic, households may be systematically exposed to the geometry of redistricting. Following Borusyak & Hull (2023), I enumerate all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, using predetermined 1991 county borders as building blocks, I compute each household's expected instrument μ_i across these counterfactual redistrictings. Three results, presented in Appendix G, support the design. Controlling for this expected exposure leaves the main estimates stable or slightly larger, and weak-instrument-robust Anderson–Rubin confidence intervals continue to exclude zero for the main service outcomes. Randomization inference across counterfactual redistrictings places the realized first stage, and the reduced forms for electricity, district health and central-government water, in the tails of their placebo distributions.

[12] The control group for the cropland analysis is based on 79 not-yet-treated parishes which became district headquarters between 2001 and 2023. Appendix Figure A4 presents the corresponding event study for built-up area using the GHS-BUILT-S product, available from 1990 to 2025.



A. Night lights 2013, before the split

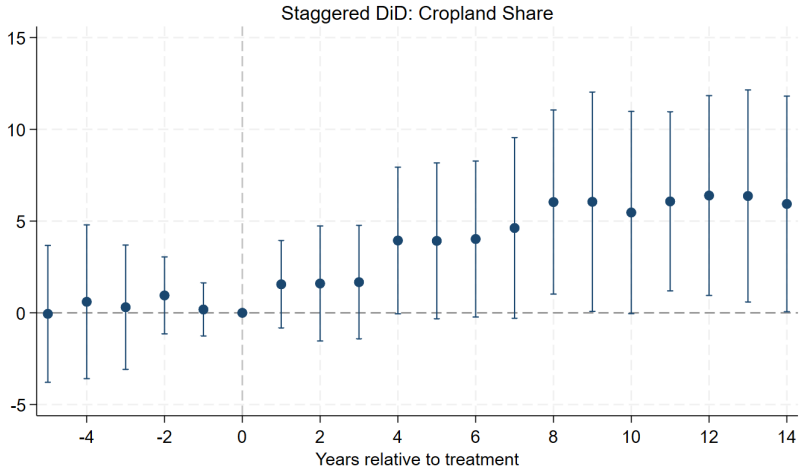
B. Night lights 2013, after the split

C. Population density 2010, before the split

D. Population density 2010, after the split

FIGURE 3: Pre-existing development and the location of the district centroid: Gulu and the creation of Omoro (2016). Gulu's 17 sub-counties on their 2014 boundaries, shaded by their rank within Gulu on night lights in 2013 (Panels A and B) and population density in 2010 (Panels C and D). Panels A and C: before the split, with the headquarters of the mother district in Gulu town (star). Panels B and D: the new district of Omoro outlined, with its geographic centroid (cross), the point on which the instrument is anchored, and its new headquarters (triangle).

FIGURE 4: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND SHARE



Notes: Event study graph of the estimated coefficients of the de Chaisemartin & D'Haultfœuille (2020) staggered difference in difference estimator of the parish cropland share on headquarters creation.

5.3 Exclusion Restriction: Specificity

A remaining concern is that the instrument captures broader geographical centrality or proximity to government institutions rather than proximity to the district administration itself. I examine whether instrumented proximity to the district headquarters affects households' knowledge of or distance to lower-tier local government institutions (LC1, LC2, and LC3), which operate below the district level and should not be directly affected by the creation of a new district headquarters. Appendix Table A6 shows no systematic effects on these outcomes. Eleven of the twelve coefficients are statistically indistinguishable from zero at the 5 percent level, and the only exception does not survive correction for multiple hypothesis testing. This supports the interpretation that the identifying variation is specific to the district level rather than reflecting a general reduction in distance to government institutions.

Appendix E reports additional identification and robustness exercises. These show that the first stage is unchanged when controlling for proximity to major cities and that proximity to the district headquarters does not predict placebo outcomes unrelated to administrative proximity. I also consider specifications excluding households around headquarters towns, GIS-based measures of distance, a comparison of OLS and 2SLS estimates, alternative constructions of the instrument, estimates restricted to newly created districts, and spatially correlated standard errors using Conley cutoffs of 50 and 100 kilometres, as well as standard errors clustered by 1991 mother district. The main service results remain robust across these exercises and replicate when re-estimating the specification using data from 922,930 households in the 2024 Population and Housing Census. Finally, a complementary geographic regression discontinuity design following Blattman et al. (2025) reaches similar estimates for participatory engagement and RDC satisfaction by comparing enumeration areas across newly created district boundaries

(Appendix I).

6 Results

6.1 Main Results: Effects on Public Service Delivery

The National Service Delivery Survey offers a rich set of variables on public goods and services, allowing the analysis of a broad range of outcomes. As predicted by the conceptual framework (Proposition 1), proximity to district headquarters should improve access to public goods and services.

The outcomes cover education, health, water, and roads, each constructed separately for district and central government provision, while electricity is provided only by the central government and agricultural inputs only by districts. I follow Anderson (2008) and construct summary indices to account for multiple hypothesis testing. To create each index, I standardize the component variables to have a mean of 0 and a standard deviation of 1, and compute the index as the average of the standardized components available for each household. The components entering each index are listed in Appendix Table A2.

Table 3 presents the causal effects of household proximity to the district headquarters on these summary indices. Panel A reports the estimates for services under the responsibility of district governments, while Panel B reports those for services under the responsibility of the central government, where the RDC plays a coordinating role at the district level. The table also reports Anderson's sharpened False Discovery Rate (FDR) q-values to account for multiple hypothesis testing across outcome groups.

The results show that proximity to new district headquarters improves access to several key public services, with effects that are stronger and more precisely estimated for centrally provided services. Among district-provided services (Table 3, Panel A), households living closer to the district headquarters have significantly better health access (Column 2), and water (Column 3). The indices for education and roads point in the expected direction but are not statistically significant. The agricultural inputs index shows a small negative and insignificant coefficient, which likely reflects the fact that input provision operates primarily at the sub-county level, making it less sensitive to district-level administrative proximity.

The effects are most consistent and precisely estimated for centrally provided services (Table 3, Panel B). Households closer to district headquarters have significantly better access to water (Column 3), roads (Column 4), and electricity (Column 5). These significant results are robust to multiple hypothesis testing corrections, with FDR q-values below 0.10. Disaggregated results for the individual components underlying each index are presented in Appendix Tables A18–A23. The individual tables show that the improvements detected in the indexes correspond to increases in several indicators and are not driven by specific components. In addition,

TABLE 3: Effects on Services and Proximity to District Headquarters (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agricultural inputs (5)
Panel A: NSDS 2021 District services					
Log Proximity to District HQ	0.282	0.387***	0.279*	0.089	-0.085
	(0.214)	(0.121)	(0.157)	(0.110)	(0.145)
p-val	[0.187]	[0.001]	[0.076]	[0.416]	[0.556]
q-val	{0.234}	{0.008}	{0.179}	{0.452}	{0.452}
Observations	7137	7595	8417	8417	8417
Number of clusters	958	951	967	967	967
Mean of DV	-0.003	-0.016	-0.004	0.008	0.030
IV F-stat (KP rk Wald)	22.243	33.349	28.745	28.745	28.745
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: NSDS 2021 Central govt services					
Log Proximity to District HQ	0.151	0.233	0.372**	0.706***	0.428***
	(0.192)	(0.201)	(0.146)	(0.261)	(0.137)
p-val	[0.432]	[0.248]	[0.011]	[0.007]	[0.002]
q-val	{0.209}	{0.142}	{0.015}	{0.014}	{0.009}
Observations	5553	8417	8417	8417	8417
Number of clusters	951	967	967	967	967
Mean of DV	-0.001	0.013	-0.017	-0.070	-0.043
IV F-stat (KP rk Wald)	22.621	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

Appendix Table A17 contains results for service availability from the Afrobarometer adding evidence to the higher service access in areas proximate to district headquarters.

As alignment enters the framework through the appointed RDC, the complementarity predicted by Proposition 3 should be strongest for services in which the RDC plays a larger coordinating role. In the Ugandan setting, this applies primarily to services financed by the central government (Section 2), since the RDC represents the central government at the district level and helps coordinate the implementation of centrally funded programs. Within this group,

the effects of alignment should be particularly pronounced for infrastructure such as roads and electricity, where the spatial allocation of investments is relatively discretionary. By contrast, district-provided services operate mainly through the elected council's own administrative structure and should therefore respond to proximity in both aligned and non-aligned districts.

Table 4 provides evidence consistent with these predictions. For district-provided services (Panel A), proximity improves health access in both aligned and non-aligned districts, while none of the interaction terms is statistically significant. Among central government services (Panel B), the interaction is largest for roads, where the effect of proximity is 0.81 standard deviations larger in aligned districts (significant at the 10 percent level). The interaction is also positive for electricity (0.17), although it is not statistically significant. These are the two services for which the pooled effects of proximity are strongest in Table 3. By contrast, the interactions are close to zero for central schooling, health, and water. Agricultural inputs are the only district-provided service with a sizeable interaction (0.38), although this pattern is driven by a negative effect of proximity in non-aligned districts rather than by a larger positive effect in aligned ones. Overall, the evidence for Proposition 3 is concentrated in central infrastructure, where the RDC's coordinating role and discretion over implementation are likely to matter the most.

Complementing the service delivery findings, Table 5 shows that households living closer to new district headquarters report higher satisfaction with local administrators. While all three types of leaders (the elected district politician (Local Council V), the centrally appointed Resident District Commissioner (RDC), and civil servants) show positive coefficients, only satisfaction with the RDC is statistically significant. This result is consistent with the stronger effects found for centrally provided services and with the RDC's coordinating role described above. Households living closer to the headquarters may therefore not only benefit more from these services, but also be more likely to recognize this role. In terms of magnitude, a 10 percent increase in proximity raises the likelihood of being satisfied with the RDC's performance by 1.2 percentage points, equivalent to 3.7 percent of the sample mean. The geographic regression discontinuity design of Appendix I points the same direction. Satisfaction with the RDC rises by 11 percentage points per 10 kilometres of relative proximity gain across the new lines. (Appendix Table A34).

Table 6 shows that the higher satisfaction with the RDC documented in Table 5 is concentrated in aligned districts. In these districts, a 10 percent increase in proximity raises the probability of reporting satisfaction with the RDC by 2.1 pp, while it has a negative non-statistically significant coefficient in non-aligned districts. The interaction term is 0.26 and is significant at the 5 percent level. Ratings of the elected LCV chairperson display the opposite pattern. Proximity increases satisfaction with the LCV only in non-aligned districts, although the interaction term is not statistically significant. The framework does not generate a specific prediction for this outcome, but the contrast is informative. In districts where the two leaders are not politi-

TABLE 4: Effects on Services and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	School (1)	Health (2)	Water (3)	Roads (4)	Agricultural inputs (5)
Panel A: NSDS 2021 District services					
Proximity, non-aligned	-0.018	0.383**	0.308	0.007	-0.277*
	(0.275)	(0.157)	(0.203)	(0.150)	(0.165)
Proximity × Aligned	0.432	-0.058	-0.121	0.094	0.382
	(0.412)	(0.229)	(0.298)	(0.217)	(0.255)
Proximity, aligned (sum)	0.413	0.324*	0.187	0.101	0.105
	(0.307)	(0.167)	(0.218)	(0.157)	(0.195)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)
One-sided *p*-value	0.147	0.600	0.658	0.333	0.067
Observations	7,137	7,595	8,417	8,417	8,417
Number of clusters	958	951	967	967	967
Mean of DV	-0.003	-0.016	-0.004	0.008	0.030
KP F (aligned / non-aligned)	9.9 / 16.7	17.0 / 11.8	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: NSDS 2021 Central govt services					
Proximity, non-aligned	-0.026	0.070	0.337	0.202	0.254
	(0.244)	(0.344)	(0.213)	(0.274)	(0.167)
Proximity × Aligned	-0.000	0.087	-0.096	0.811*	0.170
	(0.393)	(0.436)	(0.282)	(0.472)	(0.255)
Proximity, aligned (sum)	-0.026	0.157	0.241	1.013***	0.424**
	(0.309)	(0.268)	(0.184)	(0.384)	(0.193)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)
One-sided *p*-value	0.500	0.421	0.634	0.043	0.253
Observations	5,553	8,417	8,417	8,417	8,417
Number of clusters	951	967	967	967	967
Mean of DV	-0.001	0.013	-0.017	-0.070	-0.043
KP F (aligned / non-aligned)	9.4 / 9.7	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Each column is a 2SLS regression of the outcome on log proximity to the district headquarters and its interaction with an indicator for politically aligned districts. The rows report the effect of proximity in non-aligned districts, the interaction, and their sum, the effect in aligned districts. The predicted sign is the sign of the interaction implied by Proposition 3, and the one-sided p-value tests the interaction against that sign. Standard errors clustered at the parish level are in parentheses. Observations and the mean of the dependent variable refer to the pooled sample; the indices have mean 0 and standard deviation 1 in the full sample. The Kleibergen–Paap rk Wald F-statistic is reported for the aligned and the non-aligned first stage separately. Anderson–Rubin 95 percent confidence sets for every coefficient in this table are reported in Appendix Table A14. Controls and weights as in Table 3; the excluded instruments are the one used there and its interaction with the alignment indicator.

cally aligned, households closer to the headquarters appear to attribute improvements more to the elected LCV rather than to the appointed RDC, while in aligned districts households closer show higher satisfaction with the RDC work.

TABLE 5: Satisfaction with Local Leaders and Proximity to District Headquarters (2SLS)

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
Panel A: NSDS 2021			
Log Proximity to District HQ	0.084	0.122**	0.103
	(0.076)	(0.059)	(0.085)
Observations	8417	8417	7451
Number of clusters	967	967	966
Mean of DV	0.428	0.329	0.547
IV F-stat (KP rk Wald)	28.745	28.745	30.047

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of satisfaction with local leaders' variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE 6: Satisfaction with Local Leaders and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
NSDS 2021			
Proximity, non-aligned	0.237*	-0.056	0.062
	(0.130)	(0.084)	(0.107)
Proximity × Aligned	-0.182	0.264**	0.046
	(0.164)	(0.120)	(0.161)
Proximity, aligned (sum)	0.055	0.208**	0.108
	(0.101)	(0.086)	(0.120)
Observations	8,417	8,417	7,451
Number of clusters	967	967	966
Mean of DV	0.428	0.329	0.547
KP F (aligned / non-aligned)	14.4 / 12.5	14.4 / 12.5	14.7 / 12.6

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 4. The framework makes no prediction for these outcomes, so no predicted sign is reported. Anderson–Rubin 95 percent confidence sets for every coefficient in this table are reported in Appendix Table A14.

Taken together, the main results describe a consistent pattern. Reducing the distance between households and their district administrative center improves access to public goods and services

and raises satisfaction with appointed local leaders, who are responsible for coordinating centrally financed programs. The following sections turn to the mechanisms: citizen engagement with the government and accountability.

7 Mechanisms

To investigate the underlying mechanisms behind the main findings, this study focuses on two potential channels through which the distance to new district headquarters may impact the provision of public goods and services. First, the level of citizen engagement with local governments could influence policymakers' awareness of the most pressing issues within the community. Second, individuals' perceptions of corruption may reflect problems associated with corruption in public service delivery, potentially leading to increased demands for accountability.

7.1 Citizens' Engagement with the Local Government

One of the main objectives behind the creation of new administrative units was to empower local populations and enhance their participation in decision-making. As discussed in the conceptual framework, reducing the geographic gap between citizens and government lowers the cost of formal engagement with district institutions, shifting citizens from contentious to participatory channels (Propositions 4–5). By minimizing information and communication frictions between administrators and citizens, closer proximity to administrative centers may foster greater awareness of state institutions, shift citizens away from reliance on traditional alternatives, and ultimately translate into more effective engagement with formal government. This section studies whether proximity to new district headquarters influences these dimensions of citizen-state relations.

Before turning to direct measures of citizen engagement, I first analyze whether households living closer to district headquarters are more aware of district-level institutions and whether proximity reshapes the balance between formal state institutions and traditional ones. Table 7, Panel A shows that proximity to headquarters significantly increases citizens' knowledge of important state institutions. Households living closer are more likely to know about magistrate courts, the police, the Office of the Auditor General, and the judiciary system, reflecting that the creation of new headquarters is strengthening state presence and legitimacy in previously remote areas. The institutional knowledge index in Column (6), which aggregates the five indicators, confirms that this is a broad-based effect rather than the result of any single institution (the coefficient of 0.36 is significant at the 1 percent level). Additionally, households closer to headquarters are more likely to have a family member employed as a civil servant, suggesting a broader deepening of state-society linkages (Panel B, Column 1).

Panel B further reveals that this increased awareness of formal institutions is accompanied by a substitution away from traditional ones. The NSDS measures whether any household member

TABLE 7: Institutional Knowledge and State-Traditional Substitution (2SLS)

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)	Knowledge index (6)
Panel A: NSDS 2021						
Log Proximity to District HQ	0.273***	0.088*	0.096	0.088**	0.162**	0.361***
	(0.090)	(0.053)	(0.063)	(0.039)	(0.080)	(0.123)
Observations	8413	8415	8413	8417	8417	8415
Number of clusters	967	967	967	967	967	967
Mean of DV	0.375	0.930	0.156	0.092	0.384	0.001
IV F-stat (KP rk Wald)	28.592	28.639	28.592	28.745	28.745	28.639

	Family member civil servant (1)	Conflict resolved by government (2)	Conflict resolved by trad. leader (3)	Contacted MP (4)	Contacted trad. leader (5)	State reliance index[a] (6)
Panel B: NSDS 2021 and Afrobarometer R10						
Log Proximity to District HQ	0.077**	0.096	-0.080	0.076	-0.181*	0.240
	(0.031)	(0.144)	(0.110)	(0.071)	(0.106)	(0.176)
Observations	8414	2131	2131	2178	2146	2184
Number of clusters	967	271	271	271	271	271
Mean of DV	0.057	0.556	0.211	0.168	0.277	-0.001
IV F-stat (KP rk Wald)	28.655	19.705	19.705	19.462	19.819	19.446

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of state/traditional institutions variables on self-reported household's distance to the district's capital. Column (6) of Panel A is a summary index of Columns (1)–(5). [a] Column (6) of Panel B is a state-reliance index of Columns (2)–(5), with the two traditional-institution indicators reversed so that higher values indicate greater reliance on the state; Column (1) is excluded because it comes from the NSDS. Indices are constructed to have mean 0 and standard deviation 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

is employed as a civil servant, while the Afrobarometer captures the institutions citizens turn to for conflict resolution and representation. Among respondents who experienced a dispute in the past year, I distinguish whether they first sought resolution from the police, a court, or a government official, rather than from a traditional or religious leader, elders, or family. I also measure whether respondents contacted a Member of Parliament or a traditional leader at least once in the past year. Proximity increases the probability that a household includes a civil servant, while the coefficients on state-based dispute resolution and contacting a Member of Parliament are positive but imprecisely estimated. More importantly, respondents living closer to the headquarters are significantly less likely to have contacted a traditional leader, pointing to a shift in the institutions citizens use when seeking for conflict resolution or representation.[13] Together, these results indicate that administrative proximity makes the state more visible and accessible, gradually displacing traditional institutions.

Table 8 shows that the greater visibility of the state near district headquarters is concentrated in aligned districts. Knowledge of the magistrate court and the police rises with proximity in aligned districts (0.39 and 0.12, respectively) but not in non-aligned districts, and the corresponding interaction terms are positive and significant at the 5 percent level. The exception

[13]The state-reliance index in Column (6) combines the four Afrobarometer indicators, with the two traditional-institution items entering with reversed sign, so that higher values denote greater reliance on formal state channels; it points in the same direction (0.24) but is imprecisely estimated ($p = 0.17$).

is knowledge of the Auditor General, which increases with proximity only in non-aligned districts, although the difference across groups is not statistically significant. Importantly, the summary index in column 6 shows that the increase in knowledge is differentially more concentrated in aligned districts. The results in Panel B also point toward stronger substitution away from traditional leaders in aligned districts. Proximity reduces contact with traditional leaders by 0.28 in aligned districts, compared with -0.01 in non-aligned districts, with an interaction of -0.26 (with a one-sided p-value of 0.107). At the same time, the share of households with a family member in the civil service rises with proximity only in aligned districts (0.09, $p < 0.05$). The state-reliance index, which combines the four Afrobarometer indicators with the traditional-institution items reversed, is larger in aligned districts (0.36 against 0.11) but neither coefficient nor their difference is statistically significant.

TABLE 8: Institutional Knowledge and State-Traditional Substitution (2SLS) – Aligned versus Non-Aligned Districts

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)	Knowledge index (6)
Panel A: NSDS 2021 – Knowledge of state institutions						
Proximity, non-aligned	0.010	-0.054	0.032	0.125**	-0.039	0.049
	(0.120)	(0.041)	(0.062)	(0.061)	(0.078)	(0.123)
Proximity × Aligned	0.375**	0.173**	0.086	-0.072	0.188	0.364*
	(0.172)	(0.088)	(0.106)	(0.079)	(0.129)	(0.203)
Proximity, aligned (sum)	0.385***	0.120	0.119	0.052	0.149	0.413**
	(0.123)	(0.077)	(0.086)	(0.050)	(0.102)	(0.161)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)	(+)
One-sided *p*-value	0.015	0.024	0.208	0.821	0.072	0.036
Observations	8,413	8,415	8,413	8,417	8,417	8,417
Number of clusters	967	967	967	967	967	967
Mean of DV	0.375	0.930	0.156	0.092	0.384	0.001
KP F (aligned / non-aligned)	14.3 / 12.4	14.3 / 12.5	14.3 / 12.4	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5

	Family member civil servant (1)	Conflict resolved by government (2)	Conflict resolved by trad. leader (3)	Contacted MP (4)	Contacted trad. leader (5)	State reliance index (cols 2-5) (6)
Panel B: NSDS 2021 and Afrobarometer R10 – State-traditional substitution						
Proximity, non-aligned	0.044	0.152*	-0.011	0.043	-0.011	0.113
	(0.059)	(0.085)	(0.050)	(0.073)	(0.073)	(0.085)
AR 95% set	$[-0.08, 0.21]$	$[-0.01, 0.46]$	$[-0.14, 0.12]$	$[-0.19, 0.19]$	$[-0.18, 0.21]$	$[-0.12, 0.33]$
Proximity × Aligned	0.041	-0.145	-0.147	0.125	-0.264	0.249
	(0.071)	(0.240)	(0.201)	(0.144)	(0.212)	(0.331)
Proximity, aligned (sum)	0.086**	0.007	-0.158	0.168	-0.275	0.362
	(0.040)	(0.224)	(0.195)	(0.125)	(0.199)	(0.320)
AR 95% set	$[0.01, 0.21]$	$[-0.64, 1.29]$	$[-1.78, 0.22]$	$[-0.07, 1.16]$	$[-2.38, 0.04]$	$[-0.19, 3.29]$
Predicted sign of interaction	(+)	(+)	(−)	(+)	(−)	(+)
One-sided *p*-value	0.280	0.727	0.232	0.193	0.107	0.226
Observations	8,414	2,131	2,131	2,178	2,146	2,184
Number of clusters	967	271	271	271	271	271
Mean of DV	0.057	0.556	0.211	0.168	0.277	-0.001
KP F (aligned / non-aligned)	14.4 / 12.4	6.3 / 20.7	6.3 / 20.7	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 4. The predicted sign is the sign of the interaction implied by Proposition 5, and the one-sided p-value tests the interaction against that sign. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A14.

Table 9 presents indicators of citizen participation with district-level governmental actors, differentiating between participatory and contentious forms of engagement. The results show that increased proximity reduces contentious engagement (Panel B). A 10 percent increase in

proximity to district headquarters significantly reduces the likelihood of participating in protests and contacting the media by 0.7 and 1.2 percentage points, respectively (equivalent to 19 and 13 percent of their respective sample means). The contentious engagement index confirms that this is not driven by a single indicator, with a negative coefficient of -0.38, significant at the 5 percent level. Together with the results on higher state presence above, these results are consistent with citizens having less need to engage in protests to raise their worries. Adding evidence to this interpretation, all coefficients on participatory engagement in Panel A are positive, indicating that proximity increases formal engagement with district officials, even if neither the individual coefficients nor the participatory engagement index (0.17, $p = 0.13$) are statistically significant.[14] The complementary geographic regression discontinuity design provides stronger evidence for participatory engagement. It finds a significant increase in meetings with the RDC and an effect on the participatory engagement index of 0.095 per 10 kilometres of relative proximity gain, equivalent to 0.14 to 0.18 per log point of proximity depending on the first stage used, compared with 0.17 in the main specification (Appendix I, Table A33).

The engagement results provide strong support for the predictions of the theoretical framework. Table 10, Panel B, shows that the reduction in contentious engagement documented in Table 9 is concentrated in aligned districts. In these districts, a 10 percent increase in proximity lowers protest participation by 1.6 pp and collective civic action by 5.0 pp (equivalent to 44 and 14 percent of their respective means), while the coefficients in non-aligned districts are small and not statistically significant. The corresponding interaction terms are -0.19 for protest and -0.56 for collective action, both significant at the 5 percent level. For media contact, the interaction is -0.17 and is significant at the 10 percent level in the one-sided test. These results are also reflected in the summary index, which shows that the decrease in contentious engagement is driven by aligned districts. Importantly, these patterns are not driven by the weak first stage in the aligned Afrobarometer sample. The Anderson–Rubin confidence intervals for the aligned coefficients exclude zero for all three outcomes, while the non-aligned confidence intervals are narrow and contain zero.

The evidence on participatory engagement (Panel A) points towards stronger effects of proximity for aligned districts. Involvement in LCV meetings rises with proximity only in aligned districts (0.18, significant at the 10 percent level), with the interaction term reaching 0.19 and having a one-sided p-value of 0.07. The interactions for meeting the RDC and attending an RDC baraza also have the predicted sign, although they are not statistically significant. The participatory engagement index shows the same pattern. It rises with proximity only in aligned districts (0.32, significant at the 10 percent level), with an interaction of 0.32 and a one-sided p-value of 0.07, while it is close to zero and not statistically significant in non-aligned districts.

Additional evidence suggests that this shift toward formal engagement is translating into pub-

[14] Importantly, Appendix Table A6 confirms that proximity to headquarters does not systematically affect engagement with LC1 leaders (no coefficient is significant at the 5 percent level), reinforcing that the effects documented here are specific to institutions belonging to the districts.

TABLE 9: Citizen Engagement and Proximity to District Headquarters (2SLS)

	Involved in LCV meetings (1)	Met the RDC (2)	Attended RDC baraza (3)	Participatory index (4)
Panel A: NSDS 2021				
Log Proximity to District HQ	0.086	0.045	0.033	0.169
	(0.066)	(0.043)	(0.027)	(0.112)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.255	0.102	0.052	0.020
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745
	Collective civic action (1)	Protest (2)	Contacted media (3)	Contentious index (4)
Panel B: Afrobarometer R10				
Log Proximity to District HQ	-0.176	-0.071**	-0.116**	-0.381**
	(0.112)	(0.034)	(0.058)	(0.160)
Observations	2182	2184	2184	2184
Number of clusters	271	271	271	271
Mean of DV	0.360	0.037	0.093	-0.000
IV F-stat (KP rk Wald)	19.413	19.446	19.446	19.446

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' involvement with the government variables on self-reported household's distance to the district's capital. Column (4) of each panel is a summary index of Columns (1)–(3), constructed to have mean 0 and standard deviation 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

lic investments that match local demands. The NSDS community questionnaire asks each community to rank the projects it considers most important and, for each ranked project, whether it was implemented in the village or parish in the past three years. Appendix Table A24 uses this information at the community level. The resutls show that communities closer to district headquarters are significantly more likely to have received the classroom, health facility, and road projects they prioritized, and a 10 percent increase in proximity raises the share of top-three priorities implemented by about 3 percentage points, roughly 10 percent of the sample mean. The coefficients for water and electricity are positive but lack statistical significance, and there is no effect for new schools. This pattern suggests that greater interaction with district officials helps channel local preferences into tangible projects. This responsiveness is concentrated in aligned districts (Appendix Table A25). In these districts, a 10 percent increase in proximity raises the share of top-three priorities implemented by about 7 percentage points, roughly 22 percent of the sample mean, while the coefficient in non-aligned districts is negative and not statistically significant. The interaction is significant at the 1 percent level, and the same pattern holds for health facilities and water. In line with Proposition 5, proximity translates local

TABLE 10: Citizen Engagement and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Involved in LCV meetings	Met the RDC	Attended RDC baraza	Participatory index
	(1)	(2)	(3)	(4)
Panel A: NSDS 2021 – Participatory engagement				
Proximity, non-aligned	-0.011	0.010	0.000	0.004
	(0.083)	(0.043)	(0.040)	(0.126)
Proximity × Aligned	0.190	0.062	0.065	0.317
	(0.128)	(0.079)	(0.058)	(0.213)
Proximity, aligned (sum)	0.179*	0.072	0.065	0.321*
	(0.097)	(0.066)	(0.042)	(0.172)
Predicted sign of interaction	(+)	(+)	(+)	(+)
One-sided *p*-value	0.068	0.218	0.134	0.068
Observations	8,417	8,417	8,417	8,417
Number of clusters	967	967	967	967
Mean of DV	0.255	0.102	0.052	0.020
KP F (aligned / non-aligned)	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	Collective civic action	Protest	Contacted media	Contentious index
	(1)	(2)	(3)	(4)
Panel B: Afrobarometer R10 – Contentious engagement				
Proximity, non-aligned	0.058	0.024	-0.025	0.054
	(0.059)	(0.026)	(0.049)	(0.116)
AR 95% set	$[-0.11, 0.21]$	$[-0.03, 0.11]$	$[-0.17, 0.09]$	$[-0.24, 0.36]$
Proximity × Aligned	-0.555**	-0.187**	-0.170	-0.913**
	(0.256)	(0.080)	(0.119)	(0.400)
Proximity, aligned (sum)	-0.498**	-0.164**	-0.195*	-0.860**
	(0.249)	(0.076)	(0.108)	(0.383)
AR 95% set	$[-3.60, -0.19]$	$[-1.06, -0.07]$	$[-1.50, -0.06]$	$[-5.98, -0.43]$
Predicted sign of interaction	(−)	(−)	(−)	(−)
One-sided *p*-value	0.015	0.010	0.075	0.011
Observations	2,182	2,184	2,184	2,184
Number of clusters	271	271	271	271
Mean of DV	0.360	0.037	0.093	-0.000
KP F (aligned / non-aligned)	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 4. The predicted sign is the sign of the interaction implied by Proposition 5, and the one-sided p-value tests the interaction against that sign. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A14.

priorities into projects only where the state is responsive to citizen input.

Taken together, these results support the engagement channel described in the conceptual framework (Proposition 4). Greater proximity to the district administrative center increases

state presence and institutional knowledge, reduces reliance on traditional institutions, lowers contentious political engagement, and increases participation through formal channels. The alignment patterns in Tables 10 and 8 further suggest that these effects depend on the responsiveness of the state to formal input. When responsiveness is higher, citizens living closer to the headquarters are more likely to recognize state institutions and engage through participatory rather than contentious channels. When responsiveness is lower, proximity has little effect on how citizens interact with the state.

7.2 Corruption Perceptions

A further potential mechanism through which proximity affects public service delivery is accountability. The conceptual framework predicts that proximity strengthens citizen monitoring, deterring corruption (Proposition 6). Table 11 presents two sets of measures: Panel A present results on households' corruption awareness and their direct experience of victimization, while Panel B contains bribery experiences across different service contexts.

TABLE 11: Corruption and Proximity to District Headquarters (2SLS)

	Aware of nepotism	Aware of embezzlement	Aware of absenteeism	Awareness index[a]	Victim of nepotism	Victim of embezzlement	Victim of absenteeism	Victimisation index[b]
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: NSDS 2021								
Log Proximity to District HQ	0.084	0.044	-0.011	0.078	-0.031*	-0.029	-0.086*	-0.219**
	(0.082)	(0.077)	(0.081)	(0.139)	(0.019)	(0.025)	(0.045)	(0.098)
Observations	8412	8412	8413	8413	8417	8417	8417	8417
Number of clusters	967	967	967	967	967	967	967	967
Mean of DV	0.494	0.647	0.487	0.000	0.034	0.038	0.075	0.006
IV F-stat (KP rk Wald)	28.613	28.613	28.656	28.656	28.745	28.745	28.745	28.745

	Bribe for medical care	Bribe for police help	Bribe to avoid police	Bribe for ID document	Bribery index
	(1)	(2)	(3)	(4)	(5)
Panel B: Afrobarometer R10					
Log Proximity to District HQ	-0.287**	-0.274	-0.151	0.092	-0.458**
	(0.118)	(0.183)	(0.166)	(0.151)	(0.211)
Observations	1604	345	904	396	1886
Number of clusters	271	196	257	208	271
Mean of DV	0.409	0.771	0.523	0.487	-0.047
IV F-stat (KP rk Wald)	20.664	19.947	12.130	17.949	18.161

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of corruption variables on self-reported household's distance to the district's capital. Columns (4)[a] and (8)[b] of Panel A are summary indices of Columns (1)–(3) and (5)–(7), and Column (5) of Panel B of Columns (1)–(4), where each bribery indicator is defined only for respondents who sought the service. Indices are constructed to have mean 0 and standard deviation 1; higher values indicate more corruption. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

The results reveal that proximity to district headquarters is associated with slightly higher awareness of nepotism and embezzlement, though neither coefficient is statistically significant. This suggests that households living closer to administrative centers are not necessarily exposed to more corruption. Importantly, when examining direct victimization, the results show that households closer to headquarters are significantly less likely to have personally experienced nepotism and absenteeism, with a 10 percent increase in proximity reducing the probability of experiencing nepotism or absenteeism by approximately 0.3 and 0.8 percentage

points respectively, equivalent to 9 and 11 percent of their sample means. The victimization index, which aggregates the three experiences, is negative and significant at the 5 percent level (-0.22). Consistent with these findings, Panel B reveals that proximity significantly reduces the likelihood of paying a bribe to access health services by approximately 3 percentage points (about 7 percent of the sample mean). The bribery index is also negative and significant at the 5 percent level, reflecting the general pattern in columns (1-4) of Panel B.

These results suggest that administrative proximity strengthens accountability by reducing direct corruption experiences even as awareness of corruption remains high or increases slightly. The combination of high awareness, together with lower victimization, is consistent with a monitoring mechanism. Households closer to district headquarters engage more frequently with local officials and are better able to observe and report misconduct, which in turn may deter corrupt behavior.

The framework links proximity to corruption through two channels. Citizens living closer to the headquarters monitor administrators more intensely, while in aligned districts the RDC internalises the welfare cost of diversion, extending the range of distances over which corruption is deterred (Proposition 7). Table 12 analyzes these channels, distinguishing awareness of corruption and direct experiences of it. As above, Panel A reports whether households have heard of specific corrupt practices, which proxies for monitoring intensity in the model, alongside whether they have been victims of these practices in the past year. Panel B complements these measures with bribery experiences from the Afrobarometer.

The awareness results provide evidence of stronger monitoring close to responsive district headquarters. In aligned districts, households closer to the headquarters are more likely to have heard of nepotism (0.22, significant at the 10 percent level), embezzlement (0.17), and absenteeism (0.06). In non-aligned districts, the pattern reverses: all three coefficients are negative and the index falls by 0.32 (significant at the 10 percent level). The interaction terms are positive for every item and are significant at the 1 percent level for nepotism and at the 5 percent level for embezzlement and for the summary index. Proximity to a responsive headquarters therefore seems to make citizens better informed about misconduct, while proximity to a non-aligned administration does not generate the same increase in awareness. The positive interaction is nevertheless what one would expect if citizens acquire information about misconduct through the participatory channels that Proposition 5 predicts to be more active in aligned districts.[15]

Greater awareness goes together with fewer direct experiences of corruption, in line with Proposition 6, although this reduction is not concentrated in aligned districts. The victimisation index falls with proximity in both groups (-0.18 in aligned and -0.28 in non-aligned districts,

[15]The framework predicts higher awareness closer to the headquarters, since monitoring intensity $\mu(d_i)$ depends on distance alone, but it does not generate a prediction for the interaction, which is why the awareness columns carry no predicted sign in the table.

the latter significant at the 10 percent level), leaving the interaction close to zero.

The bribery results in Panel B provide stronger support for Proposition 7. The bribery index falls by 0.82 in aligned districts (significant at the 10 percent level) and is unchanged in non-aligned districts, with an interaction of -0.82 and a one-sided p-value of 0.06. The effect is concentrated in bribes for medical care, which is the outcome with the largest sample. The aligned first stage is weak for these outcomes (Kleibergen–Paap F between 3 and 7), so the Anderson–Rubin confidence intervals are wide. They nevertheless exclude zero for the medical-care bribe and for the index, while the corresponding confidence intervals for non-aligned districts are narrow and contain zero. Bribes for police assistance fall with proximity in both groups, while the remaining two items are imprecisely estimated in smaller samples.[16]

Taken together, the panels are consistent with the monitoring mechanism of the framework rather than with worse governance near the headquarters. In aligned districts, citizens living closer to the headquarters are more aware of corruption, report fewer direct experiences of it, and pay fewer bribes. In non-aligned districts, proximity lowers both awareness and reported victimisation. This pattern is consistent with weaker exposure to (or observation of) an administration that is less responsive to local citizens.

[16]The aligned bribery results should be taken with care as the first-stage F-statistics is low, and the Anderson–Rubin confidence interval for the index is unbounded below. The evidence supports the sign of the effect and its concentration in aligned districts, but not its magnitude.

TABLE 12: Corruption and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Aware of nepotism (1)	Aware of embezzlement (2)	Aware of absenteeism (3)	Awareness index (4)	Victim of nepotism (5)	Victim of embezzlement (6)	Victim of absenteeism (7)	Victimisation index (8)
Panel A: NSDS 2021								
Proximity, non-aligned	-0.225**	-0.138	-0.106	-0.321*	0.002	-0.080*	-0.108	-0.277*
	(0.104)	(0.113)	(0.094)	(0.188)	(0.022)	(0.043)	(0.083)	(0.159)
Proximity × Aligned	0.447***	0.308**	0.169	0.630**	-0.050	0.070	0.047	0.093
	(0.157)	(0.155)	(0.146)	(0.268)	(0.037)	(0.056)	(0.099)	(0.206)
Proximity, aligned (sum)	0.221*	0.170	0.063	0.308	-0.047	-0.011	-0.061	-0.184
	(0.117)	(0.106)	(0.112)	(0.190)	(0.030)	(0.036)	(0.055)	(0.132)
Predicted sign of interaction	n.p.	n.p.	n.p.	n.p.	(−)	(−)	(−)	(−)
One-sided *p*-value					0.092	0.893	0.683	0.674
Observations	8,412	8,412	8,413	8,413	8,417	8,417	8,417	8,417
Number of clusters	967	967	967	967	967	967	967	967
Mean of DV	0.494	0.647	0.487	0.000	0.034	0.038	0.075	0.006
KP F (aligned / non-aligned)	14.3 / 12.5	14.3 / 12.5	14.3 / 12.5	14.3 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5

	Bribe for medical care (1)	Bribe for police help (2)	Bribe to avoid police (3)	Bribe for ID document (4)	Bribery index (5)
Panel B: Afrobarometer R10					
Proximity, non-aligned	-0.019	-0.299**	0.034	-0.034	-0.008
	(0.066)	(0.121)	(0.112)	(0.141)	(0.112)
AR 95% set	$[-0.18, 0.16]$	$[-0.81, -0.09]$	$[-0.25, 0.42]$	$[-0.47, 0.32]$	$[-0.28, 0.29]$
Proximity × Aligned	-0.464*	-0.100	-0.149	0.317	-0.816
	(0.256)	(0.404)	(0.329)	(0.281)	(0.512)
Proximity, aligned (sum)	-0.483*	-0.398	-0.115	0.284	-0.823*
	(0.248)	(0.385)	(0.309)	(0.243)	(0.500)
AR 95% set	$[-3.42, -0.15]$	$[-3.78, 0.26]$	$(-\infty, \infty)$	$[-0.42, 2.13]$	$(-\infty, -0.20]$
Predicted sign of interaction	(−)	(−)	(−)	(−)	(−)
One-sided *p*-value	0.035	0.403	0.326	0.871	0.056
Observations	1,604	345	904	396	1,886
Number of clusters	271	196	257	208	271
Mean of DV	0.409	0.771	0.523	0.487	-0.047
KP F (aligned / non-aligned)	6.5 / 26.1	6.8 / 16.8	2.9 / 16.0	6.6 / 12.9	4.8 / 24.2

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 4. The predicted sign is the sign of the interaction implied by Proposition 7, and the one-sided p-value tests the interaction against that sign. n.p. marks outcomes for which the framework makes no prediction. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A14.

8 Democratic Perceptions

The evidence so far suggests that bringing the administration closer changes what the state delivers and how citizens interact with it. This section examines whether these changes also affect political attitudes, moving from what the state does to how citizens perceive it. The theoretical case for decentralization has been based on a sequence running from proximity to performance to legitimacy: a government that is closer is easier to observe and hold accountable, performs better as a result, and thereby earns citizens' trust and support for the political system (Oates et al., 1972; Bardhan, 2002; Faguet, 2023). The first links of this chain have empirical support. Decentralization can improve delivery and responsiveness (e.g. (Faguet, 2004)), as the previous sections confirm for Uganda. However, the final link is often assumed rather than established. Credible evidence is scarce because political attitudes and state presence are jointly determined. Governments may expand where support is high or where contestation is politically costly, and communities that trust the state may both demand and receive more of it. As a result, observed correlations between government performance and political trust have limited causal content. Moreover, in hybrid regimes, there are reasons to expect this link to fail. Decentralization

can be designed to extend central control rather than empower citizens (Martinez-Bravo et al., 2022), and improvements in government performance not necessarily translate into political trust where democratic institutions are weak (Besley et al., 2026). The setting of this paper helps address this gap. The proximity induced by redistricting provides an exogenous shock to citizens' exposure to the state, one that improved services, engagement, and accountability. This section studies whether these improvements translate into democratic attitudes, using the same specification as before.

Table 13 details the relationship between proximity to district headquarters and a broad set of democratic perception indicators (Panel A), rule of law perceptions (Panel B), and trust in government (Panel C). The last column of each panel reports a summary index of the outcomes in that panel. The results show that across all fourteen outcomes (including satisfaction with democracy, perceived freedom to vote and organize, evaluations of judicial independence, official accountability, and trust in local government, parliament, the president, the ruling party, the opposition, and the courts) the estimated coefficients are small and statistically insignificant. The three summary indices confirm that this is not a matter of individual outcomes being too noisy. The coefficients on the democratic perceptions, rule of law, and trust indices are small and all have p-values above 0.75. This pattern is robust across specifications and holds for both local and national institutions, suggesting that the gains in service delivery and accountability documented above do not translate into measurable shifts in how citizens evaluate the democratic system or the state more broadly.[17]

Interestingly, while trust in government remains unchanged, Appendix Table A26 reveals that proximity does affect other dimensions of trust and collective capacity. Households closer to new district headquarters are significantly more likely to trust people of other religions and other ethnicities, and are more likely to engage in collective action to solve local problems. These findings suggest that administrative proximity generates an environment that foster social capital even if political trust remains unaffected.

While the framework does not generate a prediction for democratic perceptions, the heterogeneity by political alignment is informative. In aligned districts, proximity leaves assessments of democracy, perceived political freedoms, rule-of-law evaluations, and trust in political institutions largely unchanged (Table 14). In contrast, in non-aligned districts proximity lowers perceived freedom to vote (-0.11, significant at the 1 percent level), increases the perception that the president ignores the law (0.15, $p < 0.05$), and reduces trust in both local government and the ruling party (both -0.13, $p < 0.10$). The interaction terms for these four outcomes are significant at the 5 or 10 percent level, while the corresponding effects in aligned districts are close to zero. These results for non-aligned districts are supported by strong first stages, with

[17] Appendix Table A27 shows that the same null pattern holds in the 2017 Afrobarometer round. The six trust in government coefficients are small and statistically insignificant, suggesting that the null is not driven by political trust responding more slowly than service delivery, as it holds both before and after the service effects documented in Section 9 materialized.

TABLE 13: Democratic Perceptions and Proximity to District Headquarters (2SLS)

	Country is a democracy (1)	Satisfied with democracy (2)	Free to vote (3)	Free to join organisations (4)	Democracy index (5)
Panel A: Democratic perceptions					
Log Proximity to District HQ	-0.012	0.072	-0.015	-0.064	-0.031
	(0.053)	(0.085)	(0.044)	(0.060)	(0.112)
Observations	2079	2108	2159	2151	2179
Number of clusters	271	271	271	271	271
Mean of DV	0.882	0.593	0.934	0.847	0.001
IV F-stat (KP rk Wald)	19.352	19.744	20.486	21.665	20.323

	President ignores law (1)	Unequal under law (2)	Judges influenced (3)	Officials go unpunished (4)	Rule of law index (5)
Panel B: Rule of law perceptions					
Log Proximity to District HQ	-0.009	0.020	-0.100	-0.029	-0.057
	(0.098)	(0.101)	(0.108)	(0.111)	(0.180)
Observations	1859	2112	1980	2075	2161
Number of clusters	271	271	271	271	271
Mean of DV	0.366	0.640	0.601	0.607	-0.011
IV F-stat (KP rk Wald)	22.695	20.535	19.684	20.123	19.437

	Trust local government (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)	Trust index (7)
Panel C: Trust in government							
Log Proximity to District HQ	0.029	-0.066	0.043	0.044	0.017	0.024	0.028
	(0.079)	(0.084)	(0.099)	(0.084)	(0.086)	(0.083)	(0.111)
Observations	2132	2116	2184	2184	2184	2087	2184
Number of clusters	271	271	271	271	271	271	271
Mean of DV	0.507	0.695	0.503	0.589	0.315	0.526	-0.005
IV F-stat (KP rk Wald)	20.692	19.058	19.446	19.446	19.446	21.146	19.446

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of democratic perception variables on self-reported household's distance to the district's capital. The last column of each panel is a summary index of the outcomes in that panel, constructed to have mean 0 and standard deviation 1; higher values indicate more favourable democratic perceptions (Panel A), more negative rule-of-law perceptions (Panel B), and more trust in government (Panel C). All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

F-statistics between 17 and 21. The Anderson–Rubin confidence intervals also exclude zero for freedom to vote, the perception that the president ignores the law, and trust in local government, while the set for trust in the ruling party, marginally includes zero. Taken together, these patterns suggest that proximity does not build political legitimacy even where administrative coordination is stronger and may instead erode it in non-aligned districts. One possible interpretation is that citizens living closer to the headquarters are more exposed to frictions between the elected and appointed administrators.

TABLE 14: Democratic Perceptions and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Country is a democracy (1)	Satisfied with democracy (2)	Free to vote (3)	Free to join organisations (4)	Democracy index (5)
Panel A: Democratic perceptions					
Proximity, non-aligned	-0.002	0.031	-0.113***	-0.033	-0.109
	(0.038)	(0.063)	(0.032)	(0.050)	(0.078)
AR 95% set	$[-0.11, 0.09]$	$[-0.15, 0.18]$	$[-0.21, -0.04]$	$[-0.19, 0.08]$	$[-0.35, 0.06]$
Proximity × Aligned	-0.013	0.108	0.160**	-0.067	0.138
	(0.097)	(0.173)	(0.082)	(0.114)	(0.213)
Proximity, aligned (sum)	-0.015	0.138	0.047	-0.099	0.029
	(0.089)	(0.161)	(0.075)	(0.102)	(0.198)
AR 95% set	$[-0.39, 0.30]$	$[-0.15, 1.43]$	$[-0.09, 0.54]$	$[-0.55, 0.15]$	$[-0.45, 1.09]$
Observations	2,079	2,108	2,159	2,151	2,179
Number of clusters	271	271	271	271	271
Mean of DV	0.882	0.593	0.934	0.847	0.001
KP F (aligned / non-aligned)	6.4 / 21.1	6.4 / 21.5	7.1 / 18.9	7.8 / 19.6	6.9 / 20.1

	President ignores law (1)	Unequal under law (2)	Judges influenced (3)	Officials go unpunished (4)	Rule of law index (5)
Panel B: Rule of law perceptions					
Proximity, non-aligned	0.151**	-0.113	-0.014	-0.014	-0.011
	(0.067)	(0.089)	(0.086)	(0.088)	(0.131)
AR 95% set	$[0.02, 0.41]$	$[-0.43, 0.06]$	$[-0.30, 0.16]$	$[-0.31, 0.17]$	$[-0.44, 0.26]$
Proximity × Aligned	-0.327*	0.222	-0.194	-0.072	-0.145
	(0.182)	(0.175)	(0.219)	(0.204)	(0.331)
Proximity, aligned (sum)	-0.176	0.108	-0.208	-0.087	-0.156
	(0.169)	(0.150)	(0.201)	(0.184)	(0.303)
AR 95% set	$[-1.00, 0.13]$	$[-0.34, 0.70]$	$[-1.95, 0.11]$	$[-1.18, 0.34]$	$[-2.21, 0.47]$
Observations	1,859	2,112	1,980	2,075	2,161
Number of clusters	271	271	271	271	271
Mean of DV	0.366	0.640	0.601	0.607	-0.011
KP F (aligned / non-aligned)	8.8 / 16.7	7.4 / 18.5	6.7 / 17.7	6.8 / 19.5	6.6 / 19.0

	Trust local government (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)	Trust index (7)
Panel C: Trust in government							
Proximity, non-aligned	-0.129*	-0.083	0.012	-0.126*	0.079	0.091	-0.048
	(0.077)	(0.067)	(0.079)	(0.074)	(0.059)	(0.056)	(0.092)
AR 95% set	$[-0.45, -0.01]$	$[-0.28, 0.06]$	$[-0.21, 0.20]$	$[-0.40, 0.01]$	$[-0.04, 0.28]$	$[-0.10, 0.20]$	$[-0.37, 0.13]$
Proximity × Aligned	0.249*	0.034	0.011	0.273*	-0.077	-0.059	0.142
	(0.145)	(0.160)	(0.173)	(0.156)	(0.158)	(0.154)	(0.204)
Proximity, aligned (sum)	0.121	-0.050	0.023	0.146	0.002	0.032	0.094
	(0.123)	(0.146)	(0.155)	(0.137)	(0.146)	(0.144)	(0.181)
AR 95% set	$[-0.18, 0.78]$	$[-0.64, 0.65]$	$[-0.62, 0.60]$	$[-0.15, 1.11]$	$[-0.75, 0.44]$	$[-0.37, 0.70]$	$[-0.43, 1.07]$
Observations	2,132	2,116	2,184	2,184	2,184	2,087	2,184
Number of clusters	271	271	271	271	271	271	271
Mean of DV	0.507	0.695	0.503	0.589	0.315	0.526	-0.005
KP F (aligned / non-aligned)	7.0 / 20.8	6.0 / 20.0	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8	7.0 / 20.4	6.4 / 19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 4. The framework makes no prediction for these outcomes, so no predicted sign is reported. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A14.

9 Dynamics

So far the results analyzed documented the effects of distance to headquarters on public service delivery from a static perspective. However, it is likely that these effects vary depending on the stage of development of the administrative centers. This section investigates the potential dynamic effects of distance on public service delivery using data from the NSDS 2015 and the

NSDS 2021, and then uses remote-sensing data to document how land use in the headquarters parishes themselves evolves after designation.

Given the cross-sectional nature of the NSDS, the different rounds of data collection do not capture information from the same households. Nevertheless, comparing cross-sectional coefficients of the impacts of distance to headquarters on public service delivery within a 5-year time span in the same districts can provide valuable insights of the evolution of these effects over time. To do so, I focus on the evolution of the importance of the distance to administrative headquarters for households belonging to districts created between 1991 and 2014 (one year before the earliest available dataset). To guide the analysis, I take the NSDS 2015 as the benchmark and analyze how the magnitude and significance of the distance to administrative headquarters change in both samples 2015 and 2021.

Since the NSDS 2015 does not include household geo-coordinates nor consistent parish names, I approximate a household's location using the household's sub-county centroid as its geocoordinate. Thus, I calculate the distance to the district's centroid by using the distance between the household's sub-county centroid and the district's centroid. Although this measure may contain some measurement error, its primary purpose is to capture the exogenous variation in distance introduced by the creation of a new district.

9.1 Dynamics: Main Results

Table 15 presents results for the major outcome indices in districts created between 1991 and 2014, using data from the NSDS 2015 (Panel A) and the NSDS 2021 (Panel B). Comparing the two panels reveals an informative evolution in the effects of proximity over time.

In 2015, proximity to district headquarters had limited effects on most service domains. The only statistically significant result is a large positive effect on agricultural inputs, robust to multiple hypothesis testing corrections. All remaining indices in education, health, water, electricity, and roads are small and insignificant. By 2021, this pattern changes considerably. Proximity is no longer affecting the agricultural inputs index, while it now significantly improves access to health services, water at both district and central government levels, and electricity.

These dynamics suggest that the relevance of administrative proximity for public service delivery strengthened considerably between 2015 and 2021, and that the domains in which it matters shifted over time. Two complementary forces could explain this pattern. First, the gradual consolidation of administrative capacity in newly created districts: the rapid pace of district creation between 2005 and 2010 imposed significant burdens on government capacity, generating delays in the effective establishment of district headquarters (Green, 2015; Cohen, 2024), so that many districts surveyed in 2015 may still have been in the early stages of administrative development. Second, the differential timeline of public investments: a new administration can quickly influence access to fast-turnaround services such as the distribution of agricultural

TABLE 15: DYNAMICS IMPACT OF HOUSEHOLD'S DISTANCE TO HEADQUARTERS ON INDICES OF MAJOR OUTCOMES

	Education district (1)	Education central gov (2)	Health district (3)	Water district (4)	Water central gov (5)	Electricity central gov (6)	Roads district (7)	Agr inputs district (8)
Panel A. NSDS 2015 - Period 1991-2014								
Log SR Proximity HH - District HQ	0.155	0.063	0.014	0.067	-0.115	-0.172	0.103	1.274***
SE	(0.143)	(0.122)	(0.157)	(0.197)	(0.200)	(0.154)	(0.197)	(0.391)
p-value	[0.280]	[0.605]	[0.929]	[0.734]	[0.565]	[0.264]	[0.603]	[0.001]
sharpened q-value	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{0.010}
Mean of Dependent Variable	-0.017	-0.006	0.104	0.004	0.010	-0.019	0.005	0.026
Observations	5,271	4,048	8,918	8,934	8,928	8,940	8,940	8,940
Number of Parishes	946	938	953	953	953	953	953	953
IV F-stat Kleibergen-Paap rk Wald	16.620	18.056	17.799	17.512	17.598	17.505	17.505	17.505
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel B. NSDS 2021 - Period 1991-2014								
Log SR Proximity HH - District HQ	0.262	0.173	0.397***	0.295**	0.377***	0.346***	0.085	-0.093
SE	(0.193)	(0.171)	(0.107)	(0.131)	(0.129)	(0.116)	(0.097)	(0.129)
p-value	[0.174]	[0.313]	[0.000]	[0.025]	[0.004]	[0.003]	[0.382]	[0.468]
sharpened q-value	{0.162}	{0.264}	{0.002}	{0.033}	{0.009}	{0.009}	{0.280}	{0.306}
Mean of Dependent Variable	-0.016	0.004	-0.035	-0.093	-0.036	-0.060	-0.018	0.087
Observations	6,555	5,148	6,959	7,752	7,752	7,752	7,752	7,752
Number of Parishes	885	879	878	894	894	894	894	894
IV F-stat Kleibergen-Paap rk Wald	27.382	28.469	40.570	36.997	36.997	36.997	36.997	36.997
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. All regressions control for district ever split between 1991-2014, lake Victoria district, districts' area in Km^2 as of 1991, household's parish centroid distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2014. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

inputs through existing extension services structures, but infrastructure investments require several years from planning and procurement through to completion. The pattern in Table 15 is consistent with this. While in 2015 the only significant effect was on agricultural inputs, an outcome that responds quickly to administrative capacity, by 2021 infrastructure investments near new headquarters materialized, producing the significant effects on health, water, and electricity access documented in the main results. While the evidence is suggestive, the combination of consolidating administrative capacity and the timeline of infrastructure provision goes in line with the main results and with existing research of the administrative constraints facing newly created districts in Uganda (Cohen, 2024).

To complement the NSDS results, I use Afrobarometer to provide further evidence on this consolidation pattern. The Afrobarometer records service availability at the enumeration area level, based on direct observation by enumerators rather than household self reports, and uses identical questions in the 2017 and 2024 rounds (Appendix Table A28). The results show that in 2017, proximity to district headquarters has no significant effect on five of the six availability indicators, with only a marginal effect on piped water. By 2024, the effects on piped water, paved roads, and police stations are large and statistically significant. The evidence coming from the Afrobarometer mirrors the evolution observed in the NSDS, reinforcing the interpretation that the returns to administrative proximity increased as new district institutions became more established.

9.2 Land Use Dynamics After Headquarters Establishment

The consolidation pattern documented above should also leave a physical footprint in the headquarters parishes themselves. Figure 4 shows that the share of parish land devoted to cultivated cropland, defined as pixels with more than 60 percent cultivation, increases continuously after a parish acquires headquarters status, and the effect persists up to 14 years after creation. This expansion is accompanied by a decline in the share of cropland mixed with natural vegetation, a land class that includes pixels with 40 to 60 percent cultivation (Appendix Figure A5), consistent with cultivation intensifying on land that was already partly farmed rather than with new land being cleared. Built-up area moves in the same direction. Appendix Figure A4, based on the GHS-BUILT-S product, shows that the built-up share of headquarters parishes rises steadily and monotonically after creation, by roughly 1.1 percentage points over twenty years, on the order of 40 percent of the baseline built-up share (joint nullity $p = 0.003$), with no pre-existing upward trend.

Taken together, these patterns indicate that headquarters designation spurs local agricultural activity and urbanization, and that this footprint builds up gradually over more than a decade rather than appearing at once. This is the same timing that the survey-based results above suggest: the returns to administrative proximity are limited in the years immediately after creation and strengthen as district institutions, and the towns that host them, become established.

10 The Fiscal Cost of District Creation

The former sections establish that proximity to new district headquarters improves access to public goods and services, increases citizen participation, and strengthens accountability. While these benefits are important, it is also necessary to understand the costs associated with district creation. This section provides evidence from government budgets to examine the fiscal implications of creating new districts. Central transfers to districts take two main forms. Conditional grants are earmarked for specific services such as education, health, water, and roads, and are allocated through formulas that scale with population and service needs. By contrast, the unconditional block grant finances the district administration itself, including the salaries of the LCV chairperson, the chief administrative officer, and headquarters staff, and is allocated to every district regardless of its size. Creating a new district therefore entails a fixed administrative cost, since each district requires its own administrative apparatus financed largely through the unconditional grant. District creation therefore generates a growing fiscal burden that constrains how far the strategy of bringing services closer to citizens can be efficiently extended.

To estimate the cost of district creation, I use a panel of approved local-government transfers from the Ministry of Finance, Planning and Economic Development (MoFPED), covering fiscal years 2012/13 to 2023/24. These data allow me to construct a detailed decomposition of each district's annual transfer grants. All amounts are expressed in real fiscal year 2017/18

shillings, deflating nominal budgets with the Uganda consumer price index.[18] The unit of observation is the district as a budget vote, the budget-holding unit through which MoFPED channels transfers to each local government. Municipal and city councils are excluded: they are separate urban local-government votes with their own transfers and administrative payroll, nested within districts or, since 2020, carved out of them, so including them would mix a different tier of government into the comparison and count part of a territory twice. Kampala, governed by the Kampala Capital City Authority under a separate framework, and Wakiso, the metropolitan outlier, are excluded as in the household analysis. For each of the 24 districts created during the sample period (calendar years 2015–2020), I build an event-time fiscal accounting around the split. Specifically, I compare the mother district's average budget in the two fiscal years before the split with its average budget in the two years after the split. I then add the child district's average budget during its first two years of operation to obtain the net change in total spending across the territory. The wage component of the unconditional block grant (which finances the LCV chairperson, chief administrative officer, district headquarters staff, and other political officeholders) provides the cleanest measure of duplicated administrative payroll costs.

Table 16 presents the dynamics of fiscal cost reallocation. Panel A shows an inverse relationship between the total transfers received by a district and the relative importance of unconditional grants. The unconditional block grant represents a substantially larger share of central transfers in districts with smaller budgets. This pattern appears in both the wage and non-wage components. By contrast, the share allocated to conditional grants, which are tied to service delivery formulas, is largely unrelated to district fiscal size. Similarly, Panel B shows that recently created districts devote roughly two percentage points more of their budgets to unconditional grants than established districts, with most of the difference concentrated in the wage component, while the share allocated to conditional grants remains essentially unchanged. Because conditional grants scale with service needs, whereas unconditional grants finance the administrative apparatus required to operate as a district, this contrast suggests that the larger administrative share observed in small and newly created districts reflects a fixed cost of district administration rather than differences in service provision.

Finally, it is important to address whether changes of budget are a zero-sum exercise. Appendix Table A29 therefore traces the fiscal accounting across the 24 splits in real terms and benchmarks each split against the contemporaneous budget growth of never-split 1991 territories. Relative to this counterfactual, total spending in the territory increases by 5.37 billion shillings on average, equivalent to 19 percent of the mother district's pre-split budget (12 percent at the median), with a positive excess in 19 of the 24 splits.

To complement this analysis, I aggregate the panel to fixed territories defined by the 1991 district borders. This means that transfers to a mother district and to any districts later carved out of it are added within the same geographic unit, netting out reallocations between old and

[18] World Development Indicators series FP.CPI.TOTL; each fiscal year is assigned the average of the two overlapping calendar years.

TABLE 16: Administrative overhead share and recently-created districts

	Wage bill (1)	Uncond. (admin) (2)	Uncond. wage (3)	Uncond. non-wage (4)	Cond. (services) (5)
Panel A. Cross-sectional scale (log of total transfers)					
ln(transfers_full)	0.0168	-0.0539***	-0.0419***	-0.0120***	0.0064
	(0.0129)	(0.0053)	(0.0042)	(0.0015)	(0.0103)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.387	0.584	0.485	0.650	0.516
Panel B. Recently-created district premium (created within 3 FYs)					
Created within last 3 FYs	-0.0108	0.0202***	0.0156***	0.0046***	-0.0050
	(0.0162)	(0.0060)	(0.0052)	(0.0015)	(0.0176)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.383	0.255	0.140	0.536	0.515

Notes: * p<0.1, ** p<0.05, *** p<0.01. Each column is a separate regression whose dependent variable is the share of the district's total transfers going to that grant category; unconditional grants (Columns 2–4) finance the fixed administrative apparatus, while conditional grants (Column 5) are tied to service-delivery formulas. Panel A regresses each share on the log of the district's total transfers (its fiscal size); Panel B, on an indicator for districts created within the last 3 fiscal years. Column 5 shares the same denominator as the overhead measures and serves as a placebo: its flat relationship with size rules out a mechanical correlation between the overhead share and total transfers. All columns include financial year fixed effects. Sample: district-FY observations FY2012/13–FY2023/24, excluding Kampala, Wakiso, and municipal/city councils. Standard errors clustered at the district level are in parenthesis.

new districts. I then estimate a staggered event study of log real territory transfers using the estimator of de Chaisemartin & D'Haultfœuille (2020). A territory is classified as treated in the first fiscal year in which a newly created district carved out of it receives its own budget vote. Of the 37 territories, 18 experience such a split during the sample period, while the remaining 19 either are never split or were already subdivided before the panel begins, and therefore serve as controls. Figure 5 shows that real transfers to treated territories increase by approximately 14 percent after the first split (corresponding to an average total effect of 0.135 log points, with a standard error of 0.042). The effects grow from 9 to 17 percent over the following four years, with no evidence of differential pre trends (with a joint placebo p value of 0.73). The composition of this increase mirrors the allocation mechanism documented in Table 16. The wage component of the unconditional grant, which captures the duplicated administrative payroll, rises by 26 percent, compared with only 7 percent for conditional service delivery grants (Appendix Figure A8).[19] Consistently, Table 17 shows that, holding the territory fixed, the share of transfers devoted to the administrative payroll increases with the number of districts into which the territory is divided.

These results show that there is a limit to the strategy of bringing government closer to citizens through district creation. Each new district requires its own administrative apparatus, making further subdivision increasingly costly. Importantly, these fiscal costs are different from the mechanism identified in the main analysis. The estimated benefits of proximity are not driven by higher service delivery budgets in newly created districts, since the additional resources are concentrated in administrative transfers rather than conditional grants. The evidence therefore suggests that district creation can improve service access, engagement, and accountability, but does so through a fiscally costly expansion of the state apparatus.

[19]The joint placebo test for the unconditional wage component is significant, with a p value of 0.02.

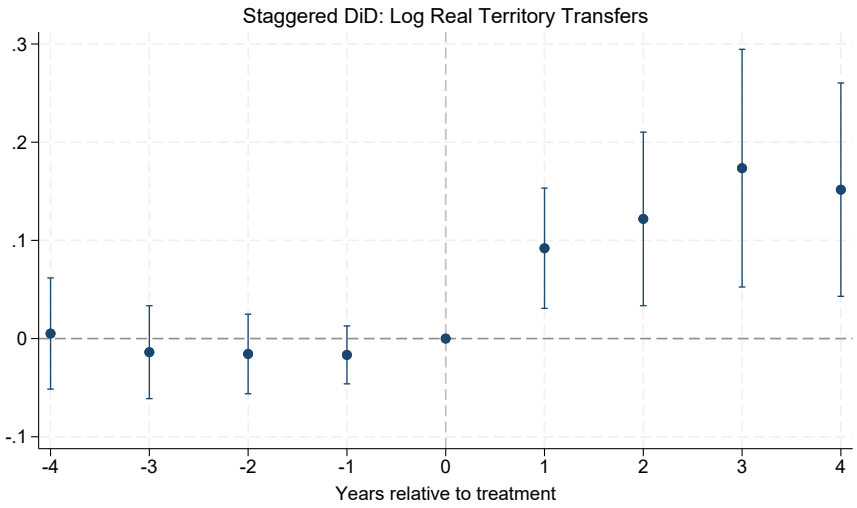


FIGURE 5: Event study: log real territory transfers around the first district split. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24 (real FY2017/18 shillings); 18 territories treated at the first fiscal year in which a newly created district receives its own budget vote, and 19 control territories with no split during the sample window (their subdivision, where any, was completed before FY2012/13). Estimates from the heterogeneity-robust estimator of de Chaisemartin & D'Haultfœuille (2020) with four pre- and four post-periods; the joint test of the placebo estimates has a p-value of 0.73. Kampala, Wakiso, and municipal/city councils excluded.

TABLE 17: Administrative Overhead and Territory Fragmentation

	UCG-wage share (1)	Unconditional share (2)
Log number of districts in territory	0.021***	0.020***
	(0.004)	(0.007)
Territory FE	Yes	Yes
FY FE	Yes	Yes
Observations	444	444
Territories	37	37

Notes: *p<0.1, **p<0.05, ***p<0.01. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24, aggregating the approved transfers of all constituent districts. The dependent variables are the shares of the territory's total approved transfers allocated to the wage component of the unconditional block grant (Column 1) and to the unconditional block grant as a whole (Column 2). The regressor is the logarithm of the number of districts in the territory in each fiscal year, where a district is counted from the first year in which it holds its own budget vote, the budget-holding unit in the MoFPED transfer data. Kampala, Wakiso, and municipal/city councils are excluded throughout. Standard errors clustered at the territory level are in parenthesis.

11 Conclusions

This paper provides empirical evidence on the impact of proximity to administrative centers on public service delivery and state-citizen relations in the context of Uganda's redistricting process. By leveraging a governmental rule governing the placement of new district headquarters through an instrumental variable approach, this study identifies the causal effects of households' proximity to new administrative centers using granular georeferenced survey data combined with detailed geographical information on districts.

The findings show that proximity to administrative centers significantly improves access to public goods and services. Households closer to new district headquarters have better access to electricity and high-quality roads, shorter travel times to health facilities, and improved access to water sources. The effects are strongest for services provided by the central government, where district administrators play an important coordinating role. Political alignment helps explain part of this pattern. Proximity generates larger gains for centrally provided infrastructure in aligned districts, particularly for roads, while the effects on district-provided services vary much less with alignment.

Proximity also changes how citizens interact with the state. Households closer to headquarters have greater knowledge of formal institutions and are less reliant on traditional alternatives. Contentious forms of engagement, including protests and contact with the media, decline, while the evidence points toward greater use of formal participatory channels. Direct corruption experiences also fall and satisfaction with the RDC increases. These changes are concentrated in politically aligned districts, where proximity produces a stronger shift in engagement and larger improvements in measures of accountability. The pattern is consistent with proximity being more effective when the elected and appointed parts of district government face compatible incentives.

However, these improvements do not translate into greater political legitimacy. Proximity has little effect on satisfaction with democracy, perceived political freedoms, rule of law, or trust in local and national institutions. This null result persists in politically aligned districts, despite the stronger gains in services, engagement, and accountability observed there. In non-aligned districts, proximity is instead associated with lower perceived freedom to vote and lower trust in local government and the ruling party. Taken together, these findings show that a state can become more present and perform better without necessarily becoming more politically trusted.

Despite these improvements, proximity to new district headquarters does not translate into higher trust in government or more favorable democratic perceptions. Satisfaction with democracy, perceived political freedoms, rule of law, and trust in both local and national institutions remain largely unchanged.

The effects of administrative proximity also take time to emerge. Comparing survey rounds shows that proximity had limited effects on most services in 2015 but substantially larger effects by 2021, particularly for health, water, and electricity. Remote-sensing evidence similarly shows gradual increases in economic activity and built-up area around new headquarters after their establishment. These patterns are consistent with the benefits of proximity strengthening as new district administrations and the towns hosting them become more established.

These gains come at a fiscal cost. Each new district requires its own administrative apparatus, and the intergovernmental transfer system finances a core set of administrative expenditures for every district. District creation raises total transfers to the original pre-split territory by

around 14 percent, with most of the increase concentrated in administrative payroll rather than conditional service delivery grants. The service improvements are therefore not mechanically explained by larger service budgets. Instead, the evidence shows that extending the spatial reach of the state through district creation requires a substantial expansion of administrative spending.

Two caveats are important when interpreting these results. First, the instrumental variable estimates identify a Local Average Treatment Effect for households whose proximity to the new district centroid affects their proximity to the headquarters. The estimates therefore speak most directly to the effects of administrative proximity generated by the creation and placement of new district centers. Second, political alignment is not randomly assigned. The estimates identify the effect of proximity separately within aligned and non-aligned districts, but differences across the two groups should not be interpreted as the causal effect of political alignment itself.

Overall, the results show that reducing the geographical distance between citizens and local administrators can improve public service delivery and reshape citizen-state interactions. These gains are larger and more consistent when elected and appointed district leaders are politically aligned, highlighting that the effects of administrative proximity depend on the institutional environment in which local governments operate. At the same time, creating smaller administrative units carries substantial fiscal costs, and better service delivery and accountability do not automatically generate greater political trust. Administrative decentralization can therefore extend the reach of the state without necessarily strengthening its political legitimacy.

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Appendix

A Conceptual Framework

This section develops a reduced-form model of public good provision with two local administrators who differ in their accountability relationships and citizens who choose how to engage with the state. The model formalizes how geographical distance to the administrative center and political alignment between local leaders shape service delivery, citizen engagement, and accountability (proxied by corruption).

A.1 Setup

Consider a district with two administrators: an elected Local Council V (LCV) chairperson, indexed by L, who is directly accountable to local voters, and a centrally appointed Resident District Commissioner (RDC), indexed by R, who represents the national government within the district.[20] A household i is located at distance $d_i \geq 0$ from the district headquarters.

Public good production and spatial delivery. Each administrator simultaneously chooses effort $e_j \geq 0$, $j \in \{L, R\}$. Effort translates into higher public goods and services for the district. However, the benefits of this effort are decreasing with distance to the headquarters. The public good available to household i at distance d_i from the headquarters is

$$g_i = \left(e_L + \alpha\, e_R\right) \phi(d_i), \tag{3}$$

where $\alpha \in (0, 1]$ is a *political alignment* parameter governing the share of RDC effort that translates into local public goods, and $\phi(d_i)$ is a *spatial attenuation function* satisfying $\phi(0) = 1$, $\phi'(d_i) < 0$, and $\phi''(d_i) \leq 0$.

The spatial attenuation function $\phi(d_i)$ captures the fact that infrastructure coverage, service quality, and administrative reach decline with distance from the district headquarters. This reflects the empirical patterns that electricity grids and national roads are less available farther from administrative centers, health workers tend to concentrate near headquarters, and oversight of service delivery weakens at the periphery (Asher et al., 2018). Importantly, the attenuation does not reflect a deliberate choice by administrators to favor nearby households. Instead, it reflects the spatial structure of service delivery: administrators choose effort once for the district, and the household-level realization of that effort depends on where the household is located relative to the headquarters.

[20] The distinction between elected and appointed local officials builds on Besley & Coate (2003), who study the trade-offs between these two selection mechanisms. A related dual-principal structure is detailed in the working paper version of Martinez-Bravo et al. (2022) with village chiefs in rural China being simultaneously accountable to villagers and to the autocratic central government.

The alignment parameter α captures the weight the RDC places on local welfare relative to a fixed outside payoff from central loyalty. When the LCV chairperson and the RDC share political affiliation (α close to 1), local welfare dominates the RDC's payoff, and the RDC exerts high effort on locally valued services. When they are not aligned (α is small), the fixed outside payoff (which captures career concerns, patronage rents, or political loyalty rewards) dominates the RDC's payoff, and effort on local services falls accordingly.[21].

Administrator preferences. Administrators choose effort to maximize their respective payoffs. The LCV, being accountable to local voters, maximizes a payoff that is increasing in aggregate household welfare and decreasing in effort cost:

$$U_L = \big(e_L + \alpha\, e_R\big)\, \bar{\phi} - \tfrac{1}{2}\, e_L^2, \tag{4}$$

where $\bar{\phi} \equiv \int \phi(d)\, dF(d)$ denotes the average spatial attenuation across the district population. The LCV cares about the welfare of the representative constituent, whose access to public goods depends on $\bar{\phi}$. The RDC internalizes local welfare with weight α and receives a fixed outside benefit $(1-\alpha)\bar{u}$ from serving the central government's agenda, capturing career concerns, patronage rents, or political loyalty rewards:

$$U_R = \alpha\big(e_L + \alpha\, e_R\big)\, \bar{\phi} + (1-\alpha)\, \bar{u} - \tfrac{1}{2}\, e_R^2. \tag{5}$$

A.2 Equilibrium Provision

Administrators choose effort simultaneously in a Nash equilibrium. The first-order conditions are:

$$\text{LCV:} \quad e_L^* = \bar{\phi}, \tag{6}$$

$$\text{RDC:} \quad e_R^* = \alpha^2\, \bar{\phi}. \tag{7}$$

The LCV's optimal effort equals $\bar{\phi}$, the average spatial return to effort in the district. The RDC's effort scales with $\alpha^2\, \bar{\phi}$. Substituting into (3), the equilibrium public good available to household i is

$$g_i^* = \big(1 + \alpha^3\big)\, \bar{\phi}\, \phi(d_i). \tag{8}$$

The equilibrium expression separates two components. The term $\bar{\phi}$ captures the district-level supply of public goods: it determines how much effort administrators exert and reflects the average accessibility of the population. The term $\phi(d_i)$ captures the household-level realization:

[21]This specification follows Martinez-Bravo et al. (2022) in modeling local officials under dual accountability, where preference congruence between vertically and locally accountable principals shapes governance outcomes, abstracting from the within-agent reallocation of effort across competing tasks between the central and the local government

it determines what share of the district's effort effectively reaches household i. Together, they imply that a household benefits from both living in a compact district (high $\bar{\phi}$) and from being located close to the headquarters (high $\phi(d_i)$). In the context of redistricting, the creation of a new district with a centrally located headquarters raises $\bar{\phi}$ (the average distance falls for the splitter district population) and, for many households, also increases $\phi(d_i)$ (the household is now closer to its headquarters). These two forces reinforce each other.

Proposition 1 (Provision decreasing in distance)**.** *Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters:*

$$\frac{\partial g_i^*}{\partial d_i} = \left(1 + \alpha^3\right) \bar{\phi}\, \phi'(d_i) < 0.$$

Households located closer to the district headquarters have better access to public goods and services. The result follows directly from the spatial attenuation of service delivery: infrastructure and administrative capacity radiate outward from the administrative center, and households farther away receive less of the public good regardless of the effort level chosen by administrators. This prediction is consistent with the government's stated objective of bringing services closer to citizens through district creation.

Proposition 2 (Alignment premium)**.** *Equilibrium provision is strictly increasing in political alignment:*

$$\frac{\partial g_i^*}{\partial \alpha} = 3\alpha^2\, \bar{\phi}\, \phi(d_i) > 0.$$

When the RDC and LCV share political incentives, a greater share of RDC effort is channeled toward locally valued services, raising total provision. The convexity in α implies that the marginal return to alignment is largest when alignment is already high, reflecting the compounding nature of incentive congruence between the two administrators.

Proposition 3 (Complementarity between proximity and alignment)**.** *The marginal benefit of proximity is strictly increasing in alignment:*

$$\frac{\partial}{\partial \alpha}\left(-\frac{\partial g_i^*}{\partial d_i}\right) = -3\alpha^2\, \bar{\phi}\, \phi'(d_i) > 0.$$

This cross-derivative establishes that distance and alignment are complements in the production of local public goods. The intuition is that proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. In non-aligned districts, even nearby households benefit less from proximity because RDC effort is partly diverted to central objectives. This predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

A.3 Citizen Engagement

The improvements in service delivery documented above depend not only on administrative effort but also on the quality of information that administrators receive about local needs. This section models how proximity shapes citizens' decisions to engage with the state and, consequently, the information available to administrators.

Citizen preferences. A citizen located at distance d_i from the headquarters derives utility from the public good g_i^* and may choose to engage with the state. Engagement takes one of two forms: *participatory* (attending district council meetings, meeting the RDC, using formal complaint and feedback channels) or *contentious* (joining protests, contacting the media, organizing collective pressure campaigns). The citizen's payoff is:

$$V_i = g_i^* + \max\Big\{ \underbrace{b_P(\alpha) - c_P(d_i)}_{\text{participatory}},\ \underbrace{b_C - \bar{c}_C}_{\text{contentious}},\ 0 \Big\}, \tag{9}$$

where $b_P(\alpha)$ is the benefit of participatory engagement, $c_P(d_i)$ is its cost, b_C is the benefit of contentious engagement, and $\bar{c}_C$ is the cost of contentious action. The citizen chooses whichever form of engagement yields the highest net benefit, or does not engage at all.

Two features of the cost structure are important. First, the cost of participatory engagement is increasing in distance ($c_P'(d_i) > 0$): attending a district council meeting, visiting the RDC's office, or filing a formal complaint all require traveling to the headquarters, with time and transport costs that rise with d_i. Second, the cost of contentious engagement $\bar{c}_C$ is independent of distance to the headquarters (e.g. protests can be organized locally).

The benefit structure includes the role of alignment. Participatory engagement is assumed to be more effective at influencing policy outcomes ($b_P(\alpha) > b_C$ for sufficiently high α) because it transmits detailed information about local priorities directly to decision-makers, while contentious action is a noisier signal. Additionally, $b_P(\alpha)$ is increasing in alignment: when the RDC internalizes local welfare, the state is more responsive to citizen input, making participatory engagement more rewarding. In non-aligned districts, citizens who engage through formal channels may find their input disregarded by an RDC pursuing central directives, reducing the returns to participation.

Engagement equilibrium. Given these assumptions, there exists a distance threshold $\hat{d}(\alpha)$, defined by $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$, such that citizens at $d_i < \hat{d}(\alpha)$ choose participatory engagement, while citizens at $d_i > \hat{d}(\alpha)$ choose contentious engagement or do not engage at all. Since $b_P'(\alpha) > 0$, the threshold $\hat{d}(\alpha)$ is increasing in alignment: in aligned districts, participatory engagement dominates over a wider range of distances.

Proposition 4 (Proximity shifts engagement from contentious to participatory)**.** *A reduction in*

distance d_i increases participatory engagement and decreases contentious engagement:

$$\frac{\partial \Pr(\textit{participatory})}{\partial d_i} < 0, \qquad \frac{\partial \Pr(\textit{contentious})}{\partial d_i} > 0.$$

The mechanism is that proximity reduces the cost of formal engagement with district institutions. When the headquarters is within reach, citizens can attend council meetings and communicate their preferences directly to local administrators, making contentious action unnecessary. When the headquarters is distant, the cost of formal engagement rises and citizens resort to protests and media contact as lower-cost alternatives for voicing demands. The overall pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it from contentious to participatory channels.

Proposition 5 (Alignment amplifies participatory engagement)**.** *The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$. In aligned districts, proximity produces stronger shifts toward participatory engagement.*

Proof. Differentiating $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$ with respect to α yields $b'_P(\alpha) = c'_P(\hat{d})\,\hat{d}'(\alpha)$. Since $b'_P(\alpha) > 0$ and $c'_P(\hat{d}) > 0$, it follows that $\hat{d}'(\alpha) > 0$. □

When the state is responsive to citizen input (high α), the returns to participatory engagement are larger, making citizens willing to bear higher travel costs to engage formally. This predicts that the reduction in contentious engagement and the increase in participatory engagement from proximity should be more pronounced in politically aligned districts. In non-aligned districts, the lower responsiveness of the state limits the returns to formal engagement, and citizens may not shift away from contentious channels even when the headquarters is nearby.

A.4 Corruption and Monitoring

A further channel through which proximity affects public service delivery is accountability. After effort choices are made, each administrator can divert a fraction $\kappa_j \in [0, \bar{\kappa}]$ of resources for private gain, yielding private benefit $b \cdot \kappa_j$. Citizens monitor administrators with intensity

$$\mu(d_i), \quad \text{with } \mu'(d_i) < 0 \text{ and } \mu(0) = \bar{\mu} \leq 1, \tag{10}$$

so that closer households are better able to observe and report misconduct. Detection triggers a penalty $P > 0$. The monitoring-through-proximity mechanism builds on Campante & Do (2014), who show that isolated capital cities are associated with higher corruption, and with Narasimhan & Weaver (2024), who show that polity size can improve public service delivery under political accountability mechanisms. In the Ugandan context, proximity to the headquarters determines how effectively citizens can exercise this informational advantage.[22]

[22]Experimental evidence from Uganda supports the accountability channel modeled here. Björkman & Svensson (2009) show that community-based monitoring significantly improves the quality of health service delivery in

The RDC's net payoff from corruption is $b\,\kappa_R - \mu(d_i)\,P\,\kappa_R - \alpha\,\lambda\,\kappa_R$, where $\lambda > 0$ is the marginal welfare cost of diversion, internalized at rate α. Since the payoff is linear in κ_R, the RDC diverts resources if and only if $b - \alpha\,\lambda > \mu(d_i)\,P$. Two forces discipline the administrator: external monitoring by citizens ($\mu(d_i)\,P$) and internal alignment with local welfare ($\alpha\,\lambda$). Both are stronger when the household is closer and when the district is politically aligned.

Proposition 6 (Proximity deters corruption)**.** *There exists a distance threshold $\hat{d}_c$ defined by $b - \alpha\lambda = \mu(\hat{d}_c)\,P$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$. Since $\mu'(d_i) < 0$, households closer to the headquarters experience less corruption.*

Proposition 7 (Alignment amplifies the accountability effect of proximity)**.** *The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial\hat{d}_c/\partial\alpha > 0$. In aligned districts, corruption is deterred over a wider range of distances.*

Proof. Implicit differentiation of $b - \alpha\lambda = \mu(\hat{d}_c)\,P$ with respect to α gives $-\lambda = \mu'(\hat{d}_c)\,P\,\frac{\partial\hat{d}_c}{\partial\alpha}$. Since $\lambda > 0$, $\mu'(\hat{d}_c) < 0$, and $P > 0$, it follows that $\frac{\partial\hat{d}_c}{\partial\alpha} = \frac{\lambda}{-\mu'(\hat{d}_c)\,P} > 0$. □

Alignment introduces an additional deterrent beyond citizen monitoring: an aligned RDC internalizes the welfare cost of corruption, making diversion less attractive even at moderate distances from the headquarters. Taken together, these two propositions predict that corruption should be lower in areas closer to the administrative center, and that this reduction should be more pronounced in politically aligned districts.

Discussion. The model predicts that proximity to district headquarters improves public service delivery, shifts citizen engagement from contentious to participatory channels, and deters corruption. Political alignment between the elected LCV and the appointed RDC amplifies each of these effects. The main results in Section 6 test these predictions.

B Proofs of Propositions

B.1 Derivation of the Equilibrium

The administrators choose effort simultaneously in a Nash equilibrium. Each administrator takes the other's effort as given and maximizes their own payoff.

LCV's problem. The LCV chooses e_L to maximize (4). The first-order condition is:

$$\frac{\partial U_L}{\partial e_L} = \bar{\phi} - e_L = 0 \quad \Longrightarrow \quad e_L^* = \bar{\phi}. \tag{11}$$

Ugandan health facilities, suggesting that citizen oversight can meaningfully discipline local officials.

RDC's problem. The RDC chooses e_R to maximize (5). The first-order condition is:

$$\frac{\partial U_R}{\partial e_R} = \alpha^2 \, \bar{\phi} - e_R = 0 \quad \Longrightarrow \quad e_R^* = \alpha^2 \, \bar{\phi}. \tag{12}$$

Equilibrium public good. Substituting $e_L^* = \bar{\phi}$ and $e_R^* = \alpha^2 \bar{\phi}$ into (3):

$$\begin{aligned} g_i^* &= (e_L^* + \alpha \, e_R^*) \, \phi(d_i) \\ &= (\bar{\phi} + \alpha \cdot \alpha^2 \bar{\phi}) \, \phi(d_i) \\ &= \bar{\phi} \, (1 + \alpha^3) \, \phi(d_i). \end{aligned} \tag{13}$$

B.2 Proof of Proposition 1: Provision Decreasing in Distance

Statement. Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters: $\frac{\partial g_i^*}{\partial d_i} < 0$.

Proof. Differentiating (13) with respect to d_i:

$$\frac{\partial g_i^*}{\partial d_i} = (1 + \alpha^3) \, \bar{\phi} \, \phi'(d_i).$$

Since $\alpha \in (0, 1]$, the term $(1 + \alpha^3) > 0$. Since $\bar{\phi} > 0$ by construction. Since $\phi'(d_i) < 0$ by assumption. Therefore:

$$\frac{\partial g_i^*}{\partial d_i} = \underbrace{(1 + \alpha^3)}_{>0} \, \underbrace{\bar{\phi}}_{>0} \, \underbrace{\phi'(d_i)}_{<0} < 0.$$

B.3 Proof of Proposition 2: Alignment Premium

Statement. Equilibrium provision is strictly increasing in political alignment: $\frac{\partial g_i^*}{\partial \alpha} > 0$.

Proof. Differentiating (13) with respect to α:

$$\frac{\partial g_i^*}{\partial \alpha} = \bar{\phi} \, \phi(d_i) \, \frac{\partial}{\partial \alpha}(1 + \alpha^3) = 3\alpha^2 \, \bar{\phi} \, \phi(d_i).$$

Since $\alpha \in (0, 1]$, $\bar{\phi} > 0$, and $\phi(d_i) > 0$ for finite d_i:

$$\frac{\partial g_i^*}{\partial \alpha} = \underbrace{3\alpha^2}_{>0} \, \underbrace{\bar{\phi}}_{>0} \, \underbrace{\phi(d_i)}_{>0} > 0.$$

B.4 Proof of Proposition 3: Complementarity Between Proximity and Alignment

Statement. The marginal benefit of proximity is strictly increasing in alignment:

$$\frac{\partial}{\partial \alpha}\left(-\frac{\partial g_i^*}{\partial d_i}\right) > 0.$$

Proof. The marginal benefit of proximity is the negative of the distance derivative (since lower d_i means higher proximity):

$$-\frac{\partial g_i^*}{\partial d_i} = -(1+\alpha^3)\,\bar{\phi}\,\phi'(d_i).$$

This expression is positive since $\phi'(d_i) < 0$. Differentiating with respect to α:

$$\frac{\partial}{\partial \alpha}\left(-\frac{\partial g_i^*}{\partial d_i}\right) = -3\alpha^2\,\bar{\phi}\,\phi'(d_i).$$

Since $\alpha^2 > 0$, $\bar{\phi} > 0$, and $\phi'(d_i) < 0$:

$$\frac{\partial}{\partial \alpha}\left(-\frac{\partial g_i^*}{\partial d_i}\right) = -\underbrace{3\alpha^2}_{>0}\,\underbrace{\bar{\phi}}_{>0}\,\underbrace{\phi'(d_i)}_{<0} > 0.$$

The positive cross-derivative establishes that distance and alignment are complements: proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed.

B.5 Proof of Proposition 4: Proximity Shifts Engagement from Contentious to Participatory

Statement. A reduction in distance d_i increases participatory engagement and decreases contentious engagement.

Setup. The citizen at distance d_i chooses among three options: participatory engagement (net benefit $b_P(\alpha) - c_P(d_i)$), contentious engagement (net benefit $b_C - \bar{c}_C$), or no engagement (net benefit 0). The citizen chooses the option with the highest net payoff.

Participatory engagement dominates contentious engagement if and only if:

$$b_P(\alpha) - c_P(d_i) > b_C - \bar{c}_C. \tag{14}$$

Rearranging:

$$c_P(d_i) < b_P(\alpha) - b_C + \bar{c}_C.$$

Define the threshold $\hat{d}(\alpha)$ as the value of d_i at which the citizen is indifferent between the two forms of engagement:

$$c_P(\hat{d}(\alpha)) = b_P(\alpha) - b_C + \bar{c}_C. \tag{15}$$

Proof. Since $c'_P(d_i) > 0$ (the cost of participatory engagement is increasing in distance), the function c_P is strictly increasing. Therefore, condition (14) holds if and only if $d_i < \hat{d}(\alpha)$. This means:

- For $d_i < \hat{d}(\alpha)$: the citizen chooses participatory engagement (c_P is low enough).
- For $d_i > \hat{d}(\alpha)$: the citizen chooses contentious engagement (if $b_C - \bar{c}_C > 0$) or does not engage.

A reduction in d_i moves the citizen from the contentious to the participatory region whenever d_i crosses $\hat{d}(\alpha)$ from above. For a population of citizens with a continuous distribution of distances $F(d)$:

$$\Pr(\text{participatory}) = F(\hat{d}(\alpha)), \qquad \Pr(\text{contentious}) = 1 - F(\hat{d}(\alpha)) - \Pr(\text{no engagement}).$$

Since F is increasing, a reduction in the distance of an individual citizen (which shifts probability mass below $\hat{d}$) increases the probability of participatory engagement and decreases the probability of contentious engagement.

B.6 Proof of Proposition 5: Alignment Amplifies Participatory Engagement

Statement. The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$.

Proof. Differentiate the indifference condition (15) with respect to α:

$$c'_P(\hat{d}(\alpha))\, \hat{d}'(\alpha) = b'_P(\alpha).$$

Solving for $\hat{d}'(\alpha)$:

$$\hat{d}'(\alpha) = \frac{b'_P(\alpha)}{c'_P(\hat{d}(\alpha))}.$$

Since $b'_P(\alpha) > 0$ (the benefit of participatory engagement is increasing in alignment, as a more responsive state makes formal engagement more rewarding) and $c'_P(\hat{d}) > 0$ (the cost of

participatory engagement is increasing in distance):

$$\hat{d}'(\alpha) = \frac{\overbrace{b'_P(\alpha)}^{>0}}{\underbrace{c'_P(\hat{d}(\alpha))}_{>0}} > 0.$$

The positive sign means that higher alignment expands the set of distances for which participatory engagement is preferred. In aligned districts, citizens are willing to bear higher travel costs to engage formally because they expect the state to be more responsive to their input.

B.7 Proof of Proposition 6: Proximity Deters Corruption

Statement. There exists a distance threshold $\hat{d}_c$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$.

Setup. After effort choices are made, the RDC can divert a fraction $\kappa_R \in [0, \bar{\kappa}]$ of resources for private gain. The RDC's net payoff from corruption is:

$$\Pi(\kappa_R) = b\,\kappa_R - \mu(d_i)\,P\,\kappa_R - \alpha\,\lambda\,\kappa_R = \big[b - \mu(d_i)\,P - \alpha\,\lambda\big]\,\kappa_R,$$

where $b > 0$ is the private benefit per unit diverted, $\mu(d_i)$ is the monitoring intensity with $\mu'(d_i) < 0$ and $\mu(0) = \bar{\mu}$, $P > 0$ is the penalty upon detection, and $\lambda > 0$ is the marginal welfare cost of diversion internalized at rate α.

Proof. Since $\Pi(\kappa_R)$ is linear in κ_R, the RDC's optimal strategy is a corner solution:

$$\kappa_R^* = \begin{cases} \bar{\kappa} & \text{if } b - \alpha\lambda > \mu(d_i)\,P, \\ 0 & \text{if } b - \alpha\lambda \leq \mu(d_i)\,P. \end{cases}$$

Corruption is deterred whenever $\mu(d_i)\,P \geq b - \alpha\lambda$, i.e., when monitoring is strong enough relative to the net temptation to divert. Assuming $b - \alpha\lambda > 0$ (corruption is tempting absent monitoring) and $\mu(0)\,P > b - \alpha\lambda$ (monitoring deters corruption at zero distance), there exists a unique threshold $\hat{d}_c$ solving:

$$\mu(\hat{d}_c)\,P = b - \alpha\lambda. \tag{16}$$

The threshold exists and is unique because $\mu(\cdot)$ is continuous and strictly decreasing. Since $\mu'(d_i) < 0$, monitoring weakens with distance, and therefore:

- For $d_i < \hat{d}_c$: $\mu(d_i) > \mu(\hat{d}_c)$, so $\mu(d_i)\,P > b - \alpha\lambda$, and corruption is deterred ($\kappa_R^* = 0$).
- For $d_i > \hat{d}_c$: $\mu(d_i) < \mu(\hat{d}_c)$, so $\mu(d_i)\,P < b - \alpha\lambda$, and corruption occurs ($\kappa_R^* = \bar{\kappa}$).

B.8 Proof of Proposition 7: Alignment Amplifies the Accountability Effect of Proximity

Statement. The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial \hat{d}_c / \partial \alpha > 0$.

Proof. Differentiate the threshold condition (16) with respect to α:

$$\mu'(\hat{d}_c)\, P \, \frac{\partial \hat{d}_c}{\partial \alpha} = -\lambda.$$

Solving for $\frac{\partial \hat{d}_c}{\partial \alpha}$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{-\lambda}{\mu'(\hat{d}_c)\, P} = \frac{\lambda}{-\mu'(\hat{d}_c)\, P}.$$

Since $\lambda > 0$, $-\mu'(\hat{d}_c) > 0$ (because $\mu'(\hat{d}_c) < 0$), and $P > 0$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{\overbrace{\lambda}^{>0}}{\underbrace{(-\mu'(\hat{d}_c))}_{>0} \underbrace{P}_{>0}} > 0.$$

A higher α increases the range of distances over which corruption is deterred. The mechanism is that an aligned RDC internalizes a larger share of the welfare cost of diversion ($\alpha\lambda$ is larger), which reduces the net temptation to engage in corruption ($b - \alpha\lambda$ falls). This means that a lower level of external monitoring is sufficient to deter corruption, extending the deterrence region to more distant households. In aligned districts, both the internal alignment cost and the external monitoring force work together to discipline administrators, generating stronger accountability effects of proximity.

C Administrative Unit Proliferation in Comparative Perspective

Table A1 places Uganda's district creation in comparative perspective. It reports, for sub-Saharan African countries with verifiable counts and for five countries outside the region, the number of subnational units of a given tier at two dates. Uganda's increase is the largest in the region, but the direction is shared by most countries listed, and the constitutional or statutory criteria for creating units typically refer to population and to bringing services closer to citizens.

TABLE A1: Administrative Unit Proliferation in Sub-Saharan Africa and Elsewhere

Country	Unit	Initial (year)	Later (year)	Change (%)	Source
Panel A: Sub-Saharan Africa					
Uganda	Districts	38 (1991)	136 (2020)	+258	This paper
Kenya	Districts	41 (1992)	158 (2009)	+285	Hassan (2016); Kenya National Bureau of Statistics (2010)
Ghana	Districts	110 (1988)	216 (2012)	+96	Resnick (2017)
Nigeria	Local government areas	301 (1976)	774 (1996)	+157	Adeyemi (2013)
DR Congo	Provinces	11 (2014)	26 (2015)	+136	Law (2026)
Benin	Départements	6 (1990)	12 (2000)	+100	Grossman & Lewis (2014)
Angola	Municípios	164 (2024)	326 (2025)	+99	Law (2026)
Togo	Préfectures	21 (1990)	35 (2010)	+67	Grossman & Lewis (2014)
Zambia	Districts	72 (2011)	116 (2018)	+61	Law (2026)
Burkina Faso	Provinces	30 (1996)	45 (1997)	+50	Billing (2019)
Senegal	Départements	30 (1990)	45 (2010)	+50	Grossman & Lewis (2014)
Ethiopia	Woredas	556 (1990)	736 (2010)	+32	Grossman & Lewis (2014)
Tanzania	Districts	119 (1990)	149 (2012)	+25	Grossman & Lewis (2014)
Mali	Cercles	40 (1990)	49 (2005)	+23	Grossman & Lewis (2014)
Mozambique	Districts	128 (2012)	151 (2013)	+18	Law (2026)
Malawi	Districts	24 (1990)	28 (2003)	+17	Grossman et al. (2017)
Sierra Leone	Districts	14 (2016)	16 (2017)	+14	Law (2026)
Consolidations					
Rwanda	Districts	106 (2005)	30 (2006)	−72	Law (2026)
South Sudan	States	10 (2011)	32 (2017); 10 (2020)	0	Law (2026)
Panel B: Other regions					
United States	Counties	292 (1790)	2,713 (1900)	+829	U.S. Bureau of the Census (1994)
Indonesia	Districts and cities	302 (1999)	514 (2014)	+70	Bazzi & Gudgeon (2021)
Vietnam	Provinces	40 (1979)	64 (2004)	+60	Law (2026)
India	Districts	466 (1991)	640 (2011)	+37	Rajan & Malghan (2022)
Brazil	Municipalities	4,124 (1989)	5,507 (1997)	+34	Dahis & Szerman (2026)

Notes: The table reports the number of subnational units of the tier indicated at two dates, and the percentage change between them. Panel A lists sub-Saharan African countries for which counts could be verified against an academic source, a census, or the change history compiled by Law (2026); Grossman & Lewis (2014) document that almost half of sub-Saharan African countries increased their number of administrative units by at least 20 percent between 1990 and 2010. Tiers are not comparable across countries: some countries subdivided the top tier (Nigeria, DR Congo, South Sudan) and others the local government tier (Ghana, Kenya, Uganda, Zambia, Tanzania). Ghana had 65 districts before the 1988 reform that created 110, and further waves raised the count to 260 by 2019. Kenya's districts were replaced by 47 counties under the 2010 Constitution. Ethiopian woreda counts are sensitive to definitional changes across censuses. Rwanda consolidated its districts in 2006 on fiscal-sustainability grounds; South Sudan's 2015 and 2017 state creations were reversed in 2020. Uganda's counts correspond to the analysis sample.

D Variables Construction

Education. The first set of outcomes relates to education. Like many other public services, education in Uganda is provided at both the central and district levels. While districts oversee primary education, secondary education falls under the jurisdiction of the central government. Therefore, the analysis includes indicators for both education levels.

Although the NSDS collects school attendance data, it does not specify the grade level of children. To address this limitation, I construct two school attendance variables based on

schooling age brackets. *School attendance (prim. age)* measures attendance for 4,726 households with children aged 6–12, while *School attendance (second. age)* captures attendance for 3,663 households with children aged 13–18.

Following the same criteria, I construct variables for *Distance to school*, *School travel times*, and measures of perceived quality including *High quality of teachers*, *High quality of facilities*, and *School has an auditor*. The variable *Public school* is an indicator variable that takes the value of one if the household has any school-age children attending a public institution. *School travel time* measures the amount of time, in minutes, that families spend traveling to school.

Health. Healthcare provision in Uganda is divided between the central government and district governments, each responsible for different levels of service. Districts manage health centers, whereas the central government operates hospitals.

At the district level, I analyze several indicators, including *Government health center*, which measures access to public health centers for 2,361 households that vaccinated their children in the past year. *HH member sick/injured* captures the proportion of household members who reported illness or injury within the last 30 days. *Health center travel time* records the number of minutes taken to reach a health center, considering only households that visited a health facility in the last 30 days. At the community level, I assess access to Government health centers, focusing on their proximity *Government health center is within walking distance*, staffing (availability of *Full-time staff*), resources (*Ambulance presence*), management challenges (*Absenteeism concerns*), and oversight (*District-level supervision on a monthly basis*).

For central government provision, the analysis focuses on three indicators: *Government hospital*, *Government hospital within walking distance*, and *Insufficient PHC funds*. The latter captures whether central government funds for primary health care are insufficient in the community.

Water. The third set of outcomes examines water services, which are also provided by both district and central governments. District governments are responsible for supplying water for household consumption, while the central government provides water for productive activities.

Key district-level water indicators include *Improved water source*, which identifies households with access to safe drinking water in both dry and wet seasons, following World Health Organization guidelines (WHO, 2017). *Water distance* records the self-reported distance (in kilometers) from households to their water source. *Water committee* captures whether a functioning water committee exists within the household's sub-county. Additionally, community-level indicators include access to *Piped water*, *Boreholes*, and *Protected wells*.

For central government-provided water, the dataset includes indicators for *Improved water* and *Water distance* for sources used for production (e.g., agriculture, livestock, and fisheries) in both wet and dry seasons. Additional community-level indicators assess access to *Water*

for crop production, *Water for cattle production*, and *Water for fish production* within walking distance.

Electricity. Electricity provision falls under the jurisdiction of the central government. The analysis includes *HH has electricity*, which measures household access to electricity, as well as *HH connected to grid* and *HH interested in grid*, which indicate household access to and demand for grid electricity. Community-level indicators include *Community has electricity* and *Community connected to the grid*, measuring electrification at the community level.

Roads. The fifth set of outcomes examines road infrastructure, which is provided at both the central and district levels. The central government focuses on inter-district connectivity, while district governments are responsible for roads linking sub-counties and villages.

Although the NSDS does not explicitly classify roads by provider, the provider type can be inferred from road quality. Central government roads are typically paved (*Tarmac*), whereas district roads are predominantly unpaved (*Murram*). Additionally, district roads are categorized as *Feeder* or *Communal* roads. These household-level information is complemented by community-level information on community leaders' perceptions of *Road quality* and the extent of *Road maintenance* by different levels of government.

Agricultural Inputs. The final set of outcomes assesses the availability and use of agricultural inputs, including *Hybrid seeds*, *Planting materials*, *Pesticides*, and *Fertilizer*. Although the main national input distribution programmes, the National Agricultural Advisory Services (NAADS) and Operation Wealth Creation (OWC), are coordinated by the Ministry of Agriculture, Animal Industry and Fisheries, agricultural extension is a district mandate under the Local Governments Act, and inputs are distributed through district production departments and sub-county extension officers. Household-level variables measure whether these inputs were used in the past year and their availability in the household's sub-county. Additional community-level indicators capture access to *Agricultural markets*, *Livestock markets*, *General markets*, *Agricultural extension services*, *Maize/rice and coffee mills*, and *Milk cooling systems*.

Institutional Knowledge and Traditional Institutions. To measure households' awareness of district-level state institutions, I construct indicator variables from the NSDS capturing whether the household knows the location of the *Magistrate court*, the *Police*, the *District land office*, the *Office of the Auditor General*, and the *Judiciary*. *Family civil servant* indicates whether a household member is employed as a civil servant. I complement these with measures from the Afrobarometer to capture the balance between formal and traditional institutions. *Resolved by government* and *Resolved by traditional* indicate whether households resolve disputes through government or traditional institutions, respectively. *Contacted MP* and *Contacted traditional leader* indicate whether the respondent contacted a member of parliament or a traditional leader at least once during the past year. Column (6) of each panel in Table 7 reports a summary index constructed as described under *Summary Indices* below. The *Institutional knowledge* index

aggregates the five NSDS knowledge indicators. The *State reliance* index aggregates the four Afrobarometer indicators, with *Resolved by traditional* and *Contacted traditional leader* entering with reversed sign, so that higher values indicate greater reliance on formal state channels; *Family civil servant* is excluded because it comes from the NSDS.

Citizen Engagement. Participatory engagement with the district government is measured with three NSDS indicators. *Involved in LCV meetings* captures whether the household is involved in LCV decision-making and the management of district resources. *Met RDC* indicates whether the household has ever attended an activity organized by the Resident District Commissioner, and *RDC Baraza* whether the RDC has held a baraza (a public accountability meeting) in the community. Contentious and collective forms of engagement come from the Afrobarometer: *Collective civic action* indicates whether the respondent joined others to request government action, while *Protest* and *Contacted media* indicate participation in a demonstration or protest march and contacting the media about politics or community affairs, each measured as any occurrence during the past year. Column (4) of each panel in Table 9 reports a summary index of the three indicators in that panel (*Participatory engagement* for the NSDS and *Contentious engagement* for the Afrobarometer), constructed as described under *Summary Indices* below; higher values indicate more engagement of the corresponding type.

Satisfaction with Local Administrators. Three NSDS indicators capture households' evaluations of district leaders: *LCV performance*, which indicates a positive evaluation of the elected LCV chairperson's performance; *RDC satisfaction*, which indicates that the household is satisfied or very satisfied with the RDC's work; and *Civil servants performance*, which captures a positive evaluation of district civil servants.

Corruption. Corruption outcomes distinguish awareness from direct experience. From the NSDS, *Nepotism*, *Embezzlement*, and *Absenteeism* indicate whether the household has heard of each form of corruption, while *Victim nepotism*, *Victim embezzlement*, and *Victim absenteeism* indicate whether the household has been a victim of the corresponding practice. Bribery experiences come from the Afrobarometer: *Bribe health*, *Bribe police*, *Bribe avoid police*, and *Bribe ID* indicate whether the respondent paid a bribe at least once during the past year to obtain medical care, to obtain police assistance, to avoid problems with the police, or to obtain an identity document, in each case conditional on having sought the corresponding service. Table 11 also reports three summary indices constructed as described under *Summary Indices* below: a *Corruption awareness* index over the three NSDS awareness indicators, a *Corruption victimisation* index over the three NSDS victimisation indicators, and a *Bribery* index over the four Afrobarometer bribery indicators. Because each bribery indicator is defined only for respondents who sought the corresponding service, the bribery index is computed for respondents with at least one non-missing component. Higher values of all three indices indicate more corruption.

Democratic Perceptions and Trust. All variables in this group come from the Afrobarome-

ter. *Is democracy* indicates that the respondent considers Uganda a democracy (whether full or with problems), and *Satisf democracy* that the respondent is fairly or very satisfied with the way democracy works in Uganda. *Free vote choice* and *Free join orgs* indicate that the respondent feels somewhat or completely free to choose whom to vote for and to join any political organization. The rule of law perceptions (*President ignores law*, *Unequal law*, *Judges influenced*, and *Officials impunity*) take the value of one when the respondent believes the corresponding situation occurs often or always. The trust variables (*Trust localgov*, *Trust president*, *Trust parliament*, *Trust ruling party*, *Trust opposition*, and *Trust courts*) indicate that the respondent trusts the corresponding institution somewhat or a lot. The social capital variables in Appendix Table A26 follow the same construction for trust in relatives, neighbours, other citizens, people of other religions, and people of other ethnicities. The last column of each panel in Table 13 reports a summary index of the outcomes in that panel, constructed as described under *Summary Indices* below: a *Democratic perceptions* index (Panel A), a *Rule of law perceptions* index (Panel B), and a *Trust in government* index (Panel C). The indices follow the orientation of their components, so that higher values indicate more favourable democratic perceptions, more negative rule-of-law perceptions, and more trust in government, respectively.

Political Alignment. A district is classified as aligned when the elected LCV chairperson and the centrally appointed RDC share political affiliation. Since RDCs are appointed by the president, alignment is measured with an indicator taking the value of one when the district's LCV chairperson elected in the 2016 local elections belongs to the ruling party (NRM). Districts created after 2016 are assigned the affiliation of the LCV chairperson recorded at their creation.

Summary Indices. Table A2 lists the components entering each of the ten service indices used in the main analysis and the ten mechanism and perception indices in Tables 7, 9, 11, and 13. Components are standardized to mean zero and standard deviation one over the full survey sample and averaged with equal weights; variables for which higher values indicate worse access (or reliance on traditional rather than state institutions) enter with reversed sign, so that higher values of every service index correspond to better service access, and higher values of the mechanism and perception indices to more institutional knowledge, greater reliance on the state, more participatory or contentious engagement, more corruption awareness, victimisation, and bribery, more favourable democratic perceptions, more negative rule-of-law perceptions, or more trust in government.

E Additional Identification and Robustness Checks

This appendix collects the additional identification and robustness exercises referred to in the main text. Table A3 reports the logit regressions on where the district centroid falls relative to pre-creation development, discussed in Section 5.2.1, with Figure A1 showing the Gulu–Omoro example of Figure 3 on 2014 living standards, and Figure A2 shows the reduced-form

TABLE A2: Components of the Summary Indices

Index	Components
Education (district)	School attendance (prim. age); distance to school (rev.); school travel time (rev.); high quality of teachers; high quality of facilities; school has an auditor — all for primary-age children
Education (central gov.)	Same six components, for secondary-age children
Health (district)	Gov. health facility for immunization; HH member sick/injured in last 30 days (rev.); health center travel time (rev.); full-time staff at health center; absenteeism in health center (rev.); motorized ambulance in health center; monthly district supervision of health center
Health (central gov.)	Government hospital available; gov. hospital within walking distance; PHC funds not enough (rev.)
Water (district)	Improved water source (dry and wet seasons); water distance (dry and wet seasons, rev.); water committee; piped water available; borehole available; protected well available
Water (central gov.)	Improved water source for productive use (dry and wet seasons); water distance for productive use (dry and wet seasons, rev.); water for crop, cattle, and fish production within walking distance
Electricity (central gov.)	HH has electricity; HH connected to grid; HH interested in grid; community has electricity
Roads (district)	Sub-county has murram road; sub-county has feeder road; sub-county has communal road (rev.); high quality murram, feeder, and community roads; murram, feeder, and community roads mainly repaired by the district
Roads (central gov.)	Sub-county has tarmac road; high quality tarmac road; tarmac roads mainly repaired by the central government
Agricultural inputs (district)	Use and sub-county availability of hybrid seeds, planting materials, pesticides, and fertilizer
Institutional knowledge (NSDS)	Knows the magistrate court; the police; the district land office; the Office of the Auditor General; the judiciary
State reliance (Afrobarometer)	Dispute resolved by government; contacted an MP; dispute resolved by traditional institutions (rev.); contacted a traditional leader (rev.) — contacts in the past year
Corruption awareness (NSDS)	Has heard of nepotism; of embezzlement; of absenteeism
Corruption victimisation (NSDS)	Household was a victim of nepotism; of embezzlement; of absenteeism
Bribery (Afrobarometer)	Paid a bribe in the past year for medical care; for police assistance; to avoid problems with the police; for an identity document — each conditional on having sought the service
Democratic perceptions (Afrobarometer)	Uganda is a democracy; satisfied with democracy; free to choose whom to vote for; free to join any political organization
Rule of law perceptions (Afrobarometer)	President often ignores the law; people are treated unequally under the law; judges are influenced; officials go unpunished — higher values indicate more negative perceptions
Trust in government (Afrobarometer)	Trust in local government; the president; parliament; the ruling party; the opposition; the courts
Participatory engagement (NSDS)	Involved in LCV decision-making; ever attended an RDC activity; RDC held a baraza in the community
Contentious engagement (Afrobarometer)	Joined others to request government action; participated in a protest or demonstration; contacted the media — all in the past year

Notes: Components of each Anderson (2008) summary index. Each component is standardized to mean zero and standard deviation one before averaging; components marked (rev.) enter with reversed sign so that higher values of every service index indicate better service access; higher values of the mechanism indices indicate more institutional knowledge, greater reliance on formal state channels, more corruption awareness, victimisation, or bribery, more favourable democratic perceptions, more negative rule-of-law perceptions, more trust in government, or more engagement of the corresponding type. Household-level components come from the NSDS household questionnaire and community-level components (e.g., walking-distance and availability measures) from the community questionnaire.

binned scatters for the main outcome indices referred to in Section 5.1. The remaining subsections show that the instrument does not proxy for proximity to large cities, that proximity does not predict placebo outcomes unrelated to administrative access, and that the main estimates survive the exclusion of households around the headquarters town, spatially correlated (Conley) standard errors, a GIS-based measure of proximity, the comparison with OLS, an alternative construction of the instrument, and a sample restricted to newly created districts. Weak-instrument-robust confidence sets for the alignment subsamples are reported in Section 5. Finally, Section E.5 shows that the service results replicate on the 2024 Population and Housing Census.

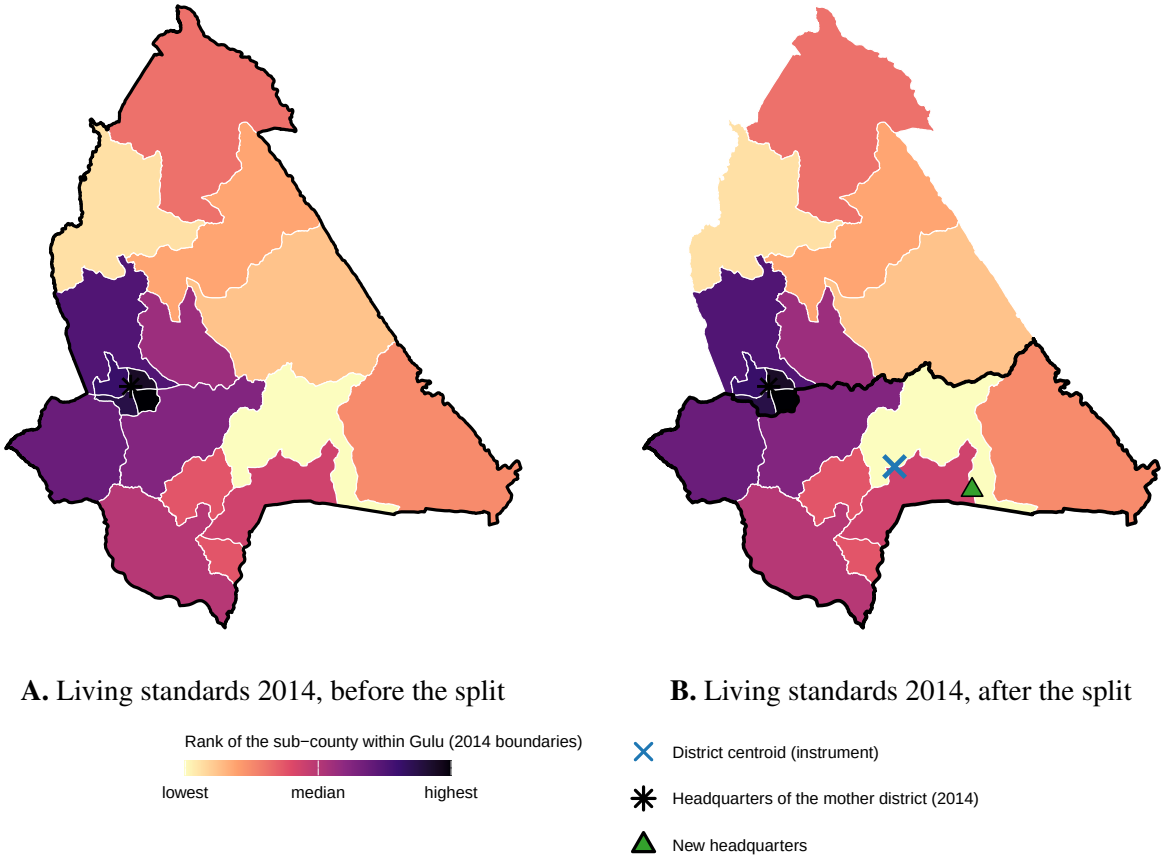
TABLE A3: Does the District Centroid Fall in a More Developed Sub-County? One Logit per Column

Pre-creation measure:	Dependent variable: sub-county contains the district centroid					
	Living standards 2014		Log pop. density 2010		Log night lights 2013	
	(1)	(2)	(3)	(4)	(5)	(6)
Pre-creation measure (SD units)	-0.400***	-0.786*	-0.510***	-1.106**	0.154	0.205
	(0.114)	(0.419)	(0.123)	(0.439)	(0.191)	(0.291)
Mean of Dependent Variable	0.136	0.136	0.136	0.136	0.136	0.136
Observations	176	176	176	176	176	176
District FE	No	Yes	No	Yes	No	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Each column reports a separate logit regression. The sample is the 176 sub-counties (2014 boundaries, with census data) of the 24 districts created between 2014 and 2020. The dependent variable is an indicator for the sub-county that contains the new district's geographic centroid, the point on which the instrument is anchored (for Butebo, whose centroid falls outside the district, the sub-county closest to it). The single regressor is one pre-creation measure of development, standardized to mean zero and unit variance across sub-counties: the 2014 census living-standards index (housing materials, services, assets and adult education, averaged by sub-county; columns 1–2), log population density in 2010 from the Gridded Population of the World (columns 3–4), and log VIIRS night lights in 2013 (columns 5–6). Even-numbered columns include district fixed effects. Robust standard errors clustered at the district level in parentheses.

E.1 Distance to Large Cities

One potential concern is that the instrument captures households' proximity to large urban centers rather than to administrative headquarters. If distance to the district centroid simply reflects distance to large cities, it would not provide independent variation in proximity to administrative centers. To address this, I include three alternative measures of urban proximity as mediators in the first stage: the household's distance to the closest large city (defined as the top 20 most populous cities in Uganda), the household's distance to the closest large city within its region, and the household's distance to Kampala. Table A4 shows that the instrument remains strong and stable across all specifications. The coefficient on the interaction term ranges from 0.373 to 0.437 and remains highly significant throughout, while the F-statistic stays above 19 in all columns.



A. Living standards 2014, before the split

B. Living standards 2014, after the split

FIGURE A1: Where the district centroid lands: Gulu and the creation of Omoro (2016), 2014 census living standards. Gulu's 17 sub-counties on their 2014 boundaries, shaded by their rank within Gulu on the 2014 living-standards index (the measure of Appendix Table A3), the same district as Figure 3. Panel A: before the split, with the headquarters of the mother district in Gulu town (star). Panel B: the new district of Omoro outlined, with its geographic centroid (cross), the point on which the instrument is anchored, and its new headquarters (triangle).

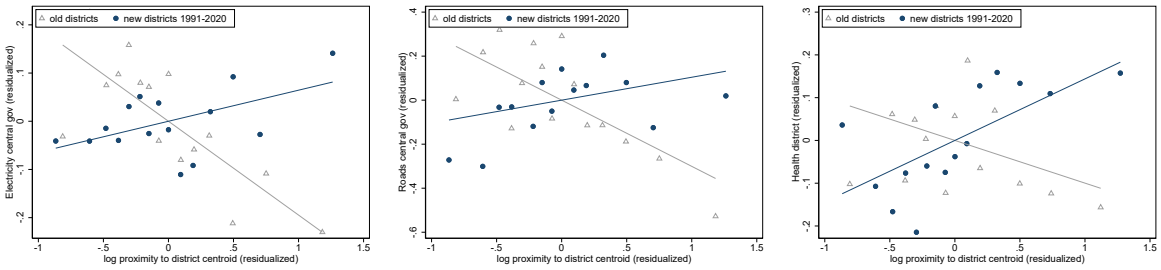


FIGURE A2: Reduced forms: binned scatters of the residualized electricity (left), roads — central government (centre), and health — district (right) indices against residualized proximity to the district centroid, separately for households in new districts (1991–2020) and old districts. Construction as in Figure 2.

E.2 Placebo Tests

I perform placebo regressions to test whether the instrument captures unobserved geographical features rather than administrative proximity. If instrumented proximity to the district head-

TABLE A4: FIRST STAGE: DISTANCE HOUSEHOLD - LARGE TOWN

	Log SR Proximity Household - District Headquarters Improved water source				
	(1)	(2)	(3)	(4)	(5)
Log Proximity HH-Centroid × New District 1991-2020	0.429***	0.383***	0.437***	0.425***	0.373***
	(0.080)	(0.081)	(0.082)	(0.082)	(0.085)
Log Proximity HH - District's centroid	-0.051	-0.002	-0.057	-0.047	0.007
	(0.063)	(0.065)	(0.064)	(0.065)	(0.069)
New district between 1991-2020	1.696***	1.565***	1.712***	1.677***	1.516***
	(0.220)	(0.221)	(0.222)	(0.227)	(0.236)
Log Distance Household - Closest Large City		-0.090***			-0.099***
		(0.027)			(0.029)
Log Distance Household - Closest Large City within Region			-0.063**		-0.057*
			(0.030)		(0.032)
Log Distance Household - Kampala				0.065	0.174**
				(0.073)	(0.073)
Constant	-0.857***	-0.622**	-0.760***	-1.086***	-1.126***
	(0.270)	(0.266)	(0.264)	(0.321)	(0.307)
Mean of Dependent Variable	-2.686	-2.686	-2.686	-2.686	-2.686
Observations	8,417	8,417	8,417	8,417	8,417
Number of Sub-counties	967	967	967	967	967
IV F-stat Kleibergen-Paap rk Wald	28.745	22.546	28.509	27.138	19.266
Controls	Yes	Yes	Yes	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Controls include: district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Lake Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. Standard errors are clustered at the Parish level.

quarters predicted outcomes that are plausibly unrelated to it, the instrument would likely be picking up unobservable characteristics, indicating omitted variable bias. Suitable placebo outcomes are therefore ones that would respond to such characteristics but are not coordinated by district administrations. The three outcomes available in the NSDS have this property and cover different channels: the use of improved cookstoves depends on household wealth and local fuel markets, the use of air transportation, and involvement in mining on the location of mineral deposits, which is fixed by geology and unrelated to district borders. If the instrument captured local development, remoteness, or natural geography rather than administrative proximity, effects should appear on at least one of these outcomes. Table A5 reports the 2SLS estimates. All three are close to zero and statistically insignificant. Table A6 reports the corresponding test for lower-tier local government institutions (LC1, LC2, and LC3), discussed in Section 5.3.

E.3 Distance to the Headquarters Town

Another potential concern is that the estimated effects reflect the growth of the headquarters town itself rather than improved administrative access for the surrounding population. If the results were driven entirely by households located in the immediate vicinity of the new headquarters, they would speak to urban agglomeration around the administrative center rather than to the reach of the state. To address this, Table A7 re-estimates the main specification excluding households progressively closer to their district headquarters: those in the same parish as the headquarters point, and those within 2 and 5 kilometres. The electricity results are essentially unchanged across all three donuts, and the roads coefficients become larger the more of the headquarters' surroundings are excluded, indicating that these effects are not an artifact of headquarters-town growth. The health index attenuates by roughly a fifth when households within 5 kilometres are excluded but remains significant at the 10 percent level, suggesting

TABLE A5: Placebo Outcomes and Proximity to District Headquarters (2SLS)

	Cookstove (1)	Air transport (2)	Minerals (3)
Log Proximity to District HQ	-0.016	-0.002	-0.022
	(0.033)	(0.007)	(0.018)
Observations	8415	8417	8415
Number of clusters	967	967	967
Mean of DV	0.929	0.006	0.019
IV F-stat (KP rk Wald)	28.749	28.745	28.683

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of placebo variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A6: Contact with Local Government Institutions and Proximity to District Headquarters (2SLS)

	Knows LCI (1)	Distance LCI (2)	Knows LCII (3)	Distance LCII (4)	Knows LCIII (5)	Distance LCIII (6)
Panel A: State presence						
Log Proximity to District HQ	0.022	0.112	0.107	-0.851**	0.065	-0.065
	(0.036)	(0.160)	(0.104)	(0.404)	(0.090)	(0.090)
p-val	[0.528]	[0.482]	[0.304]	[0.035]	[0.471]	[0.471]
q-val	{0.785}	{0.785}	{0.785}	{0.268}	{0.785}	{0.785}
Observations	8414	8044	8413	4286	8413	8413
Number of clusters	967	966	967	901	967	967
Mean of DV	0.956	0.651	0.509	1.945	0.485	1.515
IV F-stat (KP rk Wald)	28.597	25.897	28.592	27.234	28.592	28.592

	HH decision involvement (1)	Ever in LC1 (2)	LC1 meeting freq (3)	Attends LC1 (4)	LC1 performance (5)	LC1 quality change (6)
Panel B: Citizen participation						
Log Proximity to District HQ	0.136*	0.009	-0.129*	-0.013	0.064	0.011
	(0.070)	(0.045)	(0.075)	(0.076)	(0.069)	(0.025)
p-val	[0.050]	[0.843]	[0.085]	[0.865]	[0.357]	[0.658]
q-val	{0.344}	{1.000}	{0.344}	{1.000}	{0.909}	{1.000}
Observations	8414	8417	8417	8417	8217	8143
Number of clusters	967	967	967	967	967	967
Mean of DV	0.471	0.097	0.190	0.306	0.835	0.962
IV F-stat (KP rk Wald)	28.655	28.745	28.745	28.745	29.514	29.509

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' contact with local government institutions on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values, computed within each panel, are in curly brackets.

that part (though not most) of the health effect is concentrated near the headquarters town, consistent with the location of district health facilities.

TABLE A7: Donut Specifications: Excluding Households Near District Headquarters

	Baseline (1)	Excl. HQ parish (2)	Excl. <2 km (3)	Excl. <5 km (4)
Panel A: Electricity (central government)				
Log Proximity to District HQ	0.428***	0.413***	0.453***	0.464**
	(0.137)	(0.136)	(0.164)	(0.188)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
Panel B: Roads (central government)				
Log Proximity to District HQ	0.706***	0.689***	0.984***	1.115***
	(0.261)	(0.262)	(0.305)	(0.357)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
Panel C: Health (district)				
Log Proximity to District HQ	0.387***	0.376***	0.377***	0.301*
	(0.121)	(0.120)	(0.136)	(0.156)
IV F-stat (KP rk Wald)	33.35	34.48	27.14	22.79
Observations	7,595	7,378	6,838	5,700

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column re-estimates the main 2SLS specification excluding households near their district headquarters: Column 2 drops enumeration areas located in the same parish as the district headquarters point (35 enumeration areas); Columns 3 and 4 drop enumeration areas within 2 and 5 kilometres of the headquarters (125 and 304 enumeration areas, respectively). All regressions use the main specification's controls, weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

E.4 Spatial Correlation: Conley Standard Errors

The main tables cluster standard errors by enumeration area, allowing arbitrary correlation among households within the same enumeration area but assuming independence across them. However, nearby enumeration areas may share unobserved shocks, particularly since both the instrument and the outcomes vary spatially. Table A8 therefore reports Conley spatial HAC standard errors (Conley, 1999) for the ten service regressions in Table 3, allowing for spatial correlation within 50 and 100 kilometres. The results show that the coefficients for district health, centrally provided water, roads, and electricity remain statistically significant at the 5 percent level under both cutoffs. The only change is for district water, which loses its statistical significance. The first stage also remains strong, with F-statistics between 22 and 29. Finally, Table A8 also reports standard errors clustered by 1991 mother district, the pre-split administrative unit. The conclusions are similar: district health, centrally provided water, roads, and electricity remain significant at the 5 percent level and the first stage remains strong.

E.5 External Validity: The 2024 Population and Housing Census

The NSDS covers around 10,000 households. To check whether the service results extend beyond this sample, I re-estimate the main specification using the 10 percent users' file of the 2024 National Population and Housing Census, which contains 922,930 households after excluding Kampala and Wakiso. The census does not provide household coordinates or self-

TABLE A8: Spatial Correlation: Conley Standard Errors for the Main Service Results

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
Panel A: District services					
Log Proximity to District HQ	0.282	0.387***	0.279*	0.089	-0.085
SE, clustered by enumeration area	(0.214)	(0.121)***	(0.157)*	(0.110)	(0.145)
SE, Conley 50 km	(0.239)	(0.110)***	(0.193)	(0.124)	(0.172)
SE, Conley 100 km	(0.269)	(0.099)***	(0.198)	(0.125)	(0.183)
SE, clustered by 1991 mother district (37)	(0.244)	(0.095)***	(0.190)	(0.131)	(0.172)
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: Central government services					
Log Proximity to District HQ	0.151	0.233	0.372**	0.706***	0.428***
SE, clustered by enumeration area	(0.192)	(0.201)	(0.146)**	(0.261)***	(0.137)***
SE, Conley 50 km	(0.191)	(0.220)	(0.169)**	(0.298)**	(0.165)***
SE, Conley 100 km	(0.157)	(0.197)	(0.165)**	(0.294)**	(0.152)***
SE, clustered by 1991 mother district (37)	(0.162)	(0.233)	(0.169)**	(0.281)**	(0.160)**
Observations	5,553	8,417	8,417	8,417	8,417
First-stage F, clustered by enumeration area			28.75		
First-stage F, Conley 50 km			21.77		
First-stage F, Conley 100 km			29.02		
First-stage F, clustered by 1991 mother district			31.70		

Notes: *p<0.1, **p<0.05, ***p<0.01; the stars on each standard-error row are computed from that row's standard error. Each column reproduces the 2SLS estimate of Table 3 (same outcome, controls, weights and sample; Kampala and Wakiso excluded) and reports four standard errors for the same coefficient: clustered by enumeration area as in the main tables, Conley spatial HAC standard errors that allow arbitrary correlation between enumeration areas within 50 and 100 kilometres, and standard errors clustered by 1991 mother district. First-stage F: Wald F-statistic on the excluded instrument (equal to the Kleibergen–Paap rk Wald F with a single instrument) under each variance estimator, for the 8,417-observation sample; for the smaller samples of columns (1) and (2) of Panel A and column (1) of Panel B it is 22.2, 34.1 and 20.6 at 100 kilometres.

reported distances. I therefore locate households at the centroid of their 2024 parish when the parish can be matched to a geographic polygon (74 percent of households), and at the sub-county centroid otherwise. Using the administrative unit's centroid, I calculate households' distance to the district headquarters and the district centroid and estimate the same specification as in as in Section 5 with standard errors clustered at the 2024 sub-county level (2,172 clusters).

Since the census measures access rather than service quality, I construct outcome indices using household coverage indicators, grouped as in Table 3. District services include primary schooling (attendance, ever attended, completion, literacy, and grade-for-age of primary-school-age members), health (vaccination records of children under five and deaths at a health facility), and water (point source, distance to the source, and protection of the drinking-water source). Central government services include secondary schooling, government-distributed mosquito nets, piped water, and electricity. As in the NSDS, each index is the mean of its standardised components. The census does not contain information on roads or agricultural inputs.

The results in Table A9 show a strong first stage with a Kleibergen–Paap F-statistic of 131.

All three district-provided services indices increase with proximity to the headquarters: primary schooling by 0.13, health by 0.09, and water by 0.14 standard deviations per log point of proximity. All three coefficients are significant at the 5 percent level or 1 percent level, with sharpened q-values below 0.01. Among central government services, electricity increases by 0.18 standard deviations and secondary schooling by 0.07, while piped water, shows no statistically significant effect.

The estimated effects are between one-third and one-half of those in Table 3, consistent with the census measuring access rather than the quality and satisfaction dimensions that account for a large part of the NSDS effects. In terms of individual outcomes, a 10 percent increase in proximity raises primary school attendance by 0.5 percentage points and the share of households using a protected water source by 0.9 percentage points. The results are also similar when restricting the sample to households located at the parish level.

E.6 Alternative Distance Variable: GIS Distance

While the household's self-reported distance to the district headquarters arguably captures effective proximity (it reflects not only straight-line distance but also the available transportation network) it may suffer from over- or under-reporting. To address this, Table A10 replaces the self-reported measure with the GIS straight-line proximity from the household's enumeration area to the headquarters point, keeping the instrument and all controls unchanged. The two measures are highly but imperfectly correlated (0.80; Appendix Figure A3), consistent with noisy but informative self-reports. The 2SLS estimates with the GIS measure are statistically significant for all three headline outcomes and, if anything, larger in magnitude, while the first-stage F-statistics remain above 20 for both measures. OLS estimates are three to four times smaller than the 2SLS estimates with the self-reported measure and five to six times smaller with the GIS measure, a gap discussed in the next subsection.

E.7 OLS Estimates

Appendix Table A11 compares the 2SLS estimates with their OLS counterparts for the ten service indices. The OLS coefficients are positive and statistically significant for most outcomes, confirming the descriptive association between proximity to headquarters and service access. However, for the outcomes where the 2SLS estimates are significant, the OLS coefficients are three to five times smaller. This pattern is informative as it goes against the most direct selection concern. If headquarters had been systematically placed in already better served locations, the naive OLS estimates would likely be larger than the 2SLS estimates. Instead, OLS is substantially smaller, being consistent with two explanations. First, both the self reported and GIS based measures are noisy proxies for effective proximity, so OLS is likely affected by attenuation bias, which the instrument helps correct. Second, the 2SLS estimates identify local average treatment effects for compliers, that is, households whose proximity to headquarters was shifted by redistricting. These households were previously more remote, and the marginal

TABLE A9: Effects on Services and Proximity to District Headquarters (2SLS) – 2024 Population and Housing Census

	School (1)	Health (2)	Water (3)
Panel A: 2024 Census District services			
Log Proximity to District HQ	0.132***	0.086**	0.142***
	(0.050)	(0.035)	(0.043)
p-val	[0.009]	[0.013]	[0.001]
q-val	{0.009}	{0.009}	{0.004}
Observations	575,253	428,564	922,930
Number of clusters	2,172	2,172	2,172
Mean of DV	-0.032	-0.001	0.000
IV F-stat (KP rk Wald)	126.6	122.4	131.0

	School (1)	Health (2)	Water (3)	Electricity (4)
Panel B: 2024 Census Central govt services				
Log Proximity to District HQ	0.071*	0.098**	0.028	0.180**
	(0.040)	(0.045)	(0.087)	(0.082)
p-val	[0.076]	[0.030]	[0.750]	[0.029]
q-val	{0.064}	{0.064}	{0.231}	{0.064}
Observations	542,931	915,788	922,930	922,930
Number of clusters	2,172	2,172	2,172	2,172
Mean of DV	-0.047	0.000	0.000	0.000
IV F-stat (KP rk Wald)	129.2	131.6	131.0	131.0

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from 2SLS regressions of household outcome indices from the 2024 National Population and Housing Census (10% users' sample, unweighted) on the household's log proximity to the district's headquarters. Each index is the mean of standardised household coverage indicators (primary and secondary schooling, vaccination records and health-facility deaths, government mosquito nets, drinking-water source and distance, piped water, lighting); components are listed in Appendix E.5. Controls and instrument as in Table 3; Kampala and Wakiso excluded. Households are located at their parish centroid where the 2024 parish could be matched to a parish polygon (74% of households) and at their sub-county centroid otherwise. Standard errors clustered at the 2024 sub-county level are in parenthesis. Unadjusted p-values are in brackets; Anderson's sharpened FDR q-values within each panel are in curly brackets.

returns to administrative access are plausibly larger for them. The recentering exercise points in the same direction: correcting for predictable exposure to the redistricting process moves the estimates away from zero rather than toward it (Appendix G). This suggests that naive comparisons understate, rather than overstate, the effect of proximity.

E.8 Alternative Instrument: Parish-Level Distance

As an additional robustness check, I replace the enumeration area geocoordinates with parish centroid distances to construct an alternative version of the instrument. Table A12 shows that the results are similar to the ones obtained in Table 3. If anything, several coefficients are larger in magnitude and more precisely estimated under this alternative, while the first stage remains strong with F-statistics exceeding 23 across all specifications. These results show that

TABLE A10: Alternative Endogenous Variable: Self-Reported vs. GIS Proximity to District Headquarters

	2SLS Self-reported (1)	2SLS GIS (2)	OLS Self-reported (3)	OLS GIS (4)
Panel A: Electricity (central government)				
Log Proximity to District HQ	0.428***	0.557**	0.102***	0.121***
	(0.137)	(0.240)	(0.020)	(0.028)
IV F-stat (KP rk Wald)	28.75	21.19	–	–
Observations	8,417	9,313	8,417	9,313
Panel B: Roads (central government)				
Log Proximity to District HQ	0.706***	0.924**	0.222***	0.165***
	(0.261)	(0.401)	(0.035)	(0.051)
IV F-stat (KP rk Wald)	28.75	21.18	–	–
Observations	8,417	9,303	8,417	9,303
Panel C: Health (district)				
Log Proximity to District HQ	0.387***	0.593***	0.111***	0.128***
	(0.121)	(0.205)	(0.020)	(0.033)
IV F-stat (KP rk Wald)	33.35	20.10	–	–
Observations	7,595	8,320	7,595	8,320

Notes: *p<0.1, **p<0.05, ***p<0.01. Columns 1 and 3 use the household's self-reported proximity to the district headquarters as the variable of interest; Columns 2 and 4 replace it with the GIS straight-line proximity from the household's enumeration area to the headquarters point. The instrument, controls, weights, and clustering follow the main specification; Kampala and Wakiso are excluded as in the main tables. The GIS measure removes self-reporting error but provides less variation, as it does not reflect transportation networks; first-stage F-statistics nevertheless remain above 20 for both measures. Standard errors clustered at the enumeration area level are in parenthesis.

the findings are robust to the choice of geocoordinates used to compute household distances.

E.9 Alternative Sample: Only New Districts

The main specification identifies the effect of proximity from the differential relationship between centroid proximity and headquarters proximity in new versus old districts, interacting centroid proximity with the new-district indicator and using old districts as a comparison group. Because the government placement rule applies only to newly created districts, an alternative is to restrict the sample to these districts, where the location of the centroid directly determines the location of the headquarters. In this subsample the interaction is no longer necessary: the household's proximity to the district centroid can be used on its own as an instrument for its proximity to the district capital. Because the placement rule ties the centroid to the headquarters mechanically within new districts, the instrument is considerably stronger here, with first-stage F-statistics ranging from 29 to 49, well above conventional thresholds.

Table A13 reports the results. All ten point estimates are positive, and the two most robust findings of the main specification survive: proximity significantly improves access to district-

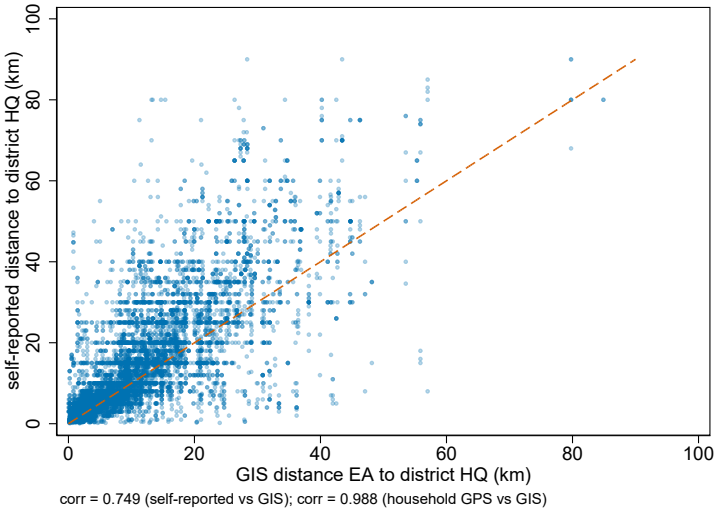


FIGURE A3: Calibration of the self-reported distance measure: household self-reported distance to the district headquarters against the GIS straight-line distance from the household's enumeration area to the headquarters point (in kilometres). The dashed line is the 45-degree line. The correlation between the two measures is 0.75; the correlation between household GPS-based and enumeration area GIS distances is 0.99, indicating that the gap reflects self-reporting rather than geocoding.

provided health services (0.317, significant at the 1 percent level) and to centrally provided electricity (0.149, significant at 5 percent). District roads (0.143) and secondary education (0.255) become marginally significant, while water and roads under central government retain the expected positive sign but lose statistical significance. The loss of precision reflects the fact that restricting the sample to new districts drops almost half of the observations and removes the identifying variation coming from the comparison with old districts. The magnitudes are also not directly comparable to the main estimates, since the within new district design identifies effects from a different mix of compliers. Overall, this exercise indicates that the main results are not an artifact of the new-versus-old district contrast, as the central findings are recovered using only the variation generated within newly created districts, where the placement rule directly applies.

E.10 Weak-Instrument-Robust Inference in the Alignment Subsamples

Tables A14 and A15 report, for every outcome in the alignment tables (Tables 4 to 14), the 2SLS coefficient, the Anderson–Rubin 95 percent confidence set and the first-stage F-statistic in the aligned and non-aligned subsamples separately. The body tables integrate these sets only for the Afrobarometer outcomes, where the subsample first stages are weakest; the full set is reported here for completeness.

TABLE A11: OLS and 2SLS Estimates: Effects on Services

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
Panel A: NSDS 2021 District services					
OLS	0.060**	0.111***	0.054***	0.003	-0.005
	(0.028)	(0.020)	(0.021)	(0.016)	(0.019)
2SLS	0.282	0.387***	0.279*	0.089	-0.085
	(0.214)	(0.121)	(0.157)	(0.110)	(0.145)
IV F-stat (KP rk Wald)	22.24	33.35	28.75	28.75	28.75
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: NSDS 2021 Central govt services					
OLS	0.090***	0.215***	0.073***	0.222***	0.102***
	(0.026)	(0.033)	(0.019)	(0.035)	(0.020)
2SLS	0.151	0.233	0.372**	0.706***	0.428***
	(0.192)	(0.201)	(0.146)	(0.261)	(0.137)
IV F-stat (KP rk Wald)	22.62	28.75	28.75	28.75	28.75
Observations	5,553	8,417	8,417	8,417	8,417

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column reports the coefficient on the household's (log) self-reported proximity to the district headquarters from two estimators: OLS, and the paper's main 2SLS specification instrumenting proximity with the interaction between the enumeration area's log proximity to the district centroid and the new-district indicator. The 2SLS rows reproduce the main results tables. Both estimators use the same controls, sampling weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

TABLE A12: Effects on Services — PA Distance Instrument (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
Panel A: District services					
Log Proximity to District HQ	0.222	0.204*	0.329**	0.019	-0.042
	(0.156)	(0.115)	(0.137)	(0.102)	(0.122)
Observations	7170	7627	8529	8529	8529
Number of clusters	974	966	984	984	984
Mean of DV	-0.002	-0.020	-0.005	0.006	0.027
IV F-stat (KP rk Wald)	25.481	26.367	24.385	24.385	24.385
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: Central govt services					
Log Proximity to District HQ	0.004	0.264	0.307***	0.886***	0.353***
	(0.188)	(0.216)	(0.119)	(0.229)	(0.113)
Observations	5582	8363	8529	8529	8529
Number of clusters	967	963	984	984	984
Mean of DV	-0.001	0.016	-0.015	-0.061	-0.038
IV F-stat (KP rk Wald)	16.993	23.989	24.385	24.385	24.385

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's parish centroid to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A13: Effects on Services — New Districts Only (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
Panel A: District services					
Log Proximity to District HQ	0.155	0.317***	0.049	0.143*	0.094
	(0.113)	(0.093)	(0.080)	(0.081)	(0.079)
Observations	3692	3973	4242	4242	4242
Number of clusters	472	470	474	474	474
Mean of DV	-0.045	-0.073	-0.067	0.063	0.126
IV F-stat (KP rk Wald)	49.487	47.274	45.888	45.888	45.888
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: Central govt services					
Log Proximity to District HQ	0.255*	0.156	0.080	0.246	0.149**
	(0.132)	(0.127)	(0.073)	(0.165)	(0.066)
Observations	2857	4242	4242	4242	4242
Number of clusters	470	474	474	474	474
Mean of DV	-0.015	-0.234	-0.080	-0.288	-0.171
IV F-stat (KP rk Wald)	28.925	45.888	45.888	45.888	45.888

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid. Standard errors clustered at the Parish level are in parenthesis.

TABLE A14: Weak-Instrument-Robust Inference in the Alignment Subsamples (Part A: Services, Engagement, Institutions)

	Aligned districts			Non-aligned districts		
	2SLS (SE) (1)	AR 95% set (2)	KP F (3)	2SLS (SE) (4)	AR 95% set (5)	KP F (6)
Panel A: Services (NSDS 2021)						
School (district)	0.413 (0.307)	$[-0.29, 1.30]$	9.9	-0.018 (0.275)	$[-0.68, 0.55]$	16.7
Health (district)	0.324* (0.167)	$[0.00, 0.75]$	17.0	0.383** (0.157)	$[0.11, 0.91]$	11.8
Water (district)	0.187 (0.218)	$[-0.34, 0.69]$	14.4	0.308 (0.203)	$[-0.04, 1.01]$	12.5
Roads (district)	0.101 (0.157)	$[-0.20, 0.53]$	14.4	0.007 (0.150)	$[-0.32, 0.40]$	12.5
Agricultural inputs (district)	0.105 (0.195)	$[-0.32, 0.58]$	14.4	-0.277* (0.165)	$[-0.75, 0.06]$	12.5
School (central)	-0.026 (0.309)	$[-1.03, 0.65]$	9.4	-0.026 (0.244)	$[-0.82, 0.48]$	9.7
Health (central)	0.157 (0.268)	$[-0.59, 0.68]$	14.4	0.070 (0.344)	$[-0.84, 0.82]$	12.5
Water (central)	0.241 (0.184)	$[-0.14, 0.73]$	14.4	0.337 (0.213)	$[-0.05, 1.03]$	12.5
Roads (central)	1.013*** (0.384)	$[0.31, 2.12]$	14.4	0.202 (0.274)	$[-0.39, 0.93]$	12.5
Electricity (central)	0.424** (0.193)	$[0.05, 0.96]$	14.4	0.254 (0.167)	$[-0.07, 0.78]$	12.5
Panel B: Citizen engagement						
Involved in LCV meetings	0.179* (0.097)	$[0.01, 0.47]$	14.4	-0.011 (0.083)	$[-0.22, 0.18]$	12.5
Met the RDC	0.072 (0.066)	$[-0.06, 0.24]$	14.4	0.010 (0.043)	$[-0.09, 0.12]$	12.5
Attended RDC baraza	0.065 (0.042)	$[-0.02, 0.18]$	14.4	0.000 (0.040)	$[-0.10, 0.10]$	12.5
Collective civic action	-0.498** (0.249)	$[-3.60, -0.19]$	6.4	0.058 (0.059)	$[-0.11, 0.21]$	19.8
Protest	-0.164** (0.076)	$[-1.06, -0.07]$	6.4	0.024 (0.026)	$[-0.03, 0.11]$	19.8
Contacted media	-0.195* (0.108)	$[-1.50, -0.06]$	6.4	-0.025 (0.049)	$[-0.17, 0.09]$	19.8
Participatory index	0.321* (0.172)	$[0.01, 0.82]$	14.4	0.004 (0.126)	$[-0.32, 0.29]$	12.5
Contentious index	-0.860** (0.383)	$[-5.98, -0.43]$	6.4	0.054 (0.116)	$[-0.24, 0.36]$	19.8
Panel C: Institutional knowledge and state-traditional substitution						
Knows magistrate court	0.385*** (0.123)	$[0.17, 0.77]$	14.3	0.010 (0.120)	$[-0.27, 0.31]$	12.4
Knows police	0.120 (0.077)	$[-0.03, 0.33]$	14.3	-0.054 (0.041)	$[-0.16, 0.04]$	12.5
Knows land office	0.119 (0.086)	$[-0.07, 0.33]$	14.3	0.032 (0.062)	$[-0.09, 0.23]$	12.4
Knows Auditor General	0.052 (0.050)	$[-0.06, 0.17]$	14.4	0.125** (0.061)	$[0.01, 0.31]$	12.5
Knows judiciary	0.149 (0.102)	$[-0.09, 0.38]$	14.4	-0.039 (0.078)	$[-0.21, 0.18]$	12.5
Family member civil servant	0.086** (0.040)	$[0.01, 0.21]$	14.4	0.044 (0.059)	$[-0.08, 0.21]$	12.4
Conflict resolved by government	0.007 (0.224)	$[-0.64, 1.29]$	6.3	0.152* (0.085)	$[-0.01, 0.46]$	20.7
Conflict resolved by trad. leader	-0.158 (0.195)	$[-1.78, 0.22]$	6.3	-0.011 (0.050)	$[-0.14, 0.12]$	20.7
Contacted MP	0.168 (0.125)	$[-0.07, 1.16]$	6.4	0.043 (0.073)	$[-0.19, 0.19]$	19.8
Contacted trad. leader	-0.275 (0.199)	$[-2.38, 0.04]$	6.4	-0.011 (0.073)	$[-0.18, 0.21]$	19.8
Knowledge index	0.413** (0.161)	$[0.10, 0.86]$	14.4	0.049 (0.123)	$[-0.21, 0.40]$	12.5
State reliance index	0.362 (0.320)	$[-0.19, 3.29]$	6.4	0.113 (0.085)	$[-0.12, 0.33]$	19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Each row reports, for the subsamples of politically aligned districts (LCV chairperson and RDC of the same party after the 2016 elections) and non-aligned districts, the 2SLS coefficient on log proximity to the district headquarters with its standard error clustered at the parish level, the Anderson–Rubin 95 percent confidence set, which is robust to weak instruments, and the Kleibergen–Paap rk Wald F-statistic of the first stage. The subsample regressions are those underlying Tables 4 to 14; specification, weights and instrument are as in Table 3. Because aligned and non-aligned districts share no clusters, the interaction between proximity and alignment reported in those tables is the difference between the two subsample coefficients; where the non-aligned first stage is strong and its coefficient is close to zero, an aligned confidence set that excludes zero implies that the interaction also differs from zero. Anderson–Rubin sets are obtained by test inversion over a grid of 500 points spanning 15 times the 2SLS confidence interval (weakiv, Finlay, Magnusson and Schaffer); an unbounded set indicates that the instrument is too weak to bound the coefficient at the 95 percent level.

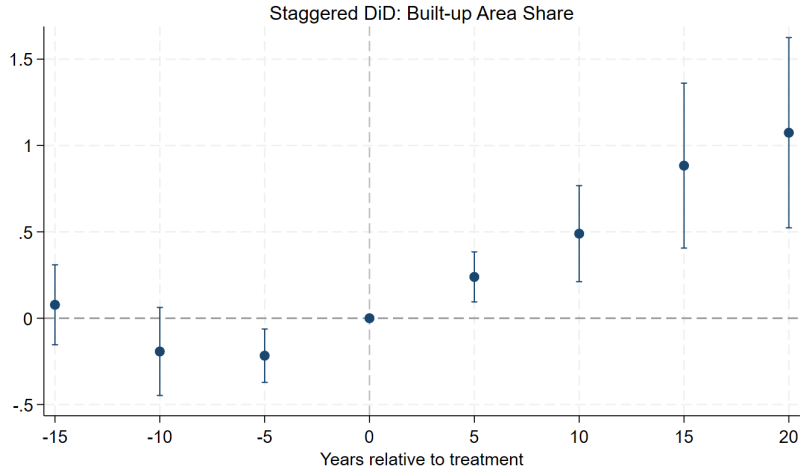
TABLE A15: Weak-Instrument-Robust Inference in the Alignment Subsamples (Part B: Satisfaction, Corruption, Democratic Perceptions)

	Aligned districts			Non-aligned districts		
	2SLS (SE) (1)	AR 95% set (2)	KP F (3)	2SLS (SE) (4)	AR 95% set (5)	KP F (6)
Panel D: Satisfaction with local leaders (NSDS 2021)						
LCV performance	0.055 (0.101)	$[-0.18, 0.29]$	14.4	0.237* (0.130)	$[0.03, 0.72]$	12.5
RDC satisfaction	0.208** (0.086)	$[0.05, 0.46]$	14.4	-0.056 (0.084)	$[-0.31, 0.11]$	12.5
Civil servants performance	0.108 (0.120)	$[-0.12, 0.43]$	14.7	0.062 (0.107)	$[-0.12, 0.41]$	12.6
Panel E: Corruption						
Aware of nepotism	0.221* (0.117)	$[-0.01, 0.53]$	14.3	-0.225** (0.104)	$[-0.56, -0.04]$	12.5
Aware of embezzlement	0.170 (0.106)	$[-0.05, 0.45]$	14.3	-0.138 (0.113)	$[-0.46, 0.09]$	12.5
Aware of absenteeism	0.063 (0.112)	$[-0.21, 0.31]$	14.3	-0.106 (0.094)	$[-0.39, 0.08]$	12.5
Victim of nepotism	-0.047 (0.030)	$[-0.14, 0.00]$	14.4	0.002 (0.022)	$[-0.05, 0.06]$	12.5
Victim of embezzlement	-0.011 (0.036)	$[-0.10, 0.07]$	14.4	-0.080* (0.043)	$[-0.21, 0.00]$	12.5
Victim of absenteeism	-0.061 (0.055)	$[-0.21, 0.05]$	14.4	-0.108 (0.083)	$[-0.33, 0.07]$	12.5
Bribe for medical care	-0.483* (0.248)	$[-3.42, -0.15]$	6.5	-0.019 (0.066)	$[-0.18, 0.16]$	26.1
Bribe for police help	-0.398 (0.385)	$[-3.78, 0.26]$	6.8	-0.299** (0.121)	$[-0.81, -0.09]$	16.8
Bribe to avoid police	-0.115 (0.309)	$(-\infty, \infty)$	2.9	0.034 (0.112)	$[-0.25, 0.42]$	16.0
Bribe for ID document	0.284 (0.243)	$[-0.42, 2.13]$	6.6	-0.034 (0.141)	$[-0.47, 0.32]$	12.9
Awareness index	0.308 (0.190)	$[-0.08, 0.79]$	14.3	-0.321* (0.188)	$[-0.91, 0.02]$	12.5
Victimisation index	-0.184 (0.132)	$[-0.56, 0.06]$	14.4	-0.277* (0.159)	$[-0.74, 0.05]$	12.5
Bribery index	-0.823* (0.500)	$(-\infty, -0.20]$	4.8	-0.008 (0.112)	$[-0.28, 0.29]$	24.2
Panel F: Democratic perceptions (Afrobarometer R10)						
Country is a democracy	-0.015 (0.089)	$[-0.39, 0.30]$	6.4	-0.002 (0.038)	$[-0.11, 0.09]$	21.1
Satisfied with democracy	0.138 (0.161)	$[-0.15, 1.43]$	6.4	0.031 (0.063)	$[-0.15, 0.18]$	21.5
Free to vote	0.047 (0.075)	$[-0.09, 0.54]$	7.1	-0.113*** (0.032)	$[-0.21, -0.04]$	18.9
Free to join organisations	-0.099 (0.102)	$[-0.55, 0.15]$	7.8	-0.033 (0.050)	$[-0.19, 0.08]$	19.6
President ignores the law	-0.176 (0.169)	$[-1.00, 0.13]$	8.8	0.151** (0.067)	$[0.02, 0.41]$	16.7
Unequal under the law	0.108 (0.150)	$[-0.34, 0.70]$	7.4	-0.113 (0.089)	$[-0.43, 0.06]$	18.5
Judges influenced	-0.208 (0.201)	$[-1.95, 0.11]$	6.7	-0.014 (0.086)	$[-0.30, 0.16]$	17.7
Officials go unpunished	-0.087 (0.184)	$[-1.18, 0.34]$	6.8	-0.014 (0.088)	$[-0.31, 0.17]$	19.5
Trust local government	0.121 (0.123)	$[-0.18, 0.78]$	7.0	-0.129* (0.077)	$[-0.45, -0.01]$	20.8
Trust president	-0.050 (0.146)	$[-0.64, 0.65]$	6.0	-0.083 (0.067)	$[-0.28, 0.06]$	20.0
Trust parliament	0.023 (0.155)	$[-0.62, 0.60]$	6.4	0.012 (0.079)	$[-0.21, 0.20]$	19.8
Trust ruling party	0.146 (0.137)	$[-0.15, 1.11]$	6.4	-0.126* (0.074)	$[-0.40, 0.01]$	19.8
Trust opposition	0.002 (0.146)	$[-0.75, 0.44]$	6.4	0.079 (0.059)	$[-0.04, 0.28]$	19.8
Trust courts	0.032 (0.144)	$[-0.37, 0.70]$	7.0	0.091 (0.056)	$[-0.10, 0.20]$	20.4
Democracy index	0.029 (0.198)	$[-0.45, 1.09]$	6.9	-0.109 (0.078)	$[-0.35, 0.06]$	20.1
Rule of law index	-0.156 (0.303)	$[-2.21, 0.47]$	6.6	-0.011 (0.131)	$[-0.44, 0.26]$	19.0
Trust index	0.094 (0.181)	$[-0.43, 1.07]$	6.4	-0.048 (0.092)	$[-0.37, 0.13]$	19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Each row reports, for the subsamples of politically aligned districts (LCV chairperson and RDC of the same party after the 2016 elections) and non-aligned districts, the 2SLS coefficient on log proximity to the district headquarters with its standard error clustered at the parish level, the Anderson–Rubin 95 percent confidence set, which is robust to weak instruments, and the Kleibergen–Paap rk Wald F-statistic of the first stage. The subsample regressions are those underlying Tables 4 to 14; specification, weights and instrument are as in Table 3. Because aligned and non-aligned districts share no clusters, the interaction between proximity and alignment reported in those tables is the difference between the two subsample coefficients; where the non-aligned first stage is strong and its coefficient is close to zero, an aligned confidence set that excludes zero implies that the interaction also differs from zero. Anderson–Rubin sets are obtained by test inversion over a grid of 500 points spanning 15 times the 2SLS confidence interval (weakiv, Finlay, Magnusson and Schaffer); an unbounded set indicates that the instrument is too weak to bound the coefficient at the 95 percent level.

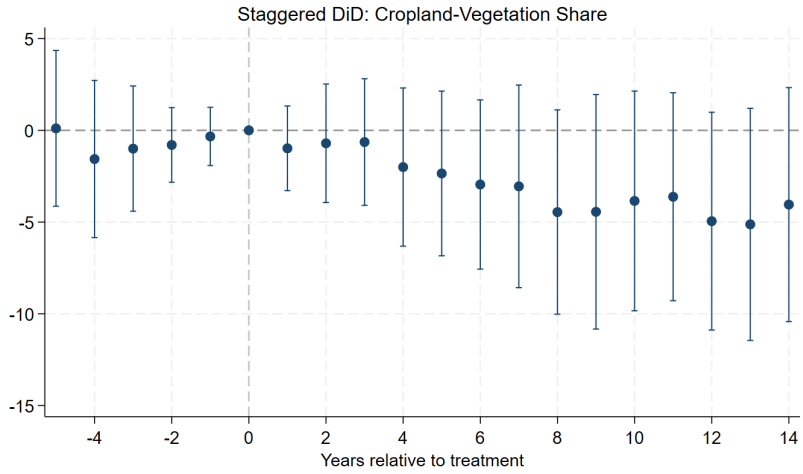
F Figures

FIGURE A4: EVENT STUDY: DISTRICT HEADQUARTERS CREATION ON BUILT-UP AREA SHARE



Notes: Event study of the de Chaisemartin & D'Haultfœuille (2020) staggered difference-in-differences estimator of the parish built-up area share on district headquarters creation. The outcome is the share of a parish that is built-up, measured from the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S; 100 m resolution), available in five-year epochs from 1990 to 2025. Event time is measured in five-year epochs and shown in years on the horizontal axis: the three points to the left of zero are placebo (pre-trend) estimates and the four points to the right are the post-creation effects. Vertical bars are 95 percent confidence intervals from standard errors clustered at the parish level.

FIGURE A5: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND-VEGETATION SHARE



Notes: Event study graph of the estimated coefficients of the de Chaisemartin & D'Haultfœuille (2020) staggered difference in difference estimator of cropland-vegetation share on headquarters creation.

G Recentered Instrument and Randomization Inference

The instrument interacts an exogenous shock, the placement of district headquarters following district creation, with non random exposure: a household's fixed location relative to the geometry of its mother district. Even if the realized partition is as good as random among the feasible partitions, households located near the geometric interior of any feasible fragment mechanically receive high instrument values under every counterfactual configuration (Borusyak & Hull, 2023). This appendix implements the correction proposed by Borusyak & Hull (2023). I compute the expected instrument μ_i across counterfactual redistricting configurations, control for it directly, construct the recentered instrument $z_i - \mu_i$, and conduct randomization inference over placebo redistrictings.

Counterfactual redistricting design

I use the 1991 county boundaries as the building blocks for counterfactual redistricting.[23] These boundaries predate the redistricting wave and therefore cannot have been affected by it. Holding fixed which mother districts split and the number of districts created by 2020, I enumerate all feasible partitions of each splitting mother district's counties into contiguous fragments. The number of feasible partitions ranges from one (for eleven mother districts with no alternative configuration at the county level) to 5,822 for Mbarara. Since the enumeration is exhaustive, μ_i is an exact expectation. For each counterfactual partition, the fragment containing the 1991 district headquarters retains the mother district's status, while the remaining fragments become new districts. I then reconstruct the instrument as in the main analysis, interacting the logarithm of household proximity to the fragment's centroid with the new-district indicator.[24] Figure A6 illustrates this procedure for Mbarara, which has the largest number of feasible partitions.

The counterfactual redistrictings substantially reshuffle households. Among enumeration areas in mother districts that split, 47 percent are assigned to a different fragment under at least one feasible alternative partition. As a result, the recentered instrument retains about two thirds of the standard deviation of the raw instrument, leaving meaningful variation to identify the effect. At the same time, a large share of the raw instrument is predictable from a household's fixed position in the geometry of the mother district, with the correlation between z_i and the expected instrument μ_i reaching 0.74.

Results

Three regressors depend on how the mother district is partitioned: the excluded instrument, proximity to the district centroid, and the new-district indicator. For each household, I compute

[23] IPUMS shapefile for the 1991 Uganda census (156 counties).

[24] As in the main analysis, centroids falling outside half-moon shaped fragments are forced to lie within the polygon. For randomization inference, I draw one partition per mother district for each of 1,000 national counterfactual configurations, using independent random streams from a single seed.

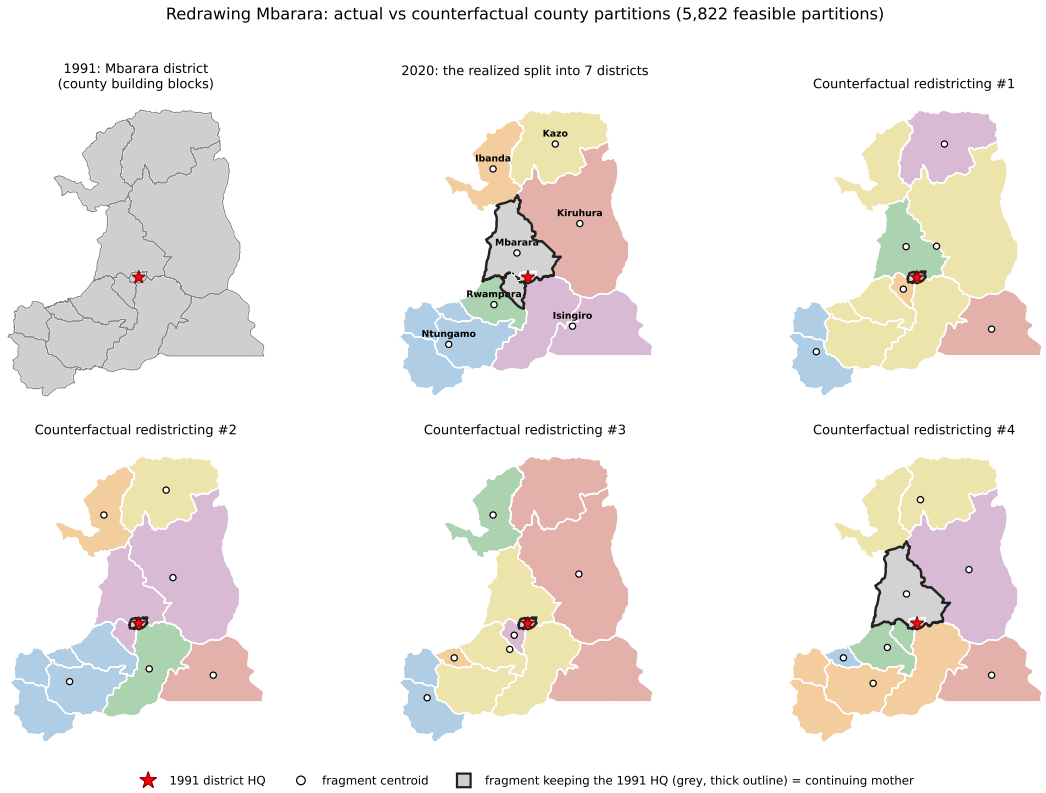


FIGURE A6: Redrawing Mbarara: actual and counterfactual county partitions. The 1991 district (top left) is composed of its county building blocks; the realized 2020 configuration (top centre) splits it into seven districts. The remaining panels show four of the 5,822 feasible contiguous partitions into seven fragments. The fragment retaining the 1991 headquarters (red star) remains the mother district (grey, thick outline); white dots mark fragment centroids, from which the instrument is computed.

the expected value of each regressor across all feasible partitions of its mother district (μ_z, μ_{prox}, and μ_{new}). Following Borusyak & Hull (2023), I consider two approaches to account for this predictable geographic exposure: controlling for the expected regressors while retaining the original instrument, and replacing the original instrument with its recentered value, $z - \mu_z$. These approaches are algebraically equivalent when μ_z is also included as a control in the recentered specification. The control approach can also improve precision by accounting for predictable variation in the outcome and endogenous variable. Table A16 reports both specifications alongside the baseline 2SLS estimates. The first adds μ_z, μ_{prox}, and μ_{new} as controls, while the second uses $z - \mu_z$ as the excluded instrument and controls for μ_{prox} and μ_{new}.

The results are similar under both specifications, with generally larger estimated magnitudes. When controlling for the expected regressors, the coefficients increase from 0.43 to 0.80 for electricity, from 0.39 to 0.52 for district health, and from 0.28 to 0.50 for district water. The corresponding first-stage F-statistics range from approximately 10 to 25. Using the recentered instrument produces even larger estimates for these outcomes (1.01, 0.58, and 0.63,

respectively), with first-stage F-statistics between 11.6 and 17.8 in the full samples. The first stage is weaker for central-government schooling ($F = 5.3$), which has a smaller sample. The signs remain unchanged across specifications, and the only change in statistical significance is for central-government health, which becomes significant at the 5 percent level under both approaches. These results suggest that the main findings are not driven by households' predictable geographic exposure to redistricting. If anything, accounting for this exposure increases the estimated effects.

Given the weaker first stages relative to the baseline, I also report weak-instrument-robust Anderson–Rubin 95 percent confidence intervals. Under both specifications these are bounded and exclude zero for the same six outcomes: district and central-government health, district and central-government water, central-government roads, and electricity (Table A16).

I complement these results with randomization inference, comparing the observed estimates with those obtained under counterfactual redistricting configurations. This approach does not rely on large-sample approximations, but instead uses the distribution of estimates generated by alternative feasible partitions of the mother districts. I draw 1,000 counterfactual national redistricting configurations. For each configuration, I recompute the instrument and the two partition-dependent controls, proximity to the district centroid and the new-district indicator, and re-estimate the reduced form of each outcome using the recentered placebo instrument and the same specification. I then compare the observed estimates with the distribution of the 1,000 placebo estimates (Figure A7).

The first stage is strongly supported by this exercise: only 2 of the 1,000 counterfactual redistrictings produce a coefficient as large as the observed one ($p = 0.002$). Among the service outcomes, electricity ($p = 0.001$), district health ($p = 0.004$), and central-government water ($p = 0.020$) are significant at the 5 percent level. District water ($p = 0.059$) and central-government roads ($p = 0.100$) are significant at the 10 percent level, while central-government health, schooling, district roads, and agricultural inputs are not distinguishable from their placebo distributions.

These results provide a complementary assessment of statistical significance based on the variation generated by feasible redistricting configurations. The counterfactual partitions vary across only 33 splitting mother districts, with eight accounting for most of the variation, highlighting the limited number of independent sources of identifying variation. Overall, the randomization inference supports the first stage and the main service results.

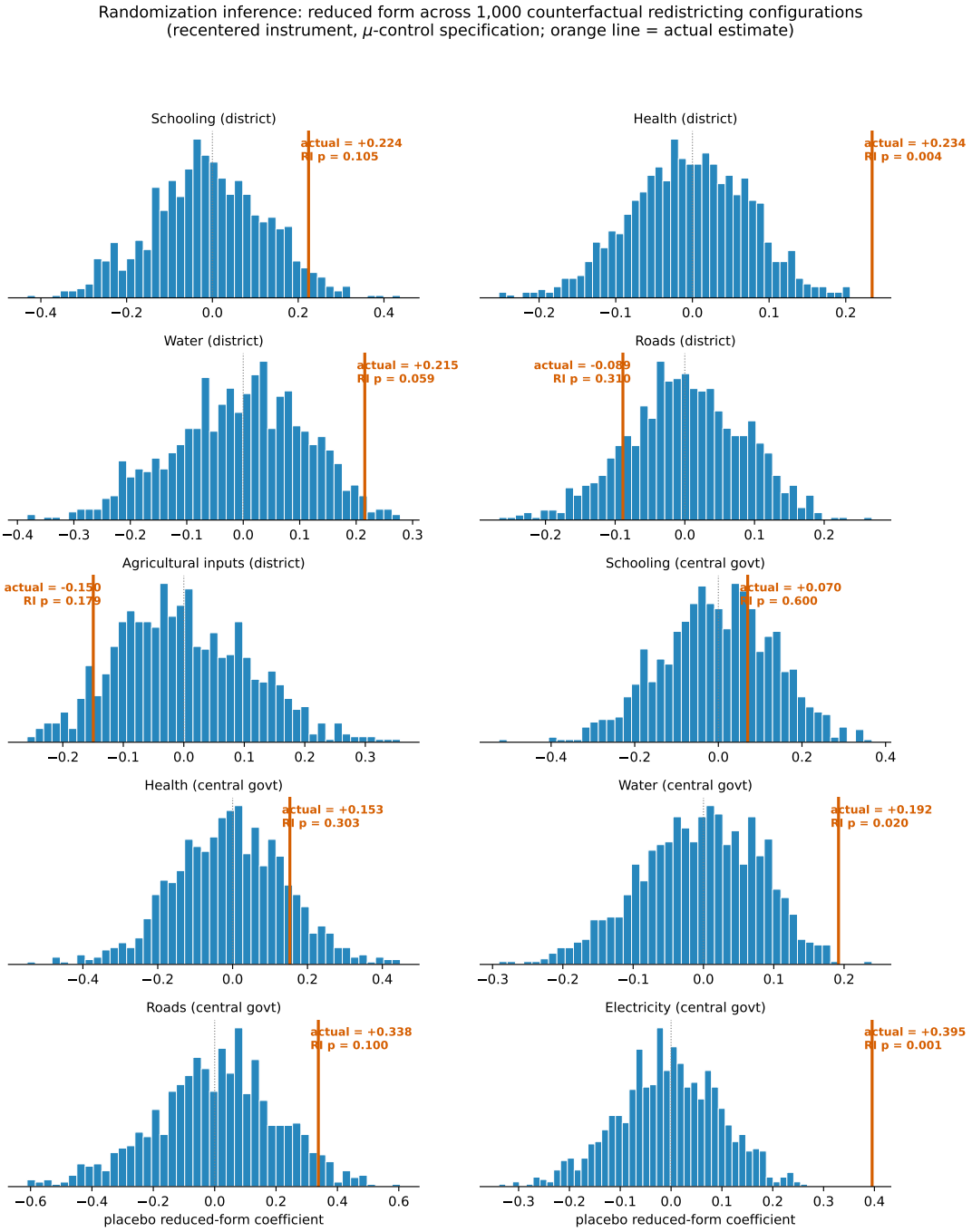


FIGURE A7: Randomization inference. Each panel shows the distribution of reduced-form coefficients on the recentered instrument across 1,000 counterfactual national redistricting configurations (μ-control specification, with the proximity and new-district controls recomputed under each configuration); the orange line marks the estimate under the realized configuration, with its two-sided randomization p-value.

TABLE A16: Recentered Instrument: Effects on Services (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
Panel A: NSDS 2021 District services					
Baseline 2SLS	0.282	0.387***	0.279*	0.089	-0.085
	(0.214)	(0.121)	(0.157)	(0.110)	(0.145)
2SLS, μ controls	0.464*	0.516***	0.496***	-0.012	-0.291
	(0.249)	(0.148)	(0.176)	(0.129)	(0.180)
AR 95% set	$[-0.00, 1.11]$	$[0.26, 0.90]$	$[0.17, 0.97]$	$[-0.31, 0.26]$	$[-0.75, 0.06]$
2SLS, recentered IV ($z - \mu_z$; μ_{prox}, μ_{new} controls)	0.546*	0.575***	0.628***	-0.094	-0.412*
	(0.301)	(0.181)	(0.231)	(0.155)	(0.231)
AR 95% set	$[0.00, 1.44]$	$[0.26, 1.11]$	$[0.25, 1.44]$	$[-0.54, 0.22]$	$[-1.13, 0.02]$
KP F (baseline)	22.24	33.35	28.75	28.75	28.75
KP F (μ controls)	17.58	24.51	17.46	17.46	17.46
KP F (recentered)	13.27	17.79	11.58	11.58	11.58
RI p (μ controls, S=1,000)	[0.105]	[0.004]	[0.059]	[0.310]	[0.179]
Observations	7,137	7,595	8,417	8,417	8,417

	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
Panel B: NSDS 2021 Central govt services					
Baseline 2SLS	0.151	0.233	0.372**	0.706***	0.428***
	(0.192)	(0.201)	(0.146)	(0.261)	(0.137)
2SLS, μ controls	0.058	0.573**	0.449***	0.970**	0.795***
	(0.261)	(0.244)	(0.154)	(0.377)	(0.205)
AR 95% set	$[-0.60, 0.68]$	$[0.08, 1.17]$	$[0.18, 0.88]$	$[0.37, 2.15]$	$[0.47, 1.45]$
2SLS, recentered IV ($z - \mu_z$; μ_{prox}, μ_{new} controls)	-0.022	0.792**	0.485***	1.150**	1.014***
	(0.368)	(0.326)	(0.185)	(0.489)	(0.297)
AR 95% set	$[-1.63, 1.05]$	$[0.18, 1.81]$	$[0.18, 1.13]$	$[0.45, 3.11]$	$[0.61, 2.27]$
KP F (baseline)	22.62	28.75	28.75	28.75	28.75
KP F (μ controls)	9.86	17.46	17.46	17.46	17.46
KP F (recentered)	5.35	11.58	11.58	11.58	11.58
RI p (μ controls, S=1,000)	[0.600]	[0.303]	[0.020]	[0.100]	[0.001]
Observations	5,553	8,417	8,417	8,417	8,417

Notes: *p<0.1, **p<0.05, ***p<0.01 (cluster-robust). Borusyak–Hull (2023) recentering exercise. The expected instrument μ_i is the exact mean of the instrument over all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, holding fixed which mothers split and into how many units; building blocks are 1991-vintage counties (IPUMS), and every mother's feasible set is enumerated exhaustively. Column blocks: the paper's 2SLS specification; adding ($\mu_z, \mu_{prox}, \mu_{new}$) as controls; and the recentered excluded instrument $z - \mu_z$ with μ_{prox} and μ_{new} as controls, so that the proximity and new-district main effects are recentered as well (adding μ_z to this column reproduces the μ-control column exactly). All regressions use the paper's controls, weights, and clustering (NSDS: EA; AB: town/village); Kampala and Wakiso excluded as in the main tables. RI p: two-sided randomization-inference p-value of the reduced form on $z - \mu$ across S=1,000 national placebo redistricting configurations (one independent draw per mother per configuration, seed 20260706; the proximity and new-district controls are recomputed under each configuration); first-stage omnibus RI p = 0.002 (NSDS) and 0.066 (AB). AR 95% set: weak-instrument-robust Anderson–Rubin confidence set (cluster-robust), reported below the standard error of the corresponding coefficient.

H Additional Results

TABLE A17: Service Availability and Proximity to District HQ (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity (4)	Paved road (5)	Police station (6)
Panel A: Afrobarometer R10						
Log Proximity to District HQ	0.006	0.166	0.495**	0.034	0.358**	0.504**
	(0.165)	(0.201)	(0.224)	(0.203)	(0.162)	(0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to public goods and services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A18: Schooling and Proximity to District Headquarters (2SLS)

	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
Panel A: District services						
Log Proximity to District HQ	-0.128*	-1.165**	-19.130***	0.021	0.046	0.343
	(0.078)	(0.536)	(6.807)	(0.067)	(0.065)	(0.267)
Observations	4566	3329	3329	4563	4564	5581
Number of clusters	951	916	916	951	951	638
Mean of DV	0.714	1.710	31.421	0.804	0.809	0.559
IV F-stat (KP rk Wald)	26.298	22.676	22.676	26.298	26.244	9.399
	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
Panel B: Central govt services						
Log Proximity to District HQ	-0.169	-0.606	-10.457	0.104	0.067	0.127
	(0.105)	(0.689)	(7.829)	(0.087)	(0.082)	(0.225)
Observations	3546	2132	2132	3539	3541	3659
Number of clusters	943	827	827	943	943	420
Mean of DV	0.700	2.007	34.856	0.769	0.780	0.695
IV F-stat (KP rk Wald)	20.727	16.442	16.442	20.948	20.697	15.054

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to schooling services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 6 of each panel is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A19: Health Services and Proximity to District Headquarters (2SLS)

	Public hospital (1)	Injury (2)	Injury travel time (3)	Health facility score (4)	HC absenteeism (5)	HC ambulance (6)	District HC supervision (7)
Panel A: District services							
Log Proximity to District HQ	-0.013	-0.040	-14.329*	29.451***	-0.265*	0.265*	0.162
	(0.090)	(0.034)	(8.670)	(9.847)	(0.138)	(0.160)	(0.161)
Observations	2291	8415	3252	6948	6821	6909	6619
Number of clusters	826	967	869	791	778	786	750
Mean of DV	0.865	0.146	36.098	26.347	0.215	0.303	0.787
IV F-stat (KP rk Wald)	26.663	28.803	23.237	33.461	30.568	30.178	29.761

	HC type 4 (1)	Distance gov hospital (2)	PHC insufficient (3)
Panel B: Central govt services			
Log Proximity to District HQ	0.146**	0.049	-0.029
	(0.066)	(0.099)	(0.157)
Observations	6979	8417	6819
Number of clusters	795	967	776
Mean of DV	0.075	0.240	0.233
IV F-stat (KP rk Wald)	33.260	28.745	32.624

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to health services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–7 of Panel A and all columns of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

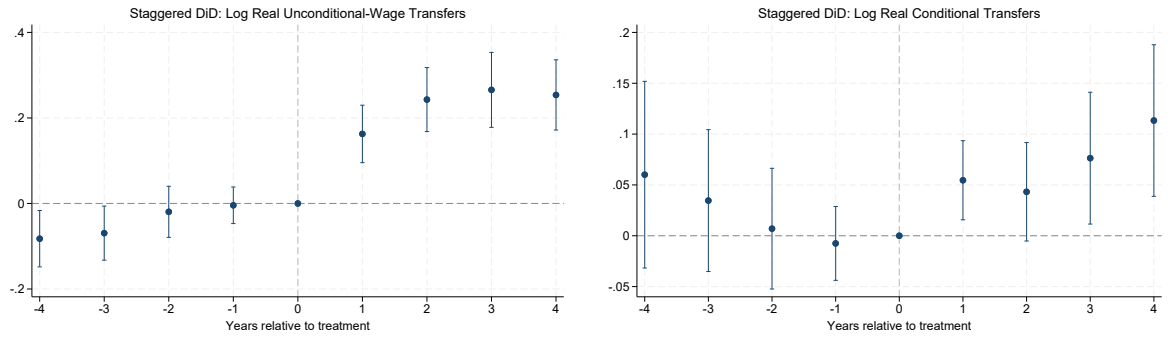


FIGURE A8: Event study: composition of the territory transfer response. Same specification and sample as Figure 5, for the wage component of the unconditional block grant (left, the duplicated administrative payroll) and conditional service-delivery grants (right), in log real FY2017/18 shillings. The joint placebo test has a p-value of 0.02 for the unconditional-wage component (with individually small placebo estimates) and 0.34 for conditional grants.

TABLE A20: Water Services and Proximity to District Headquarters (2SLS)

	Water source dry	Water distance dry	Water source wet	Piped water	Borehole	Protected well
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: District services						
Log Proximity to District HQ	0.128	-0.225	-0.029	0.212	0.184	0.293
	(0.111)	(0.194)	(0.089)	(0.136)	(0.158)	(0.188)
Observations	8417	7229	8417	8371	8371	8374
Number of clusters	967	952	967	961	961	961
Mean of DV	0.834	0.786	0.889	0.331	0.636	0.302
IV F-stat (KP rk Wald)	28.745	30.837	28.745	28.094	28.094	27.880

	Water src dry outside	Water dist dry outside	Water src wet outside	Water dist wet outside	WAP crops	WAP cattle	WAP fish
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel B: Central govt services							
Log Proximity to District HQ	0.145	-0.141	0.039	-0.065	0.042	0.084**	0.123*
	(0.110)	(0.228)	(0.075)	(0.166)	(0.030)	(0.042)	(0.063)
Observations	8417	7177	8416	5603	7815	8356	7762
Number of clusters	967	949	967	894	901	959	895
Mean of DV	0.784	0.814	0.855	0.746	0.993	0.969	0.961
IV F-stat (KP rk Wald)	28.745	28.853	28.682	28.733	25.727	27.845	25.966

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to water services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–6 of Panel A and Columns 5–7 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A21: Road Infrastructure and Proximity to District Headquarters (2SLS)

	Trunk road	Feeder road	Communal road	Murram road quality	Feeder road quality	Communal road quality	Murram maintenance	Feeder maintenance	Communal maintenance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: District services									
Log Proximity to District HQ	-0.119	0.058	-0.134**	-0.005	0.149	0.268**	-0.314**	0.075	-0.034
	(0.110)	(0.062)	(0.058)	(0.152)	(0.143)	(0.117)	(0.138)	(0.143)	(0.052)
Observations	8416	8416	8415	6220	7814	7758	8336	8364	8372
Number of clusters	967	967	967	708	894	888	957	960	961
Mean of DV	0.634	0.883	0.914	0.288	0.205	0.099	0.157	0.751	0.026
IV F-stat (KP rk Wald)	28.702	28.702	28.674	21.068	27.638	27.202	27.662	27.655	27.642

	Trunk road	Tarmac road quality	Tarmac maintenance
	(1)	(2)	(3)
Panel B: Central govt services			
Log Proximity to District HQ	0.262**	0.672**	0.117
	(0.118)	(0.317)	(0.146)
Observations	8416	6236	8362
Number of clusters	967	721	960
Mean of DV	0.436	0.775	0.866
IV F-stat (KP rk Wald)	28.702	13.092	28.125

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to roads on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–9 of Panel A and Columns 2–3 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A22: Agricultural Inputs (District) and Proximity to District Headquarters (2SLS)

	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
Panel A: Inputs Availability				
Log Proximity to District HQ	-0.080	0.005	-0.125	-0.070
	(0.073)	(0.096)	(0.094)	(0.072)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.205	0.325	0.238	0.106
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745
	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
Panel B: Inputs Use				
Log Proximity to District HQ	-0.115**	0.108	0.043	0.006
	(0.059)	(0.101)	(0.062)	(0.044)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.137	0.351	0.151	0.053
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of the reported availability (Panel A) and household use (Panel B) of agricultural inputs on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A23: Electricity (Central Government) and Proximity to District Headquarters (2SLS)

	HH electricity (1)	Connected grid (2)	Interested grid (3)	Electricity availability (4)
Panel A: Central govt services				
Log Proximity to District HQ	0.276***	0.197**	0.009	0.237*
	(0.101)	(0.091)	(0.037)	(0.127)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.463	0.175	0.055	0.824
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to electricity on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 4 is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A24: Priority Projects Implemented and Proximity to District HQ (2SLS, community level)

	Share top-3 implemented (1)	Any top-3 implemented (2)	New school (3)	Classroom (4)	Health facility (5)	Water (6)	Electricity (7)	Roads bridges (8)
Log Proximity to District HQ (EA)	0.319***	0.232*	-0.017	0.308***	0.190**	0.084	0.017	0.312**
	(0.099)	(0.136)	(0.030)	(0.105)	(0.079)	(0.126)	(0.138)	(0.152)
Observations	933	933	595	515	651	743	617	436
Mean of DV	0.304	0.501	0.022	0.064	0.055	0.229	0.209	0.135
IV F-stat (KP rk Wald)	54.747	54.747	36.911	24.773	35.653	37.159	24.678	18.671

Notes: *p<0.1, **p<0.05, ***p<0.01. Community-level regressions (one observation per enumeration area) of outcomes from the NSDS community questionnaire on the log GIS distance from the enumeration area to the district's capital. Column (1) is the share of the community's top-three priorities that were implemented in the village or parish in the past three years and Column (2) equals one if any of them was implemented. In Columns (3)–(8) the outcome equals one if the community ranked the project among its three most important and the project was implemented in the past three years; each column is estimated on the communities that ranked that project. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument is the interaction of distance from the enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Heteroskedasticity-robust standard errors are in parenthesis.

TABLE A25: Priority Projects Implemented and Proximity to District HQ (2SLS, community level) – Aligned versus Non-Aligned Districts

	Share top-3 implemented (1)	Any top-3 implemented (2)	New school (3)	Classroom (4)	Health facility (5)	Water (6)	Electricity (7)	Roads bridges (8)
NSDS 2021 communities								
Proximity, non-aligned	-0.132	-0.075	-0.018	0.209	-0.110	-0.285	-0.081	0.296
	(0.120)	(0.212)	(0.018)	(0.159)	(0.120)	(0.188)	(0.194)	(0.193)
Proximity × Aligned	0.826***	0.575*	0.052	0.314	0.600***	0.962**	0.859	0.422
	(0.217)	(0.295)	(0.056)	(0.300)	(0.208)	(0.380)	(0.574)	(0.477)
Proximity, aligned (sum)	0.694***	0.499**	0.034	0.523**	0.489***	0.677**	0.778	0.718
	(0.181)	(0.205)	(0.053)	(0.255)	(0.169)	(0.330)	(0.540)	(0.437)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)	(+)	(+)	(+)
One-sided *p*-value	0.000	0.026	0.179	0.148	0.002	0.006	0.067	0.188
Observations	933	933	595	515	651	743	617	436
Mean of DV	0.304	0.501	0.022	0.064	0.055	0.229	0.209	0.135
KP F (aligned / non-aligned)	31.3 / 15.4	31.3 / 15.4	11.7 / 17.7	7.3 / 12.9	14.5 / 13.2	12.9 / 13.0	4.5 / 11.8	5.5 / 19.5

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Community-level regressions (one observation per enumeration area) of outcomes from the NSDS community questionnaire on the log GIS proximity from the enumeration area to the district's capital and its interaction with an indicator for politically aligned districts. The rows report the effect of proximity in non-aligned districts, the interaction, and their sum, the effect in aligned districts. Outcomes, sample, controls and instrument as in Table A24; the excluded instruments are the one used there and its interaction with the alignment indicator. Heteroskedasticity-robust standard errors in parentheses; the standard error of the interaction is the square root of the sum of the squared subsample standard errors. Observations and means of the dependent variable pool the two groups. The predicted sign is the sign of the interaction implied by Proposition 5, and the one-sided p-value tests the interaction against that sign.

TABLE A26: Trust in Others and Social Capital and Proximity to District HQ (2SLS)

	Trust relatives (1)	Trust neighbours (2)	Trust citizens (3)	Trust other religion (4)	Trust other ethnicity (5)
Panel A: Afrobarometer R10					
Log Proximity to District HQ	-0.013	0.058	0.131	0.192*	0.257**
	(0.077)	(0.089)	(0.108)	(0.109)	(0.126)
Observations	2176	2179	2177	2179	2167
Number of clusters	271	271	271	271	271
Mean of DV	0.800	0.652	0.537	0.587	0.503
IV F-stat (KP rk Wald)	19.564	19.299	19.038	19.340	19.179

	Empowerment (1)	Collective action (2)	Bridging groups (3)	General trust (4)
Panel B: NSDS 2021				
Log Proximity to District HQ	0.084	0.179**	0.055	-0.050
	(0.074)	(0.080)	(0.047)	(0.075)
Observations	8416	8416	4334	8416
Number of clusters	967	967	926	967
Mean of DV	0.637	0.591	0.080	0.441
IV F-stat (KP rk Wald)	28.702	28.702	26.874	28.702

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' trust in others and social capital on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A27: Trust in Government and Proximity to District HQ: 2017 Afrobarometer (2SLS)

	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
Panel A: Afrobarometer R7 (2017)						
Log Proximity to District HQ	-0.085	0.018	-0.142	0.047	-0.092	-0.025
	(0.100)	(0.099)	(0.106)	(0.083)	(0.130)	(0.080)
Observations	995	1014	1009	1025	995	987
Number of clusters	129	129	129	129	129	129
Mean of DV	0.587	0.711	0.554	0.642	0.361	0.626
IV F-stat (KP rk Wald)	10.770	10.062	9.688	10.029	10.004	10.106

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of trust in government variables from the 2017 Afrobarometer (Round 7) on self-reported household's distance to the district's capital. Each variable indicates that the respondent trusts the corresponding institution somewhat or a lot. All regressions control for district ever split, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of the household's distance to the district's centroid and an indicator for new districts between 1991-2016, using district boundaries and headquarters as of 2017. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The first-stage Kleibergen-Paap F statistic is close to the conventional threshold of 10, so magnitudes should be read with caution; the estimates are reported to show that the absence of trust effects documented in the 2024 round already holds in 2017.

TABLE A28: Service Availability and Proximity to District HQ Across Afrobarometer Rounds (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
Panel A: Afrobarometer R7 (2017)						
Log Proximity to District HQ	-0.070	-0.293	0.268*	0.155	-0.105	-0.059
	(0.174)	(0.242)	(0.153)	(0.228)	(0.285)	(0.260)
Observations	1048	1040	1048	1048	1048	1048
Number of clusters	129	128	129	129	129	129
Mean of DV	0.809	0.692	0.191	0.275	0.489	0.366
IV F-stat (KP rk Wald)	9.621	9.806	9.621	9.621	9.621	9.621
	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
Panel B: Afrobarometer R10 (2024)						
Log Proximity to District HQ	0.006	0.166	0.495**	0.034	0.358**	0.504**
	(0.165)	(0.201)	(0.224)	(0.203)	(0.162)	(0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of enumeration-area service availability indicators, recorded by Afrobarometer enumerators (not self-reported), on self-reported household's distance to the district's capital. Outcomes are identical across the two rounds and indicate whether the enumeration area has a school, a health clinic, a piped water system, an electricity grid, a paved road surface over the last five kilometers, and a police station. All regressions control for district ever split, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of the household's distance to the district's centroid and an indicator for new districts, defined over 1991-2016 with 2017 boundaries in Panel A and over 1991-2020 with 2020 boundaries in Panel B. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The first-stage Kleibergen-Paap F statistic in the 2017 round is close to the conventional threshold of 10, so Panel A magnitudes should be read with caution.

TABLE A29: Non-zero-sum fiscal accounting around district splits

	Own pre-split counterfactual	Trend-adjusted counterfactual	Matched-trend counterfactual
Panel A. Net extra spending across the territory, $(M_{post} + C_{post}) - \widehat{M}_{cf}$			
Mean (billions UGX)	10.11	5.37	5.60
Median (billions UGX)	7.39	2.83	3.00
Splits with positive excess	20/24	19/24	19/24
Panel B. As percent of the mother's pre-split budget			
Mean (%)	35.26	19.21	20.42
Median (%)	28.34	11.84	9.99
Standard deviation (%)	37.98	30.59	32.12
Panel C. Underlying real budget levels (billions of FY2017/18 UGX)			
Mother, pre-split (M_{pre}), median		31.05	
Mother, post-split (M_{post}), median		23.01	
Child, first two years (C_{post}), median		14.38	

Notes: The table asks whether the spending a mother district loses when it splits is recovered elsewhere, or whether the split raises total spending across the original territory. The sample is the 24 districts created within the panel window (calendar 2015–2020; approved local-government transfers FY2012/13–FY2023/24), and all amounts are in real FY2017/18 shillings. For each split at creation year T, M_{pre} is the mother's average budget over the two fiscal years ending at T, M_{post} its average over $(T{+}1, T{+}2)$, and C_{post} the child's average over its first two years. The excess spending generated by the split is $(M_{post} + C_{post}) - \widehat{M}_{cf}$, the post-split total across mother and child minus the budget the undivided territory would have had, $\widehat{M}_{cf}$. The three columns differ only in that counterfactual. Column (1) sets $\widehat{M}_{cf} = M_{pre}$, asking whether post-split spending exceeds the mother's own pre-split level. Because all local budgets grew over the period, columns (2) and (3) instead let the counterfactual grow at the pace of comparable districts that never split: $\widehat{M}_{cf} = M_{pre} \times g$, where g is the median real growth factor over the same windows of the 19 territories with no split during the panel window (column 2), or of those same territories in the mother's own quartile of pre-split budget (column 3). Kampala, Wakiso, and municipal and city councils are excluded throughout. Both trend-adjusted counterfactuals leave a positive excess in 19 of the 24 splits and an average net cost near 20 percent of the mother's pre-split budget, indicating that district creation raises total spending rather than redistributing a fixed envelope.

I Alternative Identification Strategy: Geographic Regression Discontinuity Design

This appendix presents a complementary identification strategy to the instrumental variable design in Section 5. Following Blattman et al. (2025), I exploit the new boundaries created by district splits within former districts. The idea is to compare households living on opposite sides of these boundaries, who belonged to the same district before the split but experienced different changes in their distance to district headquarters afterward. By comparing how these differences vary along the same boundary, the design isolates the role of administrative proximity from average differences associated with district creation.

I.1 Design, Sample and Identifying Assumptions

District proliferation creates new boundaries within former districts. Before a split, households living on opposite sides of these boundaries belonged to the same district and were served by the same headquarters. After the split, they belong to different districts, each with its own headquarters. The resulting change in administrative proximity varies along the boundary, depending on the location of the new headquarters. For example, a household near the new headquarters may experience a substantial reduction in distance relative to its neighbour across the boundary, while a pair farther along the same boundary may experience little change or even a relative loss in proximity. Figure A9 illustrates this variation using three pairs along the same boundary. Figure A10 maps the 143 internal boundaries created within the 1991 districts, the 78 containing NSDS 2021 cross-boundary pairs, the 275 pair midpoints coloured by the magnitude of the proximity shock, and the 2020 district headquarters.

Boundary fixed effects absorb average differences between the two sides of each boundary, including differences associated with belonging to a new district rather than its mother or another newly created district. These may reflect differences in budgets, elected councils, Resident District Commissioners, administrative staff, and district-level policies. Identification therefore does not come from comparing households assigned to new and existing districts, but from variation in the relative change in headquarters proximity along the same boundary. This provides a complementary source of identifying variation to the distance to centroid instrument used in the main analysis. The two designs also differ in the populations and variation they identify from, as the boundary design focuses on households near district boundaries rather than households throughout the district.

Sample. I identify 143 internal boundaries within the 1991 mother districts, excluding Kampala and Wakiso. Each NSDS 2021 enumeration area located within 10 kilometres of a boundary is matched to its nearest enumeration area across that boundary, provided the two are no more than 20 kilometres apart. This yields 275 pairs across 78 boundaries in 26 mother dis-

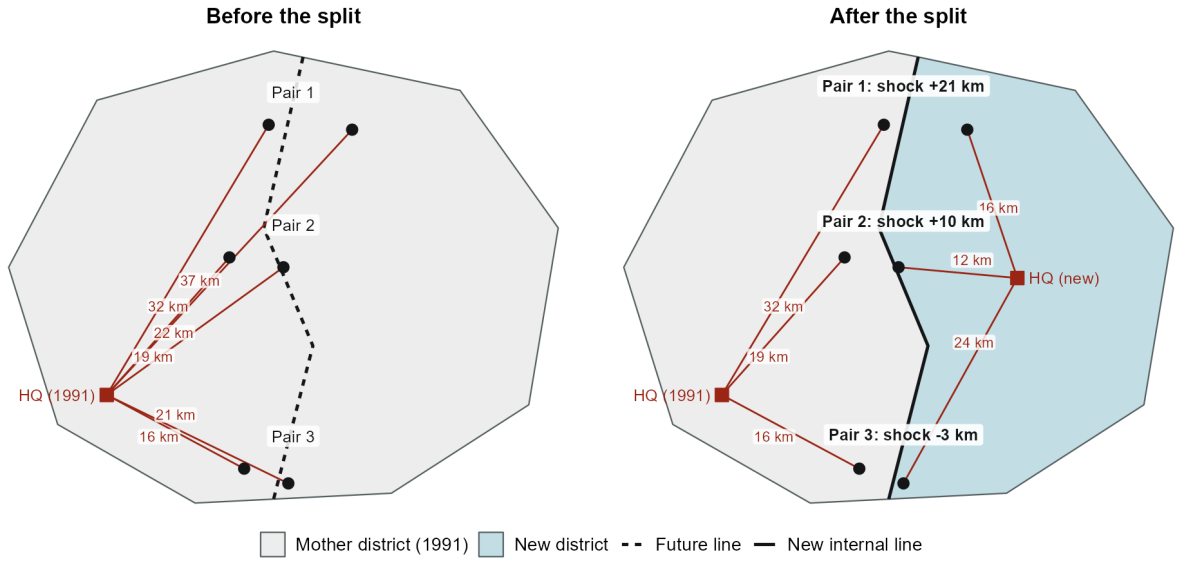


FIGURE A9: The pair design at a newly drawn internal line. Three enumeration-area pairs straddle a line that a split draws through a 1991 district. Before the split (left), both members of every pair are served by the 1991 headquarters. After it (right), the member on the new-district side is served by the new headquarters while its neighbour keeps the old one. The pair near the new headquarters receives a large relative proximity gain, the pair in the middle a moderate one, and the pair at the far end a negative one. The line fixed effect absorbs the level difference between the two sides; the variation in the shock along the line identifies the effect. Dots: enumeration areas; squares: district headquarters; red segments: distance to the assigned headquarters in kilometres.

tricts, covering 315 enumeration areas and 2,991 households. Pairs are on average 10.5 kilometres apart (median 10.3), reflecting the larger distances between settlements in rural areas.[25] The proximity shock averages $+5.6$ kilometres, with a standard deviation of 12.4 and an interquartile range from -0.7 to $+11.6$ kilometres. Overall, 47 percent of enumeration areas are closer to their headquarters after the split, while the remainder retain their 1991 headquarters. The boundaries were created between 2000 and 2019 (median 2010). I restrict the analysis to the NSDS because the Afrobarometer Round 10 sample contains only 44 cross-boundary enumeration-area pairs across 28 boundaries, providing too little variation to estimate the effects on outcomes measured exclusively in that survey (contentious engagement, bribery, and political attitudes) with enough precision.

Specification. For a pair $p = (i, j)$ at line b, where i denotes the side whose district name comes first alphabetically (an arbitrary and immaterial orientation), I estimate

$$\Delta Y_p = \alpha_b + \beta\, S_p + \lambda' B_p + \gamma' D_p + \theta' \Delta X_p + \varepsilon_p, \tag{17}$$

[25] These distances are large relative to the 300-metre city blocks studied by Blattman et al. (2025).

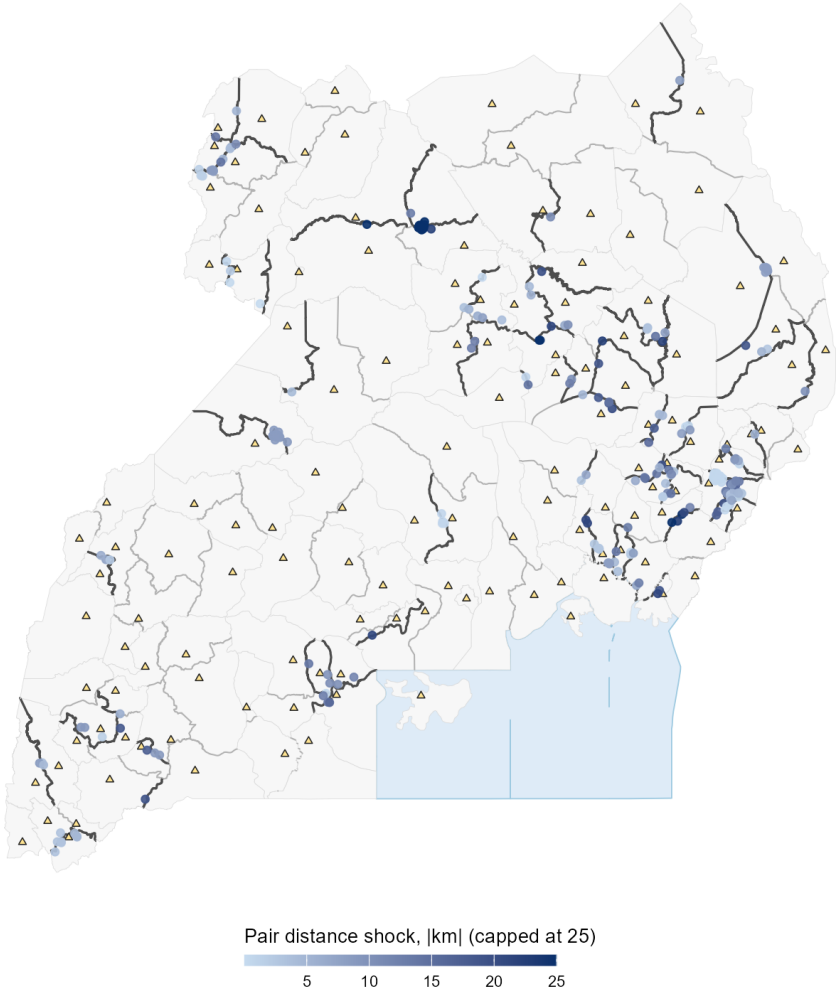


FIGURE A10: Internal lines, pairs and headquarters. Dark lines: the 78 internal 1991-district lines with NSDS 2021 cross-line enumeration-area pairs (grey: the other internal lines). Dots: the 275 pair midpoints, coloured by the absolute size of the assigned headquarters-distance shock (capped at 25 kilometres). Triangles: 2020 district headquarters. Kampala and Wakiso excluded.

where $\Delta Y_p = \bar{y}_i - \bar{y}_j$ is the difference in survey-weighted enumeration-area means; $S_p = (d_j^1 - d_j^0) - (d_i^1 - d_i^0)$ is the shock, with d^0 the distance in kilometres to the 1991 mother headquarters (shared before the split) and d^1 the distance to the own 2020 headquarters, so that $S_p > 0$ means that side i's headquarters moved closer relative to side j's and a positive β means that proximity raises the outcome; B_p contains each enumeration area's distance to a common fixed point on the line, the running variable of Blattman et al. (2025) (position along the line); D_p contains both enumeration areas' distance to the line and the distance between them; ΔX_p contains pair differences in elevation, distance to Kampala and 1990 built-up share; and α_b is the line fixed effect. Standard errors are clustered by 1991 mother district (26 clusters) and I report wild-cluster bootstrap p-values. The shock is expressed in units of 10 kilometres.

Identifying assumption and balance. Identification requires that, conditional on border fixed effects and the running variables, variation in the proximity shock along a line is unrelated to pre-existing differences between the enumeration areas in each pair. The location of the border itself may be endogenous, since district splits and boundary placement can reflect political considerations (Grossman & Lewis, 2014). However, border fixed effects absorb average differences between the two sides, while the running variables control for position along the line. The main concern is therefore that the proximity shock varies systematically with pre-existing local conditions. For example, a new headquarters may be located near a more developed area, where households on the gaining side were already better off.

Table A30 examines this concern using three sets of predetermined characteristics: geography and 1990 built-up area for all borders used, sub-county characteristics from the 2014 census for lines created after 2014, and service, engagement through LCV satisfacton, and corruption outcomes from the NSDS 2015 for borders that had not yet been created (in the spirit of a baseline balance test).[26]. Conditional on position along the new line, the proximity shock does not systematically predict differences in elevation, distance to Kampala, or 1990 built-up area. Similarly, there are no systematic differences in the 2014 census characteristics of subcounties later divided by the 2016–2020 boundaries, including electricity, water, sanitation, housing, assets, literacy, school attendance, urban share, and distances to facilities. None of these coefficients is significant at the 10 percent level using wild-cluster bootstrap inference. The 2015 NSDS outcomes also show no systematic differences. These results provide support for the identifying assumption, showing that variation in the proximity shock is not systematically related to observed pre-existing characteristics or outcomes.

[26]The NSDS 2015 does not include questions on participation in district meetings or meetings with district leaders.

TABLE A30: Internal-Border Pair Design: Balance of Predetermined Characteristics

	Mean (SD)	Pair shock (10 km)	N pairs
	(1)	(2)	(3)
Panel A: All lines — geography and 1990 built-up surface (NSDS 2021 EA pairs)			
Elevation (m)	1252	-0.2	275
	(282)	(14.5)	
		[0.99]	
Distance to Kampala (km)	216	-0.35	275
	(70)	(1.36)	
		[0.79]	
Built-up share 1990 (%)	2.69	-0.01	275
	(4.14)	(0.54)	
		[0.98]	
Panel B: Lines drawn 2016–2020 — 2014 census characteristics (NSDS 2021 EA pairs)			
Electricity, any source	0.304	-0.238	75
	(0.179)	(0.133)	
		[0.13]	
Improved drinking water	0.864	0.063	75
	(0.104)	(0.029)	
		[0.25]	
Improved sanitation	0.539	-0.207	75
	(0.249)	(0.172)	
		[0.24]	
Permanent roof	0.744	-0.021	75
	(0.264)	(0.057)	
		[0.63]	
Finished floor	0.500	-0.096	75
	(0.251)	(0.123)	
		[0.36]	
Owns radio	0.685	0.008	75
	(0.087)	(0.028)	
		[0.85]	
Literate (10+)	0.790	-0.063	75
	(0.100)	(0.053)	
		[0.24]	
Attending school (6–12)	0.935	0.045	75
	(0.029)	(0.024)	
		[0.17]	
Urban residence	0.784	-0.252	75
	(0.404)	(0.243)	
		[0.36]	
Km to public health facility	2.88	1.94	75
	(1.51)	(1.02)	
		[0.10]	
Km to public primary school	1.75	0.47	75
	(0.70)	(0.21)	
		[0.24]	
Km to police post	3.06	2.72	75
	(1.80)	(2.66)	
		[0.44]	

Notes: continued in the next table (Panel C); notes at the end of the continuation.

TABLE A30: Internal-Border Pair Design: Balance of Predetermined Characteristics (continued)

	Mean (SD)	Pair shock (10 km)	N pairs
	(1)	(2)	(3)
Panel C: Lines drawn 2016–2020 — NSDS 2015 outcomes (2015 sub-county pairs)			
Education index (district)	-0.031	0.041	49
	(0.344)	(0.077)	
		[0.50]	
Health index (district)	0.190	0.095	50
	(0.336)	(0.165)	
		[0.86]	
Water index (district)	0.096	0.002	50
	(0.397)	(0.108)	
		[0.98]	
Roads index (district)	0.118	0.011	49
	(0.481)	(0.141)	
		[0.91]	
Agricultural inputs index (district)	-0.000	-0.112*	50
	(0.399)	(0.075)	
		[0.09]	
Education index (central)	-0.010	-0.040	49
	(0.338)	(0.087)	
		[0.51]	
Water index (central)	0.074	0.113	50
	(0.421)	(0.127)	
		[0.51]	
Electricity index (central)	-0.021	0.027	50
	(0.256)	(0.117)	
		[0.89]	
LCV performs well	3.490	0.084	45
	(0.527)	(0.150)	
		[0.53]	
Aware: nepotism	0.600	0.022	50
	(0.282)	(0.040)	
		[0.58]	
Aware: embezzlement	0.726	0.016	50
	(0.212)	(0.057)	
		[0.84]	
Aware: absenteeism	0.613	-0.058	50
	(0.243)	(0.070)	
		[0.40]	
Victim: nepotism	0.090	0.036	50
	(0.101)	(0.037)	
		[0.38]	
Victim: embezzlement	0.066	0.064	50
	(0.088)	(0.024)	
		[0.27]	
Victim: absenteeism	0.139	0.018	50
	(0.146)	(0.041)	
		[0.58]	

Notes: *p<0.1, **p<0.05, ***p<0.01. Each row regresses the within-pair difference in a predetermined variable on the pair shock (10 km units) with border fixed effects, the fixed-point distances, distances to the line and the pair distance; the three covariate differences used as controls elsewhere are omitted because they are among the variables tested. Column 1 reports the mean (SD) of the variable across units. Panels A and B use the NSDS 2021 EA pairs; Panel B attaches to each EA the 2014 census mean of its sub-county and keeps the 14 lines activated after the census (75 pairs). Panel C tests the same lines in NSDS 2015 sub-county-centroid pairs (45–50 pairs), before the lines existed, with the shock implied by the 2020 headquarters; units are sub-county centroids because the 2015 survey has no enumeration-area coordinates. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets.

I.2 First Stage

The proximity shock is constructed from geographic data and measures how much closer one side's headquarters moved relative to the other. I first examine whether this shock translates into changes in the distance households actually face. To do so, I use two measures of proximity: households' self-reported distance to district headquarters and GIS-based distance from their enumeration area to the headquarters. I regress the within-pair difference in each measure (in logs) on the shock, including the same fixed effects and controls as in the outcome regressions (Table A31). Both measures respond strongly. A 10-kilometre relative gain increases log proximity by 0.69 using GIS distances and by 0.52 using self-reported distances (wild-bootstrap $p < 0.001$ in both cases). The self-reported first stage is stable across specifications: 0.58 without fixed-point distance controls, 0.53 with line-by-direction fixed effects, 0.54 when excluding pairs separated by water, and 0.53 for pairs less than 15 kilometres apart. The estimate becomes less precise within the 5-kilometre band (0.24, $p = 0.15$), which contains only 80 pairs. Figure A11 illustrates this relationship. Each line is oriented so that the side that gained proximity is on the right. The change in distance to the assigned headquarters jumps by approximately 12 kilometres at the boundary, while log GIS-based and self-reported proximity increase by 0.9 and 0.2, respectively.

TABLE A31: Internal-Border Pair Design: First Stage

	Pair shock (10 km) (1)	N pairs (2)
Within-pair difference in log proximity to the headquarters		
Self-reported distance (the paper's endogenous variable)	0.517***	246
	(0.126)	
	[0.00]	
GIS distance, household to headquarters	0.687***	275
	(0.161)	
	[0.00]	

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow. The dependent variable is the within-pair difference in log proximity to the district headquarters, self-reported (the paper's endogenous variable; 246 pairs) or GIS-measured. The table is a relevance check on the assigned shock; the shock itself, not experienced proximity, is the treatment in the pair-design tables.

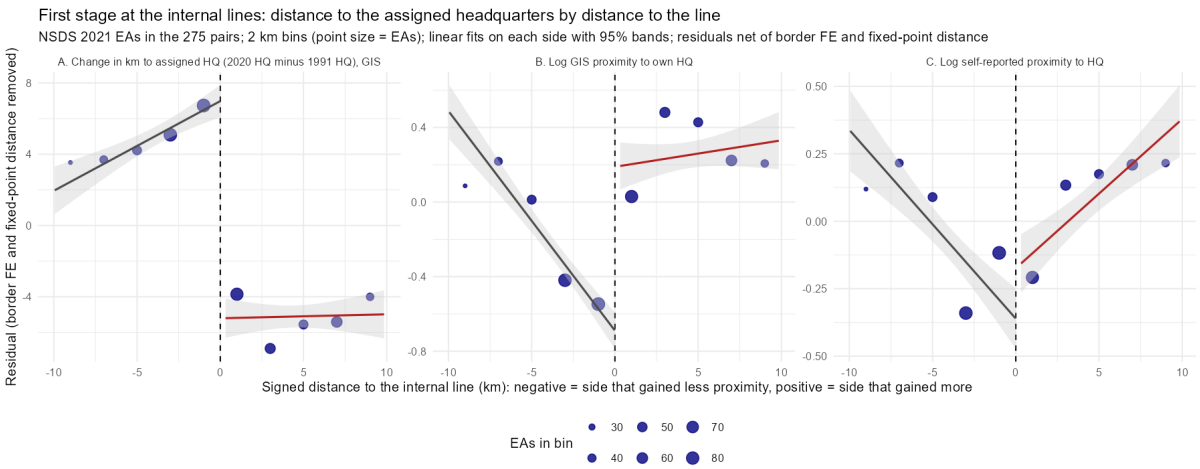


FIGURE A11: First stage at the internal lines. NSDS 2021 enumeration areas in the 275 pairs, in 2-kilometre bins (point size proportional to the number of enumeration areas), residualised on line fixed effects and the fixed-point distance; linear fits on each side with 95 percent bands. The horizontal axis is the signed distance to the line, negative on the side that gained less proximity and positive on the side that gained more. Panel A: change in kilometres to the assigned headquarters (2020 headquarters minus 1991 headquarters). Panel B: log GIS proximity to the own headquarters. Panel C: log self-reported proximity.

I.3 Results

The boundary design is particularly suited to analyze households' interactions with their own district administration, such as meeting the Resident District Commissioner, attending RDC barazas, and evaluating the performance of local officials. These outcomes are closely tied to the administration to which households belong and are therefore less likely to spill over across district boundaries. Public services are different. Boreholes, roads, and health facilities near a boundary may serve households on both sides, reducing the differences captured by the within-pair comparison. Moreover, since the design focuses on households near district boundaries, it may miss improvements concentrated closer to the headquarters. I therefore expect smaller service effects than in the main specification, even if administrative proximity improves service delivery throughout the district.

I.3.1 Public Services

Table A32 reports the results for the ten service indices in Table 3. I expect smaller effects in this design for three reasons. First, public services such as boreholes, roads, and health facilities can benefit households on both sides of a district boundary, so within-pair differences may not capture improvements shared by neighbouring households. Second, the largest proximity gains occur near the headquarters of recently created districts, which may not yet have fully developed their administrative capacity (Section 9). Third, the sample is relatively small, with 275 pairs compared with 9,313 households in the main specification. Consistent with these

limitations, none of the ten estimates is statistically significant. However, the confidence intervals are wide and generally include the magnitudes implied by the main 2SLS estimates. For example, the 95 percent confidence intervals are $[-0.26, 0.33]$ for electricity, compared with a 2SLS-implied effect of 0.22 per 10-kilometre shock, and $[-0.14, 0.33]$ for district health, compared with 0.20. The same holds for central-government roads and district water. These estimates therefore do not provide independent evidence of service improvements, but neither do they rule out the effects found in the main specification.

TABLE A32: Internal-Border Pair Design: Public Service Delivery

Panel A: District-provided services (NSDS 2021)					
	Education	Health	Water	Roads	Agricultural inputs
	(1)	(2)	(3)	(4)	(5)
Pair shock (10 km)	0.199	0.099	0.035	-0.021	-0.002
	(0.157)	(0.120)	(0.063)	(0.063)	(0.059)
	[0.22]	[0.36]	[0.58]	[0.71]	[0.98]
Mean of DV	0.097	0.133	0.083	0.016	-0.010
Pairs	274	273	275	275	275
Panel B: Central-government services (NSDS 2021)					
	Education	Health	Water	Roads	Electricity
	(1)	(2)	(3)	(4)	(5)
Pair shock (10 km)	0.200	-0.144	0.119*	0.093	0.036
	(0.140)	(0.175)	(0.081)	(0.151)	(0.150)
	[0.11]	[0.34]	[0.09]	[0.52]	[0.84]
Mean of DV	0.038	0.120	0.070	-0.135	-0.025
Pairs	270	275	275	275	275

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow. Service indices as in Table 3.

I.3.2 Citizens' Engagement and Satisfaction with the Administration

I first examine whether changes in administrative proximity affect households' engagement with their district government. Table A33 reports the results for the participatory engagement outcomes in Table 9. Households experiencing larger relative gains in proximity are more likely to have met the Resident District Commissioner (4.6 percentage points per 10 kilometres, wild-bootstrap $p = 0.02$) and report higher participatory engagement (0.095 standard deviations per

10 kilometres, $p = 0.04$). The coefficients for participation in LCV meetings and RDC barazas are also positive but not statistically significant.

To compare these magnitudes with the main specification, I scale the estimates by the first-stage effect on log proximity. A 10-kilometre relative gain increases log proximity by 0.69 using GIS distances and 0.52 using self-reported distances. This implies an effect of 0.14 to 0.18 per log point for the participatory engagement index, compared with 0.169 in Table 9, and an increase of 7 to 9 percentage points in the likelihood of meeting the RDC, compared with 4 percentage points in the main specification. Table A34 shows similar results for satisfaction with local administrators. A 10-kilometre relative gain increases satisfaction with the RDC by 11 percentage points ($p = 0.06$), equivalent to 0.16 to 0.22 per log point of proximity, compared with 0.122 in Table 5. Satisfaction with the LCV chairperson and civil servants shows no statistically significant response, consistent with the main results.

TABLE A33: Internal-Border Pair Design: Citizens' Engagement with the Local Government

Participatory engagement (NSDS 2021)				
	Involved in LCV meetings	Met the RDC	Attended RDC baraza	Participatory index
	(1)	(2)	(3)	(4)
Pair shock (10 km)	0.011	0.046**	0.022	0.095**
	(0.024)	(0.016)	(0.015)	(0.041)
	[0.61]	[0.02]	[0.16]	[0.04]
Mean of DV	0.196	0.094	0.048	-0.040
Pairs	275	275	275	275

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow. Indices are z-score means of the components in Table 9.

TABLE A34: Internal-Border Pair Design: Satisfaction with Local Leaders

NSDS 2021	LCV performs well (1)	Satisfied with the RDC (2)	Civil servants perform well (3)
Pair shock (10 km)	0.025	0.112*	-0.032
	(0.042)	(0.054)	(0.040)
	[0.45]	[0.06]	[0.39]
Mean of DV	0.392	0.322	0.597
Pairs	275	275	274

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow.

I.3.3 Corruption Perceptions

Table A35 reports the corruption outcomes from Panel A of Table 11. As with public services, the boundary design does not yield statistically significant effects on the overall indices. The estimates are close to zero, providing no evidence of either improvements or deterioration in accountability. Among individual outcomes, victimization by absenteeism increases significantly ($p = 0.04$), but this effect does not translate into a significant change in the overall victimization index. Bribery and political attitudes are measured only in Afrobarometer, where the small number of cross-boundary pairs (44) limits the precision of the estimates, so I do not report them.

TABLE A35: Internal-Border Pair Design: Corruption Awareness and Victimization

NSDS 2021	Aware: nepotism (1)	Aware: embezzlement (2)	Aware: absenteeism (3)	Awareness index (4)	Victim: nepotism (5)	Victim: embezzlement (6)	Victim: absenteeism (7)	Victimization index (8)
Pair shock (10 km)	-0.057*	0.033	-0.023	-0.030	-0.013	0.019	0.039**	0.059
	(0.038)	(0.065)	(0.040)	(0.077)	(0.016)	(0.013)	(0.020)	(0.055)
	[0.10]	[0.58]	[0.56]	[0.67]	[0.39]	[0.15]	[0.04]	[0.28]
Mean of DV	0.481	0.660	0.456	-0.020	0.031	0.034	0.046	-0.043
Pairs	275	275	275	275	275	275	275	275

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow. Indices as in Table 11.

I.4 Falsification and Robustness

Placebo before the lines existed. Panel C of Table A30 and Table A36 apply the boundary design to the NSDS 2015 at lines created between 2016 and 2020, before these boundaries

existed; the former reports the service indices, LCV performance and the corruption indicators, the latter the individual service components behind the indices. The future proximity shock does not significantly predict any of these 29 outcomes at the 5 percent level. Only one outcome, distance to school, is significant at the 10 percent level, consistent with what could arise by chance due to multiple hypothesis testing.

Robustness. Table A37 re-estimates the citizen engagement outcomes under alternative specifications. The results are similar when excluding covariate differences, dropping pairs separated by water, and clustering standard errors by line. The estimated effects are larger when excluding fixed-point distance controls.

Inference. The main specifications cluster standard errors by 1991 mother district and report wild-cluster bootstrap p-values. This allows for arbitrary correlation across boundaries belonging to the same mother district. Since the proximity shock depends on the locations of the two headquarters associated with each boundary, I also consider alternative clustering levels. Table A38 shows that clustering by boundary produces similar standard errors, while two-way clustering by the two 2020 districts reduces them by 10 to 15 percent. Conley spatial HAC standard errors with cutoffs of 50 and 100 kilometres are 25 to 35 percent smaller. The main results therefore remain robust to these alternative approaches to inference, with clustering by mother district producing the largest standard errors among those considered.

TABLE A36: Internal-Border Pair Design: NSDS 2015 Service Components at Lines Not Yet Drawn

	Mean (SD) (1)	Pair shock (10 km) (2)	N pairs (3)
School attendance, primary age	0.881	-0.053	49
	(0.138)	(0.042)	
		[0.51]	
School attendance, secondary age	0.791	0.064	49
	(0.215)	(0.035)	
		[0.12]	
Distance to school (km)	1.497	-0.427*	49
	(0.714)	(0.148)	
		[0.08]	
School travel time (min)	30.98	-2.16	49
	(14.93)	(2.80)	
		[0.30]	
Uses public hospital	0.942	-0.009	49
	(0.114)	(0.025)	
		[0.49]	
Travel time to health care	43.36	0.54	50
	(31.65)	(10.20)	
		[0.94]	
Improved water source	0.773	0.007	50
	(0.310)	(0.108)	
		[0.91]	
Distance to water (km)	0.872	0.025	50
	(0.602)	(0.102)	
		[0.77]	
Electricity	0.177	0.026	50
	(0.194)	(0.089)	
		[0.91]	
Trunk road	0.760	0.007	49
	(0.381)	(0.068)	
		[0.92]	
Feeder road	0.938	-0.006	49
	(0.201)	(0.039)	
		[0.83]	
Communal road	0.937	-0.007	49
	(0.218)	(0.019)	
		[0.49]	
Hybrid seeds	0.160	-0.031	50
	(0.148)	(0.062)	
		[0.53]	
Fertilizer	0.079	-0.042	50
	(0.118)	(0.047)	
		[0.23]	

Notes: *p<0.1, **p<0.05, ***p<0.01. The main specification estimated on NSDS 2015 outcomes at the internal lines activated in 2016–2020, before the lines existed, with the shock implied by the 2020 headquarters; units are sub-county centroids (NSDS 2015 has no EA coordinates). Column 1 reports the mean (SD) of the outcome. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets.

TABLE A37: Internal-Border Pair Design: Robustness of the Headline Estimates

	Met the RDC (1)	Attended RDC baraza (2)	Satisfied with the RDC (3)	Participatory index (4)	N pairs (5)
Main specification	0.046**	0.022	0.112*	0.095**	275
	(0.016)	(0.015)	(0.054)	(0.041)	
	[0.02]	[0.16]	[0.06]	[0.04]	
Without fixed-point distances (border FE + distances to line)	0.055***	0.031**	0.120**	0.112**	275
	(0.014)	(0.016)	(0.050)	(0.037)	
	[0.00]	[0.04]	[0.03]	[0.01]	
Without covariate differences	0.045**	0.022	0.110*	0.093**	275
	(0.017)	(0.015)	(0.056)	(0.041)	
	[0.02]	[0.17]	[0.06]	[0.04]	
Border × treatment-direction fixed effects	0.030	-0.004	0.056	0.041	275
	(0.023)	(0.013)	(0.059)	(0.061)	
	[0.17]	[0.65]	[0.27]	[0.41]	
Dropping water-separated pairs and lines	0.048**	0.022	0.107*	0.089*	265
	(0.019)	(0.017)	(0.056)	(0.047)	
	[0.04]	[0.24]	[0.10]	[0.08]	
Clustering by border (78)	0.046***	0.022	0.112*	0.095**	275
	(0.017)	(0.017)	(0.053)	(0.042)	
	[0.01]	[0.27]	[0.06]	[0.03]	
Both EAs within 5 km of the line	0.011	-0.002	0.192	0.003	106
	(0.067)	(0.043)	(0.087)	(0.164)	
	[0.85]	[0.95]	[0.11]	[0.98]	
Pair distance $\leq$15 km	0.040	0.029	0.100	0.102	230
	(0.021)	(0.017)	(0.053)	(0.055)	
	[0.10]	[0.18]	[0.12]	[0.12]	
Mean of DV	0.094	0.048	0.322	-0.040	

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column is a regression of the within-pair difference in the enumeration-area mean of the outcome on the pair shock (relative gain in proximity to the assigned headquarters, 10 km units), with border fixed effects, each EA's distance to a fixed point on the border, both EAs' distance to the line, the pair distance, and pair differences in elevation, distance to Kampala and 1990 built-up share. Standard errors clustered at the 1991 mother-district level in parentheses; wild-cluster bootstrap p-values in brackets, which the stars follow. Each row re-estimates the main specification under the named variant. Border × direction fixed effects absorb the within-line switch between the gaining and the losing side (only 9 of 78 lines contain both). Water-separated pairs are divided by a major river or lake or lie on a line that follows a river (10 pairs).

TABLE A38: Internal-Border Pair Design: Alternative Inference

	First stage (self-reported) (1)	Met the RDC (2)	Attended RDC baraza (3)	Satisfied with the RDC (4)	Participatory index (5)
Pair shock (10 km)	0.517	0.046	0.022	0.112	0.095
1991 mother district (26 clusters)	(0.126)***	(0.016)***	(0.015)	(0.054)**	(0.041)**
	[0.00]	[0.02]	[0.16]	[0.06]	[0.04]
Line (78 clusters)	(0.136)***	(0.017)***	(0.017)	(0.053)**	(0.042)**
	[0.00]	[0.01]	[0.27]	[0.06]	[0.03]
Two-way: the pair's two 2020 districts (91)	(0.110)***	(0.014)***	(0.014)	(0.047)**	(0.038)**
Conley spatial HAC, Bartlett 50 km	(0.087)***	(0.013)***	(0.012)*	(0.041)***	(0.031)***
Conley spatial HAC, Bartlett 100 km	(0.068)***	(0.013)***	(0.011)*	(0.042)***	(0.032)***

Notes: *p<0.1, **p<0.05, ***p<0.01. Standard errors for the main-specification estimates under alternative inference, with the stars they imply; wild-cluster bootstrap p-values in brackets where computed. Two-way clustering uses `reghdfe`; Conley standard errors (`acreg`, Bartlett kernel) are computed at pair midpoints after partialling out the border fixed effects.